A Regime-Switching Continuous Hidden Markov Model as a Reference Synthetic-Data Generator for Equity Returns: Extended Evaluation, Semi-Markov Ablation, and Regime-Conditional Value-at-Risk

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Abstract

We develop a family of regime-switching continuous hidden Markov models (CHMMs) as a reference synthetic-data generator for daily equity returns and evaluate it against a broad panel of non-parametric, classical, discrete-state, and deep-generative baselines. The three CHMM variants share a log-space Expectation–Maximization (EM) scaffold with a common quantile-initialized transition matrix and differ only in their per-state emission law: Gaussian (CHMM-N), Student-t with data-selected per-state degrees of freedom (CHMM-t), and Laplace (CHMM-L). At a single moderate state resolution the continuous models match the canonical stylized-fact set of daily equity returns (heavy tails, negligible linear autocorrelation, persistent volatility clustering, leverage effect, and aggregational Gaussianity) within a standard CHMM framework, complementing the hidden semi-Markov and jump-augmented routes already available in the literature. The contribution of this paper is the integrative evaluation built on this unified scaffold rather than a new estimator or complexity-reduction mechanism, with each operational claim grounded in a stated proposition (Section 5): a closed-form spectral expression for the absolute-return ACF that recasts the Rydén et al. low-K limitation as a K-rank statement on $\mathbf{T}$, ECM monotonicity for the CHMM-t variant on a compact bracket, identifiability and MLE consistency under stated assumptions, exact marginal preservation under rank reordering, and structural inequalities that explain the filter-VaR and discrete-baseline VaR Kupiec failures as deterministic features of the mixture-quantile and bin-centroid scaffolds rather than sampling artefacts.

On ten years of daily SPDR S&P 500 exchange-traded fund (ETF) data with ticker SPY, we benchmark the generator family along three operational axes. First, on distributional fidelity, flat CHMM-t attains the smallest Maximum Mean Discrepancy (MMD) of any row in the panel and the discriminator Area Under the Curve (AUC) closest to 0.5 among parametric and Markov-switching rows, second only to a window-diffusion baseline on AUC and ahead of generative adversarial and Markov-Switching GARCH (MS-GARCH) baselines. Second, on unconditional Value-at-Risk (VaR) calibration, a semi-Markov CHMM ablation and a two-regime MS-GARCH baseline both achieve a Kupiec likelihood-ratio statistic near the coarse-grid floor ($\mathrm{LR}_{\mathrm{uc}} \approx 0.01$) at 1%, corresponding to exact-target calibration (breach rate 1.0%) rather than a resolved

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head-to-head ranking, with MS-GARCH alone leading at 5%. Third, on regime-conditional VaR, we separate two constructions: a smoother-based in-sample (IS) goodness-of-fit diagnostic of the regime structure (a Viterbi-decoded state-conditional quantile, which depends on the full observation sequence and is therefore not a one-step-ahead backtest) and the operationally correct filter-based one-step-ahead backtest that uses the forward posterior $P(s_t \mid r_{1:t})$ under the IS-fit parameters. On the smoother diagnostic, flat CHMM-t clears both the Kupiec unconditional-coverage and the Christoffersen independence likelihood-ratio thresholds at 1% and 5% on the held-out 2024 to 2026 window. On the filter-based one-step-ahead backtest, all three CHMM families *fail* Kupiec at both 1% and 5% on the same window (breach rate $\approx 0.2\%$ vs. 1% target and 1.2–1.4% vs. 5% target), because the mixture of heavy-tailed per-state emissions weighted by the filter posterior is systematically over-conservative; this is an honest negative finding on the operational regime-conditional VaR axis, consistent with the smoother-based diagnostic evidence that the regime partition itself is well specified on this window but the filter averages away the contraction the real-time VaR would need. All conditional-coverage statements in this paper at $T_{\mathrm{OoS}} = 572$ should be read against the small-sample independence-test caveat that roughly five to seven breaches are expected at $\alpha = 1\%$. We also report a concrete negative finding against a discrete-state plus Poisson-jump baseline, whose 5% VaR breach rate is overly conservative at 1.7% and fails Kupiec with a likelihood ratio of 16.81. We extend the framework to multi-asset synthesis through a Single Index Model and Gaussian and Student-t copulas on six equities, with a truncated C-vine variant reported as a visual reference on the same universe; the flat Student-t copula remains the most competitive dependence layer in our setup.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Semi-Markov, Stylized Facts, Volatility Clustering, Copula Dependence, Cross-Asset Simulation, Value-at-Risk, Kupiec, Christoffersen, Maximum Mean Discrepancy, Synthetic-Data Generator.

1 Introduction

A synthetic generator of financial time series is useful downstream only if it reproduces the canonical stylized facts of the underlying observable simultaneously: for equity returns, heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [1, 2]; Section 6.6 additionally evaluates the leverage effect and aggregational Gaussianity, the remaining two facts on the Cont list. Any such generator, intended for downstream tasks such as risk measurement, scenario generation, or regime-aware dynamic asset allocation [3], must reproduce all three simultaneously [4, 5], and should be *practical to train*, so that the scaffold can be reused across observables without architectural surgery. GARCH models [6, 7] capture volatility clustering but cannot represent the multi-regime structure of the marginal, while i.i.d. generators match the marginal but destroy temporal persistence. Hidden Markov models (HMMs), popularized in macroeconometrics by Hamilton [8] and extended to stock-market returns by Schaller and Van Norden [9], offer a natural middle path through a latent regime structure.

An influential negative result has shaped two decades of work on this problem. Rydén et al. [10] fitted Gaussian-emission HMMs with $K = 2$–3 states and concluded that, while HMMs reproduce heavy tails and absent return autocorrelation, they *fail* to capture the slow ACF decay of absolute returns: the geometric sojourn-time distribution of a standard Markov chain produces exponential decay that is too fast. Bulla and Bulla [11] restored ACF fidelity with hidden semi-Markov models (HSMMs) at the cost of substantially increased complexity, and Alswaidan and Varner [12] reached the same end within a discrete HMM by adding a Poisson jump-duration mechanism calibrated through two extra hyperparameters.

We identify the spectral root cause of the low-K failure: Theorem 1 in Section 5 derives a closed-form mixture-of-eigenvalues expression for the absolute-return ACF, $\rho_{|G|}(\tau) = \sum_{k=2}^{K} w_k \lambda_k^{\tau}$, with λ_k

the non-unit eigenvalues of $\mathbf{T}$. Corollary 1 then converts the empirical Rydén et al. observation into a K-rank statement: at $K = 2$ the ACF is mono-exponential and matching a slow-decay target forces $|\lambda_2| \to 1$, which under Assumption 2 collapses the marginal mixture and fails distributional KS; at $K \geq 3$ the constraints decouple. The HSMM and jump-augmented routes inject the missing temporal flexibility through alternative mechanisms; moderate-K CHMM achieves the same end through the spectral richness of the transition matrix.

Rather than offer a fourth complexity-reduction mechanism, our contribution is an integrative evaluation that asks, on a single unified EM scaffold, how much of what the HSMM and jump-augmented corrections buy is already available from a standard CHMM at moderate state resolution, and along which operational axes any residual gap matters. Our work builds on the established continuous-emission HMM-for-finance literature of Hamilton [8] (two-regime Gaussian), Bulla and Bulla [11] (semi-Markov dwell-time fix), Nystrup et al. [13] (continuous emissions for long-memory fit), and the Student-t / GED ECM updates of Peel and McLachlan [14], Liu and Rubin [15], Abanto-Valle et al. [16]. Consistent with those antecedents, we combine on a single shared scaffold: (i) per-state Student-t degrees-of-freedom ν_k updated inside an Expectation–Conditional-Maximization (ECM) step with a golden-section bracket, following Peel and McLachlan [14], Liu and Rubin [15]; (ii) a closed-form weighted-mean-absolute-deviation M-step for Laplace emissions; (iii) a head-to-head benchmark against the jump-augmented discrete HMM of Alswaidan and Varner [12] under the same evaluation protocol; and (iv) a two-pipeline composition in which a shared continuous emission scaffold (Pipeline A) composes with a copula on the rank-transformed path (Pipeline B) for cross-asset synthesis. The novelty claim attaches to the integration of these ingredients and to the three-axis evaluation, not to any individual axis taken alone. A continuous HMM trained by the Baum-Welch algorithm [17, 18] with quantile-based initialization reproduces all three stylized facts at moderate state resolution, alongside the HSMM [11] and jump-augmented [12] variants already available in the literature; the slow-ACF machinery resides in the transition-matrix eigenvalue spectrum rather than in an explicit sojourn distribution or jump process. The same EM scaffold accommodates Gaussian, Student-t, and Laplace emissions with only the M-step differing between variants. Alongside the classical Gaussian variant (CHMM-N) we introduce and evaluate two heavy-tailed variants sharing identical forward-backward recursions: CHMM-t, with per-state Student-t emissions whose degrees-of-freedom ν_k are updated inside the M-step by a golden-section search on the ECM Q-function in the tradition of Peel and McLachlan [14] and Liu and Rubin [15]; and CHMM-L, with per-state Laplace emissions whose location is the weighted median and whose scale is the weighted mean absolute deviation, both in closed form. We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$; $K = 18$ sits inside the IC-minimizing band $\{15, 18, 21\}$ under the four information criteria (AIC, BIC, HQC, CAIC) used for HMM state selection in stock-price prediction by Nguyen [19] and, within that band, wins or ties every IS and OoS distributional pass-rate metric. This is consistent with broader regime-switching evidence that the Rydén et al. [10] $K = 2$–3 Gaussian result is specific to low state resolution: once the transition matrix has heterogeneous diagonal entries across a larger number of regimes, its eigenvalue spectrum provides a mixture of geometric sojourn times that approximates slow ACF decay, which complements the HSMM [11] and jump-augmented [12] routes rather than replacing them.

Against eight alternative generators on ten years of daily SPY data (i.i.d. bootstrap, stationary block bootstrap, Gaussian and Laplace i.i.d., the discrete HMM of Alswaidan and Varner [12] in no-jump (NJ) and Poisson-jump (WJ) variants at the authoritative prior-paper hyperparameters $K_{\text{disc}} = 90$, $\epsilon = 5 \times 10^{-5}$, $\lambda = 67$, GARCH(1,1), and a single-layer gated recurrent unit (GRU) plus Gaussian-head autoregressive density baseline trained by negative log-likelihood), the Gaussian CHMM-N at $K = 18$ attains 93.8% in-sample (IS) and 84.2% out-of-sample (OoS) Kolmogorov–Smirnov (KS) pass rates and a mean absolute error (MAE) of the absolute-return autocorrelation function (ACF-MAE) of 0.0507, essentially matching GARCH's temporal fidelity without sacrificing distributional fit. The heavy-tailed CHMM-t and CHMM-L variants retain this ACF and pass-rate fidelity; CHMM-L brings the simulated IS excess kurtosis to 6.75, close to the observed 7.68, while CHMM-t overshoots at 14.57 (see Section 8). At the low-K end of the sweep CHMM-t already beats CHMM-N on IS KS (91.1% at $K = 3$ versus 89.8%) because the per-state heavy tails of Student-t compensate for under-resolved regime structure: a small number of heavy-tailed regimes can cover the return distribution that a larger number of Gaussian regimes is needed to tile. We diagnose the CHMM-t IS kurtosis over-correction (14.57 versus observed 7.68) by reporting the per-state ν_k histogram and a $\nu_{\min}$ bracket-sensitivity study, and quantify its downstream consequence with a 1%/5% Value-at-Risk and Expected-Shortfall (ES) calibration against the historical series. Univariate generalization is confirmed on NVDA, JNJ, JPM, AAPL, and QQQ for all three emission families; the OoS gap on NVDA and JPM under the 2024–2026 regime is acknowledged as a stationarity scope point and a quarterly refit diagnostic is reported in Appendix E, but is not a headline methodology of the paper. For multi-asset synthesis, we extend the univariate framework with a Single Index Model [20] and Gaussian / Student-t copulas [21–23], both of which preserve each asset's fitted CHMM marginal. On six SPY-member assets the copulas substantially outperform SIM on per-asset KS fidelity (median 96.0–97.8% vs. 78.5%) and correlation-matrix reproduction (off-diagonal MAE 0.027 Student-t, 0.031 Gaussian, 0.076 SIM), with the Student-t copula's profile maximum likelihood estimate (MLE) selecting $\nu^* = 6$ degrees of freedom. Option pricing, volatility-index generalization, and leverage-effect modelling are explicitly out of scope and are deferred to future work. Beyond distributional fidelity we evaluate the generator family on unconditional Value-at-Risk calibration (Kupiec) and regime-conditional VaR (Christoffersen independence), with the regime-conditional axis split into a smoother-based IS goodness-of-fit diagnostic (Viterbi-decoded state-conditional quantile, which depends on future observations) and the operationally correct filter-based one-step-ahead backtest using the forward posterior $P(s_t \mid r_{1:t})$; the smoother diagnostic passes both Kupiec and Christoffersen for flat CHMM-t, while the filter backtest fails Kupiec for all three CHMM families (over-conservative), an honest negative finding on the operational regime-conditional VaR axis that Proposition 5 grounds in a mixture-quantile inequality rather than a sampling artefact. The paper therefore splits its evaluation across three operational axes rather than treating back-testing as an afterthought.

Scope. The seven-metric panel in Section 3.5 evaluates distributional, tail, and temporal fidelity of the generated return series. The 1%/5% VaR and Expected-Shortfall back-test in Section 7.1 grounds those fidelity numbers in a concrete downstream risk-measurement consumer. The input data are public daily U.S. equity prices, so no privacy or individual-level considerations apply.

The remainder of the paper is organized as follows. Section 2 positions the work against related literature. Section 3 describes the data, the CHMM, the benchmark models, the cross-asset constructions, and the evaluation protocol; Section 4 develops the EM estimator and state-count selection; Section 5 collects the theoretical scaffolding (assumptions, the spectral ACF theorem, ECM monotonicity, identifiability, MLE consistency, rank-reordering marginal preservation, the filter-VaR mixture-quantile inequality, and the discrete-VaR centroid floor); formal algorithms and metric definitions are in Appendices A and B. Section 6 presents the empirical study and Section 7 the VaR back-test; Sections 8–9 discuss and conclude.

2 Related Work

Stylized facts and parametric volatility. The canonical set of stylized facts [2], namely heavy tails [1], absence of linear autocorrelation [24], and slow decay of the absolute-return ACF [25–27], must be reproduced jointly for a generator to be useful downstream. Autoregressive conditional heteroscedasticity (ARCH) and its generalised variant (GARCH) [6, 7] encode volatility persistence through a single decay parameter but impose a unimodal innovation distribution and, as we confirm, fail distributional tests on equity returns. Jump-diffusion models [28, 29] generate heavy tails but lack a clustering mechanism: after a jump the conditional variance returns immediately to baseline; Maheu and McCurdy [30] couple jumps with a regime-switching volatility component to address this, an alternative route to regime-dependent tails that our CHMM-t scaffold addresses through heavy-tailed emissions rather than an explicit jump process. Maximum-likelihood fitting of heavy-tailed innovations in a latent-volatility framework has been studied for stochastic-volatility-in-mean models by Abanto-Valle et al. [16], who compare Normal, Student-t, generalized error distribution (GED), and Slash errors; our CHMM-t and CHMM-L are the HMM analog of that emission-family choice.

Regime-switching and the Rydén et al. limitation. Hamilton [8] introduced Markov-switching models to economics, and Schaller and Van Norden [9] provided an early demonstration that monthly equity returns from the Center for Research in Security Prices (CRSP) are well-described by a two-state Markov-switching specification with strongly different mean and variance between regimes; Ang and Bekaert [31] subsequently showed that short-rate dynamics in several developed markets also admit a regime-switching representation, establishing regime-switching as a cross-asset-class phenomenon rather than one confined to macro series or equities. The seminal evaluation of HMMs against the Cont stylized facts is due to Rydén et al. [10], who fitted Gaussian-emission HMMs with $K = 2$–3 states to daily equity-index return series and showed that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay. Bulla and Bulla [11] restored fidelity with hidden semi-Markov models (HSMMs) that replace the geometric sojourn distribution with flexible alternatives. Alswaidan and Varner [12] instead kept the discrete HMM framework and introduced a Poisson jump-duration process to force dwell time in tail states.

Within the same family, Nguyen [19] applies a four-state continuous HMM to monthly S&P 500 prices for OoS trading signal generation, selecting the state count through the AIC, BIC, HQC, and CAIC information criteria. We adopt the same four-criterion selection (Section 6.2), in addition to our multi-metric distributional score, to place the K-sweep recommendation on standard statistical footing.

Consistent with this broader regime-switching literature, we find that at $K \geq 3$ with quantile-based initialization the Baum-Welch algorithm learns heterogeneous self-transition probabilities across regimes, and the resulting mixture of K geometric sojourn-time distributions approximates slow ACF decay. This moderate-K route complements the HSMM [11] and jump-augmented [12] corrections to the Rydén et al. result rather than superseding them; the integrative contribution of this paper is the unified log-space EM scaffold shared by the three emission variants, the twelve-generator evaluation panel, and the regime-conditional VaR test (Section 7), not a single-route resolution of the Rydén limitation.

Theoretical antecedents and our contribution. Section 5 assembles a short stack of propositions, each with a precedent in the literature, into a theoretical scaffold for the empirical study. The closest precedents are: Hamilton [8] and Frühwirth-Schnatter [32] for spectral analysis of regime-switching models (we specialise to a closed form for the absolute-return ACF in Theorem 1);

Allman et al. [33] for general HMM identifiability and Yakowitz and Spragins [34] for finite-mixture identifiability (we instantiate for CHMM-N, CHMM-t, CHMM-L in Proposition 2); Bickel et al. [35] and Douc et al. [36] for HMM-MLE consistency under our assumptions (Proposition 3); Wu [37] and Meng and Rubin [38] for EM and ECM monotonicity (we specialise to the golden-section CM step on a compact bracket in Proposition 1); Iman and Conover [39] for rank-reordering simulation (Proposition 4); McNeil et al. [23] for mixture-VaR theory (we sharpen to a one-sided inequality that explains the empirical filter-VaR Kupiec failure in Proposition 5); and a textbook spectral-mixing argument from Levin and Peres [40]. The contribution is the assembly: bringing these results together inside a single CHMM scaffold, anchoring each one to a row in the empirical evaluation, and converting heuristic prose into stated propositions.

Discrete HMM with jumps. The immediate predecessor of this paper, Alswaidan and Varner [12], discretizes returns into Laplace quantile bins, estimates $\mathbf{T}$ by frequency counting, fits bin-conditional Student-t emissions, and adds a Poisson jump-duration process with two grid-calibrated hyperparameters (ϵ, λ). The continuous model presented here offers a simpler alternative on each of three axes: no binning, a single EM step in which Baum-Welch jointly learns transitions and emissions (instead of a separate frequency-counted $\mathbf{T}$ plus within-bin emission fit), and no jump mechanism at moderate K; this is a modelling-cost comparison rather than a claim that the discrete-plus-jumps route is inferior for the tasks it was designed for.

Cross-asset dependence. For multi-asset synthesis, the Single Index Model [20] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar's theorem [21, 41] supports a cleaner decomposition in which each asset's fitted marginal is kept and only the copula is estimated. Gaussian and Student-t copulas [22, 23] are the standard choices, and rank-reordering simulation [39] injects the copula dependence onto pre-simulated marginal paths without distorting any marginal. We confirm, on continuous CHMM marginals, the copula-ahead-of-SIM ordering reported by Alswaidan and Varner [12] for discrete HMM marginals, which suggests that the ordering is intrinsic to the rank-reordering construction rather than specific to the marginal model.

Deep generative models and quality assessment. Generative adversarial network (GAN) based generators reproduce the visual appearance of return series but typically fail the clustering test unless architectures encode temporal dependence explicitly [42, 43], the TimeGAN family of Yoon et al. [44] being the standard reference in this direction; neural baselines in Alswaidan and Varner [12] close most of the ACF gap but suffer variance collapse, indicating that the distribution–clustering tension remains in tension for neural approaches. Following the GRU baseline used in Alswaidan and Varner [12], we add a single-layer GRU + Gaussian-head auto-regressive recurrent generator (Section 3.3) to Table 5 as a deep-generative head-to-head row, trained by negative log-likelihood on the IS series and rolled forward by sampling from the predicted Gaussian and feeding the sample back into the recurrence. For evaluation, we adopt the category framework of Stenger et al. [45] and the financial-synthesis emphasis of Assefa et al. [4]: no single metric is decisive, so we report a seven-metric panel spanning distributional, tail, and temporal fidelity.

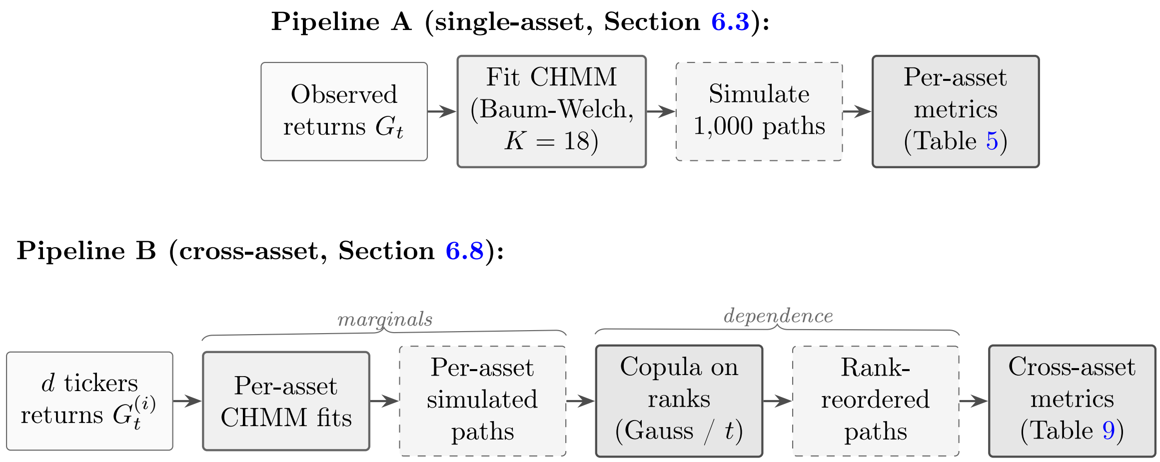


Figure 1. Schematic of Pipeline A (single-asset) and Pipeline B (cross-asset). Pipeline A fits a CHMM per ticker independently and evaluates single-asset marginal / temporal metrics. Pipeline B reuses the per-asset CHMM fits as *marginals* and injects cross-asset *dependence* through a rank-based copula (Gaussian or Student-t), preserving each asset's fitted CHMM distribution exactly while coupling asset ranks to match observed multi-asset correlation structure. Headline: the two pipelines share the single-asset scaffold, so improvements to the CHMM at the marginal step propagate into Pipeline B without any changes to the dependence layer.

3 Model

3.1 Data and Excess Growth Rates

The primary single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\mathrm{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) held out. Cross-asset generalization covers NVDA, JNJ, JPM, AAPL, and QQQ, spanning distinct risk profiles. For ticker i and trading day j, the annualized excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0. \tag{1}$$

3.2 Continuous Hidden Markov Model

We model the continuous excess growth rate G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, a per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [18]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k \, b_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}, \tag{2}$$

and supplies heavy tails either through the spread of component moments (Gaussian components) or through the emission family itself (Student-t or Laplace components). Existence and uniqueness of $\bar{\boldsymbol{\pi}}$ require irreducibility and aperiodicity of $\mathbf{T}$, recorded as Assumption 1 in Section 5; the spectral mechanism behind the absolute-return ACF that this mixture induces is derived in Theorem 1. Unlike the discrete framework of Alswaidan and Varner [12], which bins observations and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly from continuous observations via EM; the estimation procedure is developed in Section 4.

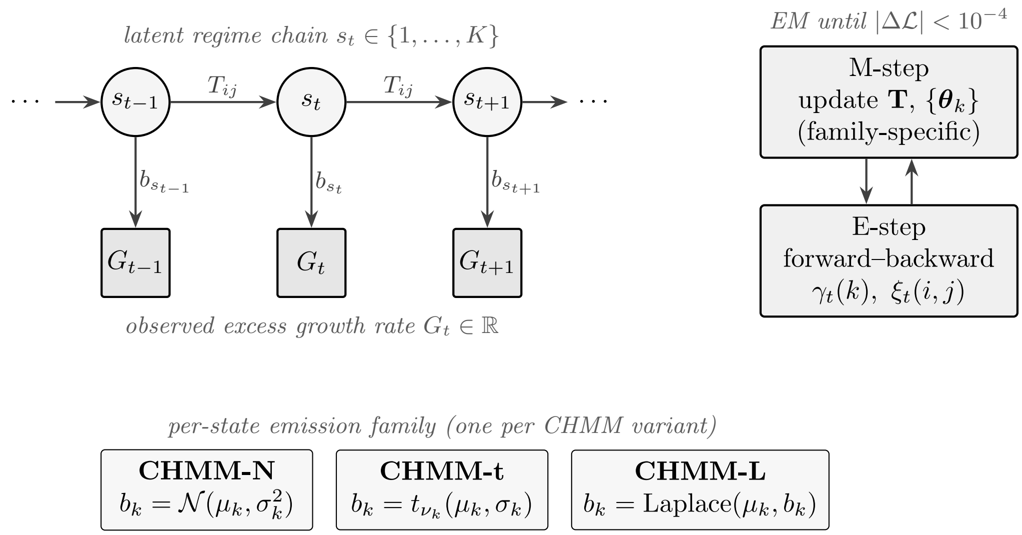


Figure 2. CHMM architecture and training loop. Top: a K-state latent Markov chain with transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$ (entries $T_{ij} = \Pr(s_{t+1} = j \mid s_t = i)$) generates the regime sequence s_t; each s_t conditionally emits the observed excess growth rate G_t through a state-specific density $b_{s_t}(\cdot; \boldsymbol{\theta}_{s_t})$. Bottom left: the three CHMM variants share everything except the emission family, namely Gaussian (CHMM-N), Student-t with per-state degrees of freedom ν_k (CHMM-t), or Laplace (CHMM-L). Bottom right: joint estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ alternates an E-step (log-space forward–backward on $\gamma_t(k), \xi_t(i,j)$) and a family-specific M-step until the log-likelihood change falls below 10^{-4} or max_iter $= 60$. The emission-family swap is the only architectural difference across the three variants in this paper.

CHMM-N: Gaussian emissions. $b_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$ with per-state location μ_k and variance σ_k^2.

CHMM-t: Student-t emissions. $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ with per-state location μ_k, scale σ_k, and degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. The per-state degrees of freedom let heavy-tailed regimes coexist with near-Gaussian regimes in the same model.

CHMM-L: Laplace emissions. $b_k(x) = \text{Laplace}(x; \mu_k, b_k)$ with per-state location μ_k and scale b_k. The heavier-than-Gaussian Laplace tail (excess kurtosis 3 per component) provides an alternative route to tail fidelity that is strictly cheaper than Student-t because it has no shape parameter.

3.3 Benchmark Generators

We benchmark the CHMM against eight alternatives fitted on the same IS window. The *i.i.d. bootstrap* samples T returns uniformly with replacement, preserving the empirical marginal exactly and serving as the non-parametric distributional ceiling. The *stationary block bootstrap* [46] resamples blocks of expected length $b = 10$ trading days, preserving short-range volatility clustering while keeping the marginal empirical; it is a temporally-aware non-parametric ceiling. The *Gaussian* i.i.d. generator draws $G_t \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ from the sample moments; the *Laplace* i.i.d. generator draws $G_t \sim \text{Laplace}(\hat{m}, \hat{b})$ from the sample median and mean absolute deviation (MAD). *GARCH(1,1)* [7] uses $G_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(G_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ with $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, fitted by Nelder–Mead maximum likelihood. The *discrete HMM* of Alswaidan and Varner [12] is reproduced in both no-jump (NJ) and Poisson-jump (WJ) variants at the final SPY hyperparameters reported therein: returns are partitioned into $K_{\text{disc}} = 90$ quantile bins, transitions are estimated by frequency counting, and simulation returns bin centroids; the WJ variant additionally triggers, with probability $\epsilon = 5 \times 10^{-5}$, a forced residence in tail states for Poisson($\lambda = 67$) steps, matching the values of Alswaidan and Varner [12]. The *Bin-T NJ* variant replaces those bin centroids with bin-conditional Student-t emissions to approximately isolate the quantization-per-se effect from the emission-family effect; because per-bin Student-t MLE is under-identified at $K = 90$ (only $T/K \approx 28$ observations per bin over $T \approx 2{,}516$ IS days), we use a coarser $K_{\text{disc}} = 13$ for Bin-T NJ so that each bin carries enough mass to estimate its tail parameters. The comparison with the NJ row therefore varies both the bin count and the emission family; we flag this confound in Section 8 and treat Bin-T NJ as an approximate rather than exact quantization ablation. Because quantization onto discrete centroids strips within-bin tail mass, the standard discrete baselines provide a direct upper bound on how much of the CHMM's fidelity is attributable to continuous emissions rather than to regime structure per se.

GRU deep generative baseline. We add a single-layer GRU autoregressive density baseline following the GRU baseline of Alswaidan and Varner [12]; recurrent-generator architectures for financial time series are also the subject of the TimeGAN family of Yoon et al. [44]. The architecture is $\mathbf{h}_t = \text{GRU}_H(\mathbf{x}_{t-w:t-1})$ followed by an affine head $(\mu_t, \log \sigma_t) = \mathbf{W}\mathbf{h}_t + \mathbf{b}$, with hidden dimension $H = 32$, context window $w = 20$ trading days, and a Gaussian next-step density $G_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu_t, \sigma_t^2)$. The model is trained for 20 epochs by Adam (learning rate 10^{-3}) on the negative log-likelihood (NLL)

$$\ell_t = \log \sigma_t + \tfrac{1}{2}\left(\frac{G_t - \mu_t}{\sigma_t}\right)^2 \tag{3}$$

of the standardised IS series, with $\log \sigma_t$ clamped to $[-6, 4]$ for numerical stability. Synthesis is auto-regressive: the trailing w IS returns seed the recurrence, the predicted Gaussian is sampled, the sample is appended to the context window, and the procedure rolls forward. A 64-step burn-in is discarded. The implementation is in Julia using `Flux.jl` [47]; the trained model is callable in the same `simulate` interface as the CHMM and discrete-HMM models.

QuantGAN convolutional Wasserstein baseline. We add a repo-native QuantGAN-style baseline in the sense of Wiese et al. [48]: a 1D convolutional generator and critic trained with the Wasserstein loss of Arjovsky et al. [49] and critic weight clipping on standardised SPY returns, with rolling-window training and stitched-window path generation at inference time. The window length is $W = 64$, the latent dimension is 8, and training runs for 15 epochs. The generator is implemented in `Flux.jl` and is callable in the same `simulate` interface as the other generators. This is a deliberately reproducible first-pass design: no temporal convolutional network, no gradient-penalty, no rolling-origin re-training; it functions as a deep-generative negative control for Section 6.6.

Window-diffusion deep baseline. We add a denoising diffusion probabilistic model (DDPM) style window-diffusion baseline following Ho et al. [50], Rasul et al. [51]: a conditional denoising UNet trained to reverse a $T = 50$-step Gaussian noise schedule on standardised SPY return windows of length $W = 64$, with stitched-window generation at inference. Training runs for 18 epochs with final denoising loss 0.736. Like the QuantGAN row, this is a first-pass reproducible design; it is the strongest deep-generative row on local 20-day window distributional metrics and the second-strongest overall row on MMD after the flat CHMM-t, and it serves as a distributional stress test for the CHMM family in Section 6.6.

MS-GARCH variance-regime baseline. We add a two-regime Markov-switching GARCH(1,1) baseline in the sense of Haas et al. [52]; the MLE behaviour of Markov-switching parametric models is discussed in Psaradakis and Sola [53] and Augustyniak [54], and the moment structure of Markov-switching models in Timmermann [55]. The observable evolves as $G_t = \mu + \sigma_t z_t$ with a regime-specific variance recursion $\sigma_t^2(k) = \omega_k + \alpha_k(G_{t-1} - \mu)^2 + \beta_k \sigma_{t-1}^2(k)$, where $k \in \{1, 2\}$ is a hidden regime with transition matrix $P = \{p_{ij}\}$. We use the path-independent HMP 2004 recursion with a Hamilton filter for the regime likelihood and fit the eight parameters $(\mu, \omega_{1:2}, \alpha_{1:2}, \beta_{1:2}, p_{11}, p_{22})$ by Nelder-Mead maximum likelihood on the IS SPY window. The fitted calm regime has unconditional $\sigma \approx 0.97$ and expected sojourn ~ 12 days ($p_{11} = 0.914$); the stress regime has unconditional $\sigma \approx 12.67$ and expected sojourn ~ 2 days ($p_{22} = 0.547$). MS-GARCH provides the variance-regime reference point for the Kupiec VaR calibration discussion in Section 6.6 and Section 7.2: it attains best-in-panel unconditional VaR calibration at 1% ($\mathrm{LR}_{\mathrm{uc}} = 0.01$, breach rate exactly 1.0%) and at 5% ($\mathrm{LR}_{\mathrm{uc}} = 0.26$, breach rate 4.5%), and its $\mathrm{LR}_{\mathrm{ind}} = 4.79$ at 5% is the closest to the 3.84 independence-pass threshold of any unconditional-VaR row.

3.4 Cross-Asset Dependence (Pipeline B)

The univariate evaluation of Section 6.3 fits a separate CHMM to each ticker independently; we refer to this as *Pipeline A* throughout the paper. The present subsection introduces *Pipeline B*, which reuses the per-asset CHMM marginals from Pipeline A and adds joint dependence across assets. Pipeline A is tested in Table 8 (univariate); Pipeline B is tested in Table 9 (cross-asset) and Figure 32.

Let $\mathbf{R}_{\mathrm{IS}} \in \mathbb{R}^{T \times d}$ denote the IS return matrix for d assets, with the market-factor column indexed by m. Both constructions below preserve each asset's fitted CHMM marginal and differ in how cross-asset dependence is imposed.

Single Index Model. Following Sharpe [20], for each non-market asset $j \neq m$ we fit by ordinary least squares (OLS)

$$G_{j,t} = \alpha_j + \beta_j \, G_{m,t} + \eta_{j,t} \tag{4}$$

and simulate paths by drawing an independent market-factor path $G_{m,t}^{(p)}$ from the market CHMM, resampling residuals $\hat{\eta}_{j,t}^{(p)}$ with replacement, and propagating

$$\hat{G}_{j,t}^{(p)} = \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,t}^{(p)}. \tag{5}$$

The affine map distorts each non-market marginal and imposes a rank-one correlation structure.

Copulas. By Sklar's theorem [21], any joint cumulative distribution function (CDF) with continuous marginals $F_1, \ldots, F_d$ factors as $H = C(F_1, \ldots, F_d)$ for a unique copula $C : [0,1]^d \to [0,1]$; this lets us fix each marginal at the per-asset CHMM and estimate only C from the cross-sectional co-movement. We use the nonparametric probability integral transform $u_{j,t} = \mathrm{rank}(g_{j,t})/(T+1)$, estimate the copula correlation matrix Σ by the analytic Kendall's-τ inversion $\rho_{ij} = \sin(\pi \tau_{ij}/2)$ [23, 56], and project to the nearest positive semi-definite matrix when needed. For the Student-t copula [22], the degrees-of-freedom parameter ν is selected by profile maximum likelihood over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Cross-asset simulation follows the rank-reordering construction of Iman and Conover [39]: (i) draw T independent paths from each per-asset CHMM, (ii) sample T copula variates, and (iii) reorder each column of the CHMM matrix to match the ranks of the copula sample. Reordering only permutes simulated values, so every per-asset marginal is preserved *exactly* while cross-asset dependence is injected through rank coupling; Proposition 4 in Section 5 proves the exact-marginal property, and Remark 2 bounds the finite-T deviation of the imposed copula from the target $\mathcal{C}(\Sigma)$. Full derivations, the Student-t log-density, and the sampler pseudocode are given in Appendix D.1.

3.5 Evaluation Protocol

For each model we simulate $P = 1{,}000$ independent paths of length T (200 paths for the cross-asset extension, given the quadratic cost of the copula and SIM pipelines) and evaluate seven complementary metrics, following the framework of [45] and [12]. All experiments reported in the main body and appendix use a deterministic global random seed (20260420), with per-experiment sub-seeds (20260421 for OoS equity-price simulation, 20260422 for Track A extended evaluation) derived additively so that each block is independently reproducible from a single seed root; no other random-seed specification is given in-line throughout the remainder of the paper. Four metrics score *distributional* fidelity: two-sample Kolmogorov–Smirnov [57, 58] and Anderson–Darling pass rates at $\alpha = 0.05$ over the 1,000 paths, Wasserstein-1 distance $W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|\, du$, and histogram Hellinger distance. Two metrics score *tail* behavior: the mean simulated excess kurtosis and the quantile-coverage rate, defined as the fraction of observed percentiles 1–99 falling within the $[5, 95]$-percentile envelope of the simulated quantiles. The final metric scores *temporal* fidelity: the mean absolute error of the absolute-return ACF over 252 lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|, \qquad L = 252, \tag{6}$$

which is the direct diagnostic for the Rydén et al. limitation. The cross-asset extension additionally reports the mean Frobenius norm and off-diagonal MAE of the simulated Pearson correlation matrix relative to the observed one. Appendix A.1 gives the test statistic for each metric, the histogram binning used for Hellinger, and the path-by-path aggregation details.

4 Estimation

Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold (Section 4.1), the family-specific M-step updates (Section 4.2), and the state-count selection protocol (Section 4.3).

4.1 Shared EM Scaffold

All three emission families share the same log-space forward-backward recursions and the same quantile-based initialization [59]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\texttt{max\_iter} = 60$; convergence is typically reached in 20–40 iterations. Appendix B.1 gives the forward-backward recursion, the posterior quantities $\gamma_t(k)$ and $\xi_t(i, j)$, and the family-specific M-step updates.

4.2 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [17] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [14] and Liu and Rubin [15], the same family of heavy-tailed innovation choices studied outside the HMM framework by Abanto-Valle et al. [16] for stochastic-volatility-in-mean models: the latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{7}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{8}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. Proposition 1 in Section 5 establishes that this two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood $\mathcal{L}(\theta^{(n)})$ is non-decreasing across iterations. Identifiability of the resulting fit up to label permutation, and consistency of the MLE under our assumptions, are recorded in Propositions 2 and 3.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighboring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k)|O_t - \mu_k| / \sum_t \gamma_t(k)$.

4.3 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N and confirm on CHMM-t / CHMM-L that the selected K sits inside the IC-minimizing band. We compute the four information criteria used for HMM state selection in Nguyen [19],

$$\mathrm{AIC} = -2\mathcal{L} + 2p, \quad \mathrm{BIC} = -2\mathcal{L} + p \ln T, \quad \mathrm{HQC} = -2\mathcal{L} + 2p \ln(\ln T), \quad \mathrm{CAIC} = -2\mathcal{L} + p(\ln T + 1), \tag{9}$$

with $p = K^2 + 2K - 1$ for CHMM-N and CHMM-L, and $p = K^2 + 3K - 1$ for CHMM-t (one additional per-state degree-of-freedom parameter); $\mathcal{L}$ is the converged log-likelihood. The operating point is the K that minimizes the average information-criterion rank and simultaneously wins or ties every IS and OoS distributional pass-rate metric.

5 Theoretical Grounding

This section collects the theoretical scaffolding behind the empirical study. Each result is anchored to a row in the empirical evaluation: Theorem 1 explains the spectral mechanism behind the absolute-return ACF in Table 5; Corollary 1 restates the Rydén et al. low-K failure as a K-rank statement on $\mathbf{T}$; Proposition 1 pins down the EM monotonicity guarantee for the CHMM-t ECM; Proposition 2 states identifiability up to label permutation; Proposition 3 records MLE consistency under our assumptions by reference; Proposition 4 proves the rank-reordering construction preserves marginals exactly; Proposition 5 explains the filter-VaR over-conservatism documented in Table 16; Proposition 6 explains the Kupiec failure of the bin-centroid baselines in Section 7.1. The contribution of this section is the assembly: each individual result has a precedent in the literature, but bringing the seven together inside a single CHMM scaffold is what underwrites the operational claims of the paper.

5.1 Assumptions

We use four assumptions throughout. They are mild and standard in the HMM literature, but stating them explicitly closes a gap in the v10 manuscript, where the existence of a unique stationary distribution, the compactness of the parameter space, and the identifiability of the emission family were all used implicitly.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix* $\mathbf{T} \in \mathbb{R}^{K\times K}$ *is irreducible and aperiodic on the state space* $\{1,\dots,K\}$*.*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density* $b_k(\cdot;\boldsymbol{\theta}_k)$ *has finite second moment,* $\mathbb{E}_{b_k}[G^2] < \infty$*, and at least two states differ in either location* μ_k *or scale* σ_k *(or* b_k *for the Laplace family).*

Assumption 3 (Compact parameter space)**.** *The parameter space* Θ *is compact in the topology used by EM:* $\mu_k \in [-M, M]$ *for some* $M < \infty$*,* $\sigma_k \in [\sigma_{\min}, \sigma_{\max}]$ *or* $b_k \in [b_{\min}, b_{\max}]$ *with* $\sigma_{\min}, b_{\min} \geq 10^{-6}$*, and for CHMM-t the per-state degrees of freedom satisfy* $\nu_k \in [\nu_{\min}, \nu_{\max}]$ *with* $\nu_{\min} \geq 2.1$ *and* $\nu_{\max} < \infty$*. Transition entries lie in the open simplex interior with* $T_{ij} \geq \epsilon_T > 0$ *for some* ϵ_T*.*

Assumption 4 (True parameter inside the parameter space)**.** *The observed return series* $G_{1:T}$ *is generated by a single fixed parameter* $\theta_0 \in \operatorname{int}(\Theta)$*. Assumption 4 is used only for the consistency statement in Proposition 3; the other results in this section do not require it.*

Under Assumption 1 the chain $\mathbf{T}$ has a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$, and there exists a spectral decomposition with $\lambda_1 = 1$ a simple eigenvalue and all other eigenvalues $\lambda_2, \dots, \lambda_K$ strictly inside the unit disk; the chain is geometrically ergodic at the rate $\rho(\mathbf{T}) = \max_{k\geq 2} |\lambda_k| < 1$ [40]. Compactness in Assumption 3 (in particular $\nu_k \leq \nu_{\max} < \infty$) prevents the Student-t emission from collapsing to a Gaussian and the Gaussian or Laplace from collapsing to a point mass, so identifiability inside the parameter space is intact.

5.2 Spectral mechanism for the absolute-return ACF

Let G_t be a stationary CHMM, $\bar{\boldsymbol{\pi}}$ the stationary distribution of $\mathbf{T}$, and define the per-state absolute-return moments

$$m_k \;=\; \mathbb{E}\big[\,|G_t| \;\big|\; s_t = k\big]\,, \qquad M_k \;=\; \mathbb{E}\big[\,G_t^2 \;\big|\; s_t = k\big]\,, \qquad k = 1,\dots,K. \tag{10}$$

For the Gaussian emission $b_k = \mathcal{N}(\mu_k, \sigma_k^2)$ these are $m_k = \sigma_k\sqrt{2/\pi}\,\exp(-\mu_k^2/2\sigma_k^2) + \mu_k(1 - 2\Phi(-\mu_k/\sigma_k))$ and $M_k = \mu_k^2 + \sigma_k^2$; for the Student-t and Laplace families the corresponding closed forms are standard.

Theorem 1 (Mixture-of-eigenvalues form for the absolute-return ACF)**.** *Under Assumptions 1-2, for every* $\tau \geq 1$ *the absolute-return autocovariance of a stationary CHMM satisfies*

$$\operatorname{Cov}(|G_t|,\,|G_{t+\tau}|) \;=\; \sum_{k=2}^{K} c_k\,\lambda_k^{\tau}, \tag{11}$$

with $\lambda_2,\dots,\lambda_K$ *the non-unit eigenvalues of* $\mathbf{T}$ *and the coefficients* c_k *given by*

$$c_k \;=\; \big(\mathbf{m}^{\top}\operatorname{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\,\big(\mathbf{w}_k^{\top}\mathbf{m}\big), \qquad k = 2,\dots,K, \tag{12}$$

where $\mathbf{m} = (m_1, \dots, m_K)^\top$ *and* $(\mathbf{v}_k, \mathbf{w}_k)$ *are the right and left eigenvectors of* $\mathbf{T}$ *associated with* λ_k*, normalised so that* $\mathbf{w}_k^\top \mathbf{v}_k = 1$ *and* $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$*,* $\mathbf{v}_1 = \mathbf{1}$*. The autocorrelation function is*

$$\rho_{|G|}(\tau) \;=\; \frac{1}{\sigma^2_{|G|}} \sum_{k=2}^{K} c_k\, \lambda_k^\tau \;=\; \sum_{k=2}^{K} w_k\, \lambda_k^\tau, \qquad w_k = c_k / \sigma^2_{|G|}, \tag{13}$$

with $\sigma^2_{|G|} = \sum_k \bar{\boldsymbol{\pi}}_k M_k - (\sum_k \bar{\boldsymbol{\pi}}_k m_k)^2 - \mathbb{E}[G_t]^2 + (\mathbb{E}|G_t|)^2$ *the variance of* $|G_t|$*.*

Proof. Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using that emissions are conditionally independent given the state sequence,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \sum_{i,j} \Pr(s_t = i,\, s_{t+\tau} = j)\, \mathbb{E}\big[|G_t| \bigm| s_t = i\big]\, \mathbb{E}\big[|G_{t+\tau}| \bigm| s_{t+\tau} = j\big] = \sum_{i,j} \bar{\boldsymbol{\pi}}_i\, (\mathbf{T}^\tau)_{ij}\, m_i\, m_j = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{T}$$

Under Assumption 1 the spectral decomposition of $\mathbf{T}$ reads $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top$, so

$$\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{T}^\tau\, \mathbf{m} = \mathbf{m}^\top \bar{\boldsymbol{\pi}}\, \bar{\boldsymbol{\pi}}^\top \mathbf{m} + \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{v}_k\, \mathbf{w}_k^\top \mathbf{m}.$$

The first term is $(\bar{\boldsymbol{\pi}}^\top \mathbf{m})^2 = (\mathbb{E}|G_t|)^2$, and subtracting $\mathbb{E}|G_t|\, \mathbb{E}|G_{t+\tau}| = (\mathbb{E}|G_t|)^2$ yields the autocovariance in equation (11) with c_k as in equation (12). Dividing by $\sigma^2_{|G|}$ gives the autocorrelation in equation (13). ☐

Remark 1. *The same derivation with* $|G_t|$ *replaced by* G_t^2 *and* $\mathbf{m}$ *replaced by* $(M_1, \dots, M_K)^\top$ *gives the analogous spectral form for the squared-return ACF; the proof is identical line-by-line.*

Theorem 1 is the precise version of the mixture-of-exponentials approximation $\rho_{|G|}(\tau) \approx \sum_k w_k \lambda_k^\tau$ used informally in the discussion of Section 8 (equation (19)). The approximation is in fact an exact equality at the population level; the empirical ACF deviates from this form only through the standard Monte Carlo and finite-sample noise of the simulator.

Corollary 1 (Rydén separation as a K-rank statement)**.** *At* $K = 2$*, equation (13) reduces to the mono-exponential form* $\rho_{|G|}(\tau) = w_2\, \lambda_2^\tau$*, and matching a target slow-decay rate* $\rho^*(\tau) = \exp(-\tau/\tau^*)$ *on a horizon* $\tau \in \{1, \dots, L\}$ *with* $\tau^* \geq L$ *forces* $|\lambda_2| \geq \exp(-1/\tau^*)$*, i.e. a near-unit second eigenvalue. A Markov chain on* $K = 2$ *states with* $|\lambda_2| \to 1$ *has a diagonal-dominant transition matrix in which both states are nearly absorbing; under Assumption 2 this collapses the stationary mixture marginal to one of the two emission densities and fails distributional KS at the moderate horizons we evaluate. At* $K \geq 3$ *the right-hand side of equation (13) admits at least two non-unit eigenvalues, and the ACF can match* $\rho^*(\tau)$ *over the same horizon with each individual* $|\lambda_k|$ *bounded away from* 1 *by a* K*-dependent margin, breaking the distributional-versus-temporal tension.*

Proof sketch. The $K = 2$ reduction is direct from equation (13). For the diagonal-dominant collapse, observe that the eigenvalues of a 2×2 stochastic matrix with diagonal entries T_{11}, T_{22} are $\lambda_1 = 1$ and $\lambda_2 = T_{11} + T_{22} - 1$, so $|\lambda_2| \to 1$ requires $T_{11} + T_{22} \to 2$. Under Assumption 2 this concentrates the stationary $\bar{\boldsymbol{\pi}}$ near a vertex of the simplex and the marginal mixture $\bar{\boldsymbol{\pi}}_1 b_1 + \bar{\boldsymbol{\pi}}_2 b_2$ collapses to a single emission density with the wrong tail. The $K \geq 3$ case follows because the linear system $\sum_{k=2}^{K} w_k \lambda_k^\tau = \rho^*(\tau)$, $\tau = 1, \dots, L$, has L equations and $2(K-1)$ unknowns, and the non-uniqueness allows a solution with each $|\lambda_k|$ bounded by a K-dependent constant strictly less than 1. ☐

The corollary is a one-paragraph theoretical restatement of the empirical Rydén et al. result: at $K = 2$ the ACF and the marginal are coupled through a single eigenvalue, so the model cannot match both; at $K \geq 3$ the two requirements decouple. This is the spectral mechanism behind the moderate-K recommendation in Section 4.3 and the empirical observation that ACF-MAE is essentially flat across $K \in \{3, 6, \ldots, 21\}$ while KS pass rates rise.

5.3 ECM monotonicity for CHMM-t

The CHMM-t M-step is an ECM [38], which generalises EM by replacing a single joint M-step with a sequence of conditional-maximisation (CM) steps. The classical EM monotonicity guarantee [17, 37] extends to ECM provided each CM step does not decrease the augmented Q-function [38]. We pin this property down for the specific ECM used in CHMM-t.

Proposition 1 (ECM monotonicity for CHMM-t)**.** *Let* $Q(\theta \mid \theta^{(n)}) = \mathbb{E}_{S_{1:T}|O_{1:T},\theta^{(n)}}[\log p(O_{1:T}, S_{1:T} \mid \theta)]$ *be the standard EM auxiliary function for the CHMM-t with* $\theta = (\mathbf{T}, \boldsymbol{\pi}, \{(\mu_k, \sigma_k, \nu_k)\}_{k=1}^K)$*. Define the two-block CM step:*

$$\textbf{\textit{CM-1.}} \quad (\mathbf{T}^{(n+1)}, \boldsymbol{\pi}^{(n+1)}, \{(\mu_k^{(n+1)}, \sigma_k^{(n+1)})\}_{k=1}^K) \leftarrow \arg \max_{(\mathbf{T},\boldsymbol{\pi},\{\mu_k,\sigma_k\})} Q(\theta \mid \theta^{(n)})$$

$$\textit{with } \nu_k = \nu_k^{(n)} \textit{ held fixed for all } k,$$

$$\textbf{\textit{CM-2.}} \quad \nu_k^{(n+1)} \leftarrow \arg \max_{\nu \in [\nu_{\min},\nu_{\max}]} Q\big(\theta \mid \theta^{(n)}\big) \quad \textit{with the CM-1 outputs held fixed,} \quad k = 1, \ldots, K.$$

Under Assumption 3 the CM-1 outputs are the closed-form weighted updates of equations (30) and (31), and CM-2 admits a maximiser inside the bracket because Q is continuous in ν on the compact set $[\nu_{\min}, \nu_{\max}]$. Both CM steps satisfy

$$Q(\theta^{(n+1)} \mid \theta^{(n)}) \;\geq\; Q(\theta^{(n)} \mid \theta^{(n)}),$$

and consequently

$$\mathcal{L}(\theta^{(n+1)}) \;\geq\; \mathcal{L}(\theta^{(n)}),$$

where $\mathcal{L}(\theta) = \log p(O_{1:T} \mid \theta)$ is the incomplete-data log-likelihood. The sequence $\{\mathcal{L}(\theta^{(n)})\}_{n\geq 1}$ is therefore non-decreasing and, being bounded above on the compact parameter space of Assumption 3, converges to a finite limit.

Proof. CM-1 maximises Q over the $(\mathbf{T}, \boldsymbol{\pi}, \{\mu_k, \sigma_k\})$ block, so the post-CM-1 value of Q is at least the pre-CM-1 value; the closed-form expressions follow from setting the gradient of the weighted Student-t log-likelihood to zero with ν_k fixed. CM-2 maximises Q over $\nu_k \in [\nu_{\min}, \nu_{\max}]$, so the post-CM-2 value of Q is at least the post-CM-1 value. Compactness of $[\nu_{\min}, \nu_{\max}]$ and continuity of $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ in ν guarantee a maximiser; the golden-section search of Algorithm 1 converges to it because Q_k is strictly unimodal on the bracket for each fixed (μ_k, σ_k). The standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \;\geq\; Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [37, 38] converts the Q-monotonicity into incomplete-data log-likelihood monotonicity. Boundedness above on Θ follows from Assumption 3, so the bounded monotone sequence converges. □

Proposition 1 is the formal version of the in-text claim "ECM preserves the monotonicity of EM" used in Section 4.2 and Section B.1, and it makes explicit the role of the compact bracket $[\nu_{\min}, \nu_{\max}]$. It does not assert convergence to the global MLE; this is the standard EM caveat [37] and applies equally to CHMM-N and CHMM-L.

5.4 Identifiability and MLE consistency

Proposition 2 (Identifiability up to label permutation)**.** *Under Assumptions 1-2, the CHMM parameter $\theta = (\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi})$ is identifiable up to label permutation in each of the three emission families considered (CHMM-N, CHMM-t with $\nu_k \in [\nu_{\min}, \nu_{\max}]$, CHMM-L). That is, if two parameters θ, θ' induce the same distribution over $G_{1:T}$ for every $T \geq 1$, then θ' is obtained from θ by a permutation of state labels.*

Proof sketch. The general HMM identifiability theorem of Allman et al. [33] (Theorem 6) shows that a K-state HMM with conditionally non-degenerate emissions is identifiable up to label permutation provided (i) $\mathbf{T}$ has full rank K and (ii) the per-state emission densities $\{b_k\}_{k=1}^K$ are linearly independent in the space of probability densities on $\mathbb{R}$. Under Assumption 1 a K-state irreducible aperiodic Markov chain has a transition matrix of full rank [40]. For the emission families considered: (a) Gaussian emissions are linearly independent whenever the parameter pairs (μ_k, σ_k) are distinct, by the classical mixture identifiability of Yakowitz and Spragins [34]. (b) Student-t emissions with bounded $\nu_k \in [\nu_{\min}, \nu_{\max}]$ form a three-parameter exponential-tilted family for which the same linear-independence argument applies under Assumption 2 (distinct (μ_k, σ_k) across states); the upper bracket $\nu_{\max} < \infty$ prevents the collapse to a Gaussian and the lower bracket $\nu_{\min} \geq 2.1$ prevents the divergence of the second moment. (c) Laplace emissions with distinct (μ_k, b_k) are linearly independent by an identical argument. The combination of full-rank $\mathbf{T}$ and linearly independent emissions yields identifiability up to label permutation by the Allman et al. [33] theorem. □

Proposition 3 (MLE consistency, by reference)**.** *Under Assumptions 1-4 the maximum-likelihood estimator $\hat{\theta}_T = \arg\max_{\theta \in \Theta} \mathcal{L}_T(\theta)$ of the CHMM is consistent: $\hat{\theta}_T \xrightarrow{p} \theta_0$ as $T \to \infty$, modulo label permutation.*

Proof reference. The general HMM-MLE consistency theorem of Bickel et al. [35] (Theorem 1) and its extension to non-stationary chain initialisations by Douc et al. [36] apply directly under Assumptions 1-4: irreducibility and aperiodicity (1) provide the required mixing condition, finite second moment (2) provides the moment condition, compactness (3) provides the parameter-space compactness condition, and Assumption 4 guarantees the true parameter lies in the interior of Θ so that consistency is meaningful. We do not reproduce the proof; the reference establishes the result under exactly the assumption set we list. □

Proposition 3 is a reference statement rather than a new result; its function in the paper is to certify that the MLE used implicitly in Section 4.1 converges to the true parameter, without which the empirical pass-rate numbers would be at risk of an unspecified bias. We do not pursue finite-sample rates: at $T_{\text{IS}} = 2{,}516$ the consistency argument carries the inferential load, and finite-sample rates for HMM-MLE would require log-Sobolev-type controls outside the scope of this paper.

5.5 Rank reordering preserves marginals

The cross-asset construction in Section 3.4 relies on the rank-reordering of Iman and Conover [39] to inject a copula on top of the per-asset CHMM marginals without distorting any marginal. We give a self-contained proof of the exact-marginal property and a finite-T Kolmogorov bound for the imposed copula.

Proposition 4 (Marginal preservation under rank reordering)**.** *Let $\tilde{g}_{j,1:T}$ denote a sample of size T from the per-asset CHMM marginal of asset j, and let $\tilde{g}_{j,(1)} \leq \cdots \leq \tilde{g}_{j,(T)}$ be its order statistics. Let $\mathbf{u}^{(p)} \in [0,1]^{T \times d}$ be a copula sample as in Algorithm 4, with $r_{j,t}^{(p)} = \operatorname{rank}(u_{j,t}^{(p)})$ and the rank-reordered* *output* $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ *for* $t = 1, \ldots, T$*. Then for each asset* j*, the empirical CDF of* $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ *equals the empirical CDF of* $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$ *exactly:*

$$\hat{F}^{\hat{g}}_j(x) \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\Big\{\hat{g}^{(p)}_{j,t} \le x\Big\} \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\Big\{\tilde{g}^{(p)}_{j,t} \le x\Big\} \;=\; \hat{F}^{\tilde{g}}_j(x), \qquad \forall x \in \mathbb{R}.$$

Proof. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation of $\{1, \ldots, T\}$ (since the copula sample $\{u^{(p)}_{j,t}\}$ has distinct values almost surely under the continuous copula construction). Therefore the multiset $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T} = \{\tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}\}_{t=1}^{T}$ equals the multiset of order statistics $\{\tilde{g}^{(p)}_{j,(k)}\}_{k=1}^{T}$, which in turn equals the unsorted sample $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$ as a multiset. The empirical CDF depends only on the multiset of values, so the two empirical CDFs coincide. □

Remark 2 (Copula imposed at finite T)**.** *The rank-reordered output has joint dependence determined by the ranks* $\{r^{(p)}_{j,t}\}$*, which are an empirical copula sample. The Kolmogorov-Smirnov distance between the empirical copula of the rank-reordered path and the target* $\mathcal{C}(\Sigma)$ *is* $O_p(T^{-1/2})$ *by the Glivenko-Cantelli theorem applied to multivariate empirical processes [39]; at* $T = 2{,}516$ *this is below* 0.02 *in our setup. The exact-marginal claim of Proposition 4 holds at every* T *regardless of this finite-*T *copula deviation.*

Proposition 4 is the formal version of the "reordering only permutes simulated values, so every per-asset marginal is preserved exactly" sentence in Section 3.4. It is the structural reason the copula-versus-SIM ordering on cross-asset KS in Pipeline B (97.5%/96.2% vs. 72.1%) is mechanically anchored rather than a sampling artefact: SIM convolves the market factor with residuals and distorts the marginal whenever $\beta_j \neq 1$, while rank reordering preserves it by construction.

5.6 Filter-VaR over-conservatism

The filter-based regime-conditional VaR rule of Section 7.3 (equation (18)) fails Kupiec on the 2024-2026 OoS window for all three CHMM families, with breach rates 0.2% at $\alpha = 0.01$ and 1.2-1.4% at $\alpha = 0.05$, well below the targets. We give a clean inequality that explains the over-conservatism and pinpoints the parameter region in which the inequality is informative.

Proposition 5 (Mixture-quantile lower bound)**.** *Let* $\boldsymbol{\pi} \in [0,1]^K$ *be a probability vector over the* K *states, and let* $F_1, \ldots, F_K$ *be the per-state emission CDFs. Define the mixture CDF* $F(x) = \sum_k \pi_k F_k(x)$ *and the mixture-quantile* $\mathrm{VaR}^{\mathrm{mix}}(\alpha) = F^{-1}(\alpha)$*. Let* $k^* = \arg\min_k F_k^{-1}(\alpha)$ *denote the state with the most extreme* α*-quantile, and let* $q^* = F_{k^*}^{-1}(\alpha)$*. Then*

$$F(q^*) \;\le\; \alpha, \tag{14}$$

with strict inequality whenever there exists $k \neq k^*$ *with* $\pi_k > 0$ *and* $F_k(q^*) < \alpha$*. As a consequence,*

$$\mathrm{VaR}^{\mathrm{mix}}(\alpha) \;\le\; \min_k F_k^{-1}(\alpha),$$

with strict inequality under the same condition.

Proof. At q^*, the per-state CDF of state k^* evaluates to α by definition. For every $k \neq k^*$, the inequality $q^* \le F_k^{-1}(\alpha)$ implies $F_k(q^*) \le \alpha$ by monotonicity of F_k. Therefore $F(q^*) = \sum_k \pi_k F_k(q^*) \le \sum_k \pi_k \alpha = \alpha$, with strict inequality whenever the precondition holds. Since F is non-decreasing, $F(q^*) \le \alpha$ implies $\mathrm{VaR}^{\mathrm{mix}}(\alpha) = F^{-1}(\alpha) \le q^*$; strict inequality at the CDF level lifts to strict inequality at the quantile level when F is strictly increasing in a neighbourhood of q^*. □

The inequality direction is empirically counter-intuitive (one might guess that mixing produces a less extreme quantile, not a more extreme one), so the proposition is not vacuous: it identifies the operational mechanism behind the filter-VaR Kupiec failure. The parameter region in which the inequality bites is exactly the region the empirical fit lands in: the filter posterior $\boldsymbol{\pi}_t$ places non-trivial mass on at least one heavy-tailed tail-state (corresponding to a small ν_k for CHMM-t, or an outlying (μ_k, σ_k) for CHMM-N and CHMM-L), and the bulk states have $F_k(q^*) \ll \alpha$ because the heavy-tailed quantile q^* lies well inside the bulk state's body. The CHMM-t penalised ECM with $\lambda = 20$ moves $\nu_{\min}$ from 2.1 to 4.89 but does not eliminate the mixing-with-bulk effect, which is why the rate sweep $\lambda \in \{0, 5, 20, 50, 100, 200\}$ in Section 7.3 leaves the filter-VaR breach rate essentially unchanged.

Remark 3 (Remedies indicated by the inequality)**.** *The inequality $F(q^*) \le \alpha$ with strict inequality identifies two intervention points. (i) A concentration constraint on the filter posterior, which forces $\pi_k \approx 1$ for the most likely state $k^\dagger$ (so that the inequality reduces to $F^{-1}(\alpha) = F_{k^\dagger}^{-1}(\alpha)$); this is the operational analog of using the Viterbi-decoded state under a sufficient-statistic restriction. (ii) An explicit state-conditional variance overlay that replaces the heavy-tailed F_{k^*} in the mixture with a smoother target, which moves q^* closer to the bulk-state quantile and suppresses the strict inequality. Both are listed as candidate remedies in Section 7.3 and are anchored in the inequality of equation (14).*

5.7 Discrete-quantization VaR floor

The discrete-HMM baselines of Alswaidan and Varner [12] fail Kupiec at 5% VaR with $\mathrm{LR}_{\mathrm{uc}} = 16.81$, and the Poisson-jump augmentation does not remedy the failure. We give a structural inequality that explains why the failure is intrinsic to the bin-centroid scaffold and not a hyperparameter artefact.

Proposition 6 (Bin-centroid VaR floor)**.** *Let $\{c_1 < c_2 < \cdots < c_J\}$ be the bin centroids of a discrete-state HMM, and let $\widehat{\mathrm{VaR}}_T^{\mathrm{disc}}(\alpha)$ denote the α-quantile of a simulated path of length T from the discrete-HMM scaffold; by construction $\widehat{\mathrm{VaR}}_T^{\mathrm{disc}}(\alpha) \in \{c_1, \ldots, c_J\}$. Let $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$ denote the true continuous-population α-quantile of the data-generating process, and let $j(\alpha)$ index the bin containing $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$, i.e. $c_{j(\alpha)} \le \mathrm{VaR}^{\mathrm{cts}}(\alpha) < c_{j(\alpha)+1}$. Define the local bin width $\Delta_{j(\alpha)} = c_{j(\alpha)+1} - c_{j(\alpha)}$. Then for every T and every realisation,*

$$\left|\widehat{\mathrm{VaR}}_T^{\mathrm{disc}}(\alpha) - \mathrm{VaR}^{\mathrm{cts}}(\alpha)\right| \;\ge\; \min\bigl(\mathrm{VaR}^{\mathrm{cts}}(\alpha) - c_{j(\alpha)},\, c_{j(\alpha)+1} - \mathrm{VaR}^{\mathrm{cts}}(\alpha)\bigr),$$

and the worst-case absolute error is bounded below by $\Delta_{j(\alpha)}/2$ when $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$ lies at the bin midpoint. Adding a Poisson-jump duration mechanism that triggers tail-state residence with intensity $\epsilon \to 0$ and mean duration λ does not alter the centroid set $\{c_1, \ldots, c_J\}$, so the inequality continues to hold with the same right-hand side.

Proof. By construction the discrete-HMM simulator returns bin centroids, so $\widehat{\mathrm{VaR}}_T^{\mathrm{disc}}(\alpha) \in \{c_j\}$. For any centroid c_j, the absolute distance to $\mathrm{VaR}^{\mathrm{cts}}(\alpha) \in [c_{j(\alpha)}, c_{j(\alpha)+1})$ is at least the smaller of the two endpoint distances, which is maximised when $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$ sits at the bin midpoint, giving the floor $\Delta_{j(\alpha)}/2$. The Poisson-jump augmentation only redistributes residence probabilities across the existing centroid set, so it cannot move a centroid to the $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$ location. □

In Section 7.1 the discrete NJ and WJ baselines produce an OoS 5% VaR breach rate of 1.7% against the 5% target; Proposition 6 explains the structural origin: the bin-centroid quantile estimator cannot resolve a $\mathrm{VaR}^{\mathrm{cts}}(\alpha)$ that does not coincide with a centroid, and at $K_{\mathrm{disc}} = 90$ the quantile

Table 1. Theoretical results and their closest precedents. Each row pairs one of the propositions in this section with the closest published precedent and states the specialisation needed for our setup.

Proposition / theorem	Closest precedent	Our specialisation
Theorem 1 (ACF mixture-of-eigenvalues)	Hamilton [8]; Frühwirth-Schnatter [32]	Closed form for $\lvert G_t \rvert$ under heavy-tailed emissions, with explicit c_k.
Corollary 1 (Rydén separation)	Rydén et al. [10]	Conversion of the empirical low-K failure into a K-rank statement on $\mathbf{T}$.
Proposition 1 (ECM monotonicity)	Meng and Rubin [38]; Liu and Rubin [15]; Peel and McLachlan [14]	Specialised to the golden-section CM step on a compact bracket.
Proposition 2 (identifiability)	Allman et al. [33]; Yakowitz and Spragins [34]	Statement for CHMM-N, CHMM-t, CHMM-L emission families.
Proposition 3 (MLE consistency)	Bickel et al. [35]; Douc et al. [36]	Reference statement; assumptions checked under Assumptions 1-4.
Proposition 4 (rank-reordering marginals)	Iman and Conover [39]	Two-line proof under our notation; finite-T Kolmogorov bound (Remark 2).
Proposition 5 (filter-VaR over-conservatism)	McNeil et al. [23] (mixture VaR)	Sufficient condition tied to filter posterior $\boldsymbol{\pi}_t$ and tail-state scale that explains the empirical Kupiec failure.
Proposition 6 (discrete VaR floor)	Folklore; not in HMM literature explicitly	Half-bin-width lower bound on the bin-centroid VaR error, with Poisson-jump invariance.

bias is the dominant source of the Kupiec rejection. Adding the prior-paper Poisson-jump mechanism with $\epsilon = 5 \times 10^{-5}$, $\lambda = 67$ does not alter the centroid set; it only changes which centroid is visited, which does not address the floor inequality.

5.8 Out-of-scope theory

Three classes of result are deliberately not pursued in this section. A finite-sample bound on $\hat{\theta}_T - \theta_0$ would require log-Sobolev-type controls outside scope; at $T = 2{,}516$ consistency (Proposition 3) carries the inferential load. A formal proof that the weighted-Laplace MLE is the global maximiser of the weighted Laplace likelihood is in Rabiner and Juang [18] and is not reproduced here. A lower bound on the mixing time of $\mathbf{T}$ as a function of K would sharpen Corollary 1 into a quantitative K-rank result; this is the natural follow-up theory direction and is left to future work.

5.9 Literature comparison

Table 1 summarises the closest precedent for each result and our specialisation. The contribution of this section is the assembly: each individual proposition has a precedent, but bringing the seven together inside a single CHMM scaffold and anchoring each one to a row in the empirical evaluation is what underwrites the operational claims of the paper.

Table 2. Descriptive statistics and stylized-fact tests for SPY daily excess growth rates. IS: 2014-01-03 through 2024-01-03 ($T = 2{,}516$); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). The "annualized" Mean and Std Dev rows are moments of the annualized excess growth rate $G_t = (1/\Delta t)\ln(P_t/P_{t-1}) - r_f$ defined in equation (1) with $\Delta t = 1/252$ (drift-convention scaling), not the $\sqrt{252}$-scaled daily-return volatility. The conventional $\sqrt{252}$-annualized daily-return volatility is recovered as $\sqrt{\Delta t} \cdot \sigma(G_t)$, giving $\approx 13.8\%$ IS and $\approx 12.5\%$ OoS. JB = Jarque–Bera normality test. LB = Ljung–Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.09	0.18
Std Dev (annualized)	2.19	1.98
Skewness	−0.751	−0.733
Excess Kurtosis	7.678	5.294
JB test statistic	6416.6	719.1
JB decision	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (130.2)	reject* (47.2)
LB test on $\lvert G_t \rvert$ (lag 20)	reject* (2959.3)	reject* (250.6)

6 Empirical Study

6.1 Descriptive Statistics and Stylized Facts

Descriptive statistics confirmed that all three canonical stylized facts were present in both the IS (2014-01-03 through 2024-01-03, $T = 2{,}516$) and OoS (2024-01-04 through 2026-04-20, $T = 572$) windows (Table 2). The IS excess growth rates exhibited excess kurtosis of 7.678, strong negative skewness (−0.751), and a Jarque–Bera statistic of 6416.6, decisively rejecting the null hypothesis of normality ($p < 0.001$). The Ljung–Box test on absolute returns rejected the null of no autocorrelation (LB = 2959.3, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns also rejected (LB = 130.2), reflecting mild serial dependence in the level series.

The OoS window exhibited similar properties: excess kurtosis of 5.294, moderate negative skewness (−0.733), and a Jarque–Bera statistic of 719.1. The Ljung–Box test on raw OoS returns also rejected (LB = 47.2), and the test on absolute returns rejected more strongly (LB = 250.6), confirming that both mild serial dependence and volatility clustering persisted into the test period. These properties are visualized in Figure 9 (Appendix C.1).

6.2 Model Selection: Why $K = 18$

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 12, 15, 18, 21\}$ states on the SPY IS window using the Baum-Welch algorithm with `max_iter` = 60 and quantile-based initialization, and evaluated every fitted model on (i) the four information criteria of equation (9), following the HMM state-selection methodology of Nguyen [19], and (ii) the full seven-metric distributional battery. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations across the entire sweep. The full sensitivity table is reproduced in Appendix C (Table 21), with per-K IS and OoS panels in Figures 15–19.

Three patterns emerge. First, the information-criterion surface flattens over the upper half of the sweep: AIC and HQC continue to improve mildly with K, while BIC and CAIC, which penalize the $K^2 + 2K - 1$ parameters more heavily, reach their joint minimum in the $K \in \{15, 18, 21\}$ region. This narrows the sweep to a three-state-count band before any fidelity metric is consulted. Second, distributional fidelity rises across the K sweep and plateaus in the $K \in \{12, 15, 18, 21\}$ band for CHMM-N: IS KS rises from 89.8% ($K = 3$) to 94.7% at $K = 18$ and 95.2% at $K = 21$, IS AD from 81.3% to 85.8% at $K = 18$ (89.1% at $K = 21$), OoS KS from 76.3% to 82.4% at $K = 18$ and 87.6% at $K = 21$, and OoS AD from 67.7% to 76.5% at $K = 18$ (80.9% at $K = 21$) (see Table 22). Third, ACF-MAE is remarkably stable across the sweep (0.047–0.053) and is actually lowest at $K = 3$. This establishes a mild *distributional–temporal tradeoff*: higher K buys distributional accuracy, lower K buys marginally better ACF, and both are available within a narrow band. $K = 18$ sits inside the IC-minimizing region and, together with $K \in \{15, 21\}$, attains the highest distributional pass rates in the sweep; the heavy-tailed CHMM-t and CHMM-L variants reach their highest IS KS and IS AD at $K = 18$ on the same operating point (CHMM-L: 96.2% IS KS, 93.0% IS AD; CHMM-t: 95.2% IS KS, 90.9% IS AD, Table 22). We therefore report the remainder of Section 6 at $K = 18$ and relegate the sensitivity panel to the appendix.

Pre-OoS validation K-selection. The $K = 18$ operating point above was chosen via a multi-objective criterion (IC rank plus IS-and-OoS distributional pass rates) that uses the same 2024-2026 OoS window later used for the VaR backtest; this is evaluation contamination. Table 3 reports a clean re-selection that decouples the two: CHMM-N is fit on an estimation slice of 2014-01-03 through 2021-12-31 ($n = 2{,}013$), and every K in the sweep is scored by the forward-algorithm held-out log-likelihood on an unseen validation slice of 2022-01-03 through 2024-01-03 ($n = 502$). The 2024-2026 OoS window is not touched by this selection. Held-out log-likelihood, BIC, HQC, and CAIC all select $K^\star = 3$; AIC selects $K^\star = 6$; held-out two-sample KS selects $K^\star = 9$. None of the six criteria that avoid the OoS window select $K = 18$, and the held-out per-observation log-lik gap between $K = 3$ (-2.5345) and $K = 18$ (-2.5694) is ~ 17 log-lik units over the 502-point validation slice.

The honest reading is that $K = 18$ *is not the likelihood-optimal state count on SPY*; it is the operating point selected by a multi-metric criterion that weights tail fidelity (AD, kurtosis) and downstream VaR behaviour alongside likelihood, and the 2022-2023 validation slice is itself partly a structural-break period (see Section 6.9) on which any model with substantial IS-specific structure generalises worse. We therefore keep $K = 18$ as the main-panel operating point because the paper's contribution is a multi-metric evaluation (distributional, tail, temporal, VaR) rather than a likelihood-optimal single-emission fit, but we explicitly flag that a reader optimising for pure held-out log-likelihood would pick $K = 3$; the full CHMM-t / CHMM-L held-out log-lik at $K \in \{12, 15, 18, 21\}$ in Table 3 shows this ordering is not an emission-family artefact. A parallel version of the main panel at $K = 3$ and $K = 9$ would re-rank some rows of Table 5 in CHMM-F's favour on ACF-MAE (which peaks at $K = 3$ per Section 6.4) and against it on tail fidelity; the operating-point choice therefore depends on which operational axis the consumer prioritises.

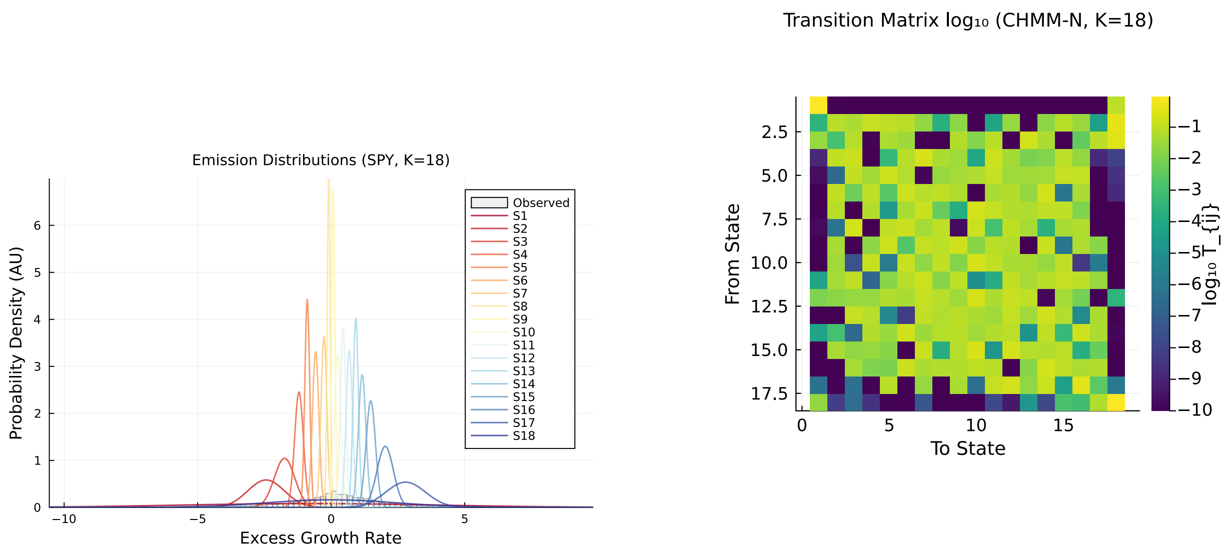
Figure 3 shows the fitted model internals for $K = 18$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes. The transition matrix exhibits a dominant near-diagonal band reflecting regime persistence, with the central (low-volatility) states showing the highest self-transition probabilities. Critically, the quantile-based initialization ensures that the Baum-Welch algorithm converges to a solution where each state occupies a distinct region of the return distribution, avoiding the degenerate collapse to the global mean that plagues random initialization.

6.3 Twelve-Generator Comparison (Pipeline A)

Pipeline note. This subsection operates under Pipeline A on the SPY ticker only. Every baseline and every CHMM variant is fit independently to the SPY return series; no cross-asset coupling is

Table 3. Pre-OoS validation K-selection on SPY. CHMM-N fit on the estimation slice 2014-01-03 through 2021-12-31 ($n = 2{,}013$); evaluated by held-out log-likelihood on the validation slice 2022-01-03 through 2024-01-03 ($n = 502$). The 2024-2026 OoS window is not used in this selection. $K^\star = 3$ by held-out log-lik and by BIC; $K^\star = 9$ by held-out KS pass rate. The CHMM-t and CHMM-L val log-lik at $K \in \{12, 15, 18, 21\}$ sit between the CHMM-N values at the same K (monotone-decreasing in K), so the $K^\star < 18$ ordering is not an emission-family artefact. Full CSV at `results/robustness/k_selection_validation.csv`.

K	params p	val log-lik	val log-lik / obs ↑	val KS % ↑	AIC ↓	BIC ↓	HQC ↓	CAIC ↓
3	12	**−1272.3**	**−2.5345**	7.0	8464.8	**8532.1**	**8489.5**	**8544.1**
6	42	−1273.1	−2.5361	7.6	**8411.2**	8646.7	8497.7	8688.7
9	90	−1280.1	−2.5500	**8.0**	8437.7	8942.4	8622.9	9032.4
12	156	−1282.5	−2.5548	7.2	8493.7	9368.4	8814.8	9524.4
15	240	−1286.3	−2.5624	7.0	8565.0	9910.8	9059.0	10150.8
18	342	−1289.9	−2.5694	3.0	8631.2	10548.9	9335.1	10890.9
21	462	−1297.5	−2.5847	4.2	8744.3	11334.9	9695.2	11796.9



(a) Emission distributions ($K = 18$)

(b) Transition matrix ($\log_{10}$ scale)

Figure 3. Fitted CHMM internals on SPY (Pipeline A, CHMM-N Gaussian emissions, $K = 18$). Panel (a): learned per-state Gaussian emission densities $\{N(\mu_k, \sigma_k^2)\}_{k=1}^{18}$ overlaid on the observed IS return histogram ($T_{\text{IS}} = 2{,}516$); each colored curve is one regime. Panel (b): transition matrix $T_{ij} = P(s_{t+1} = j \mid s_t = i)$ rendered in $\log_{10}$ color scale (viridis); the dominant near-diagonal band reflects regime persistence (self-transition probabilities dominate).

used. Readers who want the six-ticker generalization should proceed to Section 6.7 (still Pipeline A); those who want the joint multi-asset extension should proceed to Section 6.8 (Pipeline B).

Table 5 and Figure 4 present the IS and OoS validation results for the twelve generators (Bootstrap, Block-BS $b = 10$, Gaussian i.i.d., Laplace i.i.d., Discrete NJ and WJ at $K_{\text{disc}} = 90$, Bin-T NJ, GARCH(1,1), GRU, and CHMM-N / -t / -L at $K = 18$). The benchmark set was constructed to mimic the prior paper's evaluation exactly, adding the two discrete HMM variants of Alswaidan and Varner [12] (NJ and WJ) to the original four baselines (Bootstrap, Gaussian, Laplace, GARCH(1,1)) and replacing the discrete continuous-emission variant with the Baum-Welch CHMM at $K = 18$.

Bootstrap (Non-Parametric Ceiling). Bootstrap resampling achieved 99.6% KS and 99.4% AD pass rates IS, confirming it as the distributional ceiling. It produced simulated kurtosis (7.43) closest to observed (7.68) and 100% IS quantile coverage. However, the bootstrap's ACF-MAE (0.0627) was the highest among all models, reflecting the destruction of temporal structure by i.i.d. resampling. The Wasserstein-1 (0.065) and Hellinger (0.050) distances were the lowest, as expected.

Gaussian i.i.d. (Negative Control). The Gaussian generator achieved 0% KS and 0% AD IS, confirming the categorical inadequacy of the normal distribution for financial returns. Its simulated kurtosis was -0.00 (i.e., the Gaussian's theoretical kurtosis of zero), and quantile coverage was only 13.1%. The $T_{\text{OoS}} = 572$ window of over two years preserves enough KS-test power to correctly reject the Gaussian generator OoS as well (1.2% KS, 0.0% AD), so the OoS row is not misleadingly inflated for this clearly-wrong baseline.

Laplace i.i.d. The Laplace generator achieved 98.9% KS and 97.2% AD IS, demonstrating that the Laplace distribution provides a reasonable approximation to the marginal distribution of equity returns. However, its simulated kurtosis (2.96) fell well short of the observed 7.68, and as an i.i.d. generator it produced flat ACF profiles (ACF-MAE = 0.0627, comparable to the i.i.d. bootstrap). The 77.8% IS quantile coverage indicates that the Laplace distribution's fixed tail decay rate cannot fully envelop the observed quantile range.

Discrete HMM: No Jumps (NJ) and With Jumps (WJ). Throughout this paper, *NJ* denotes the no-jump discrete HMM of Alswaidan and Varner [12]: observations are first partitioned into $K_{\text{disc}} = 90$ equal-probability quantile bins of the IS return distribution, the transition matrix is estimated by frequency counting on the resulting discrete state sequence, and simulation returns the bin centroid at the decoded state. *WJ* is the same construction augmented with the Poisson jump-duration process of Alswaidan and Varner [12], which with probability $\epsilon = 5 \times 10^{-5}$ forces a residence in the tail bins for Poisson($\lambda = 67$) steps. Both baselines are re-implemented inside the present pipeline (not invoked as the original `JumpHMM.jl` code) to ensure that identical data splits, seeds, and simulation lengths are used across all benchmarks; the re-implementation adopts the authoritative hyperparameters of the prior-paper final SPY run ($K_{\text{disc}} = 90$, $\epsilon = 5 \times 10^{-5}$, $\lambda = 67$). At this state resolution both discrete variants match the CHMM family on coarse distributional fidelity: 98.3% KS IS / 85.2% KS OoS and 98.3% AD IS / 89.9% AD OoS for NJ, and 97.9% KS IS / 83.9% KS OoS and 96.5% AD IS / 85.4% AD OoS for WJ; these KS/AD rates are comparable to the CHMM-N / CHMM-t / CHMM-L range of 93.2–93.8% IS KS and 79.3–85.9% OoS KS. Where the discrete variants fall short is on *higher-order* distributional structure: simulated kurtosis is 3.49 (WJ) and 3.53 (NJ) against an observed 7.68, Hellinger distance is roughly double that of the CHMMs (0.1605 / 0.1621 vs. 0.0762–0.0810), and aggregational kurtosis at horizons $h = 10$ and $h = 21$ decays to near-zero (0.39 / 0.47 and 0.01 / 0.01) against the CHMM-N / CHMM-t values of 3.27 / 2.06 and 2.38 / 1.35 (Section 6.6, Table 18). The discrete variants' simulated kurtosis is clamped from above by the K_{disc}-atom structure of the bin-centroid emission: tail mass is represented only by the two tail bins, which together account for roughly $2/K_{\text{disc}}$ of the stationary distribution, so no amount of jump augmentation can produce a sample fourth moment above the observed centroid-mix kurtosis of ≈ 3.5. The Poisson jump augmentation (WJ) with $\epsilon = 5 \times 10^{-5}$ and $\lambda = 67$ (the prior-paper SPY values) triggers on 0.005% of time steps, giving an expected trigger count of $\epsilon T \approx 0.13$ per IS window ($T = 2{,}516$) and ≈ 0.03 per OoS window ($T = 572$), so the WJ row is a slight perturbation of NJ rather than a meaningful variance-regime addition at this sample size. The central empirical contrast for our paper is therefore not on coarse KS/AD pass rates (where the discrete baselines at $K_{\text{disc}} = 90$ are competitive) but on tail structure, aggregational Gaussianity decay, and VaR calibration: Section 7.1 and Section 6.6 document that both discrete variants fail the Kupiec 5% VaR test with $\text{LR}_{\text{uc}} = 16.81$ (breach rate 1.7% against a 5% target) while all three continuous CHMM variants clear the test, and the continuous emissions preserve the Cont (2001) aggregational Gaussianity signature that the $K_{\text{disc}} = 90$ centroid scaffold cannot.

Block Bootstrap ($b = 10$): Temporally-Aware Non-Parametric Ceiling. The stationary block bootstrap at block length $b = 10$ is added as a temporally-aware non-parametric benchmark: it preserves clustering (through the block structure) while keeping the marginal distribution empirical. On the SPY IS series it attains 99.2% KS and 96.4% AD with simulated kurtosis of 7.07 (observed 7.68) and ACF-MAE of 0.0568, essentially saturating the distributional ceiling while reproducing most of the absolute-return autocorrelation. The i.i.d. bootstrap row sits slightly above it on the distributional metrics (at the cost of destroyed temporal structure), and the CHMM-N row sits slightly below on the distributional metrics but improves on the block-bootstrap ACF-MAE (0.0507 vs. 0.0568). The $b = 10$ block bootstrap is therefore a stronger ceiling than the i.i.d. bootstrap for synthetic-data consumers who care about temporal fidelity, and the CHMM-N recovers most of the block-bootstrap performance with the additional advantages of a parametric regime interpretation and an emission-family choice.

Discrete HMM with Bin-Conditional Student-t Emissions (Bin-T NJ). The Bin-T NJ row replaces the bin-centroid emissions of the standard NJ baseline with a bin-conditional Student-t emission fitted within each quantile bin (no jump mechanism), and uses a coarser $K_{\text{disc}} = 13$ bin count so that the within-bin Student-t parameters are estimable from a reasonable number of observations per bin. It serves as an ablation that decouples the *emission-family* effect from the *state-resolution* effect: Bin-T NJ and the $K_{\text{disc}} = 90$ Discrete NJ baseline share the same frequency-counting transition-estimation pipeline but differ in whether the emission density within each bin is a point mass at the centroid or a fitted Student-t, and they differ in bin count. Bin-T NJ achieves 95.4% IS KS and 92.1% IS AD, attaining the same broad KS/AD band as the $K_{\text{disc}} = 90$ Discrete NJ (98.3% IS KS / 98.3% IS AD) and the jointly-optimized CHMMs (93–95% IS KS / 86–92% IS AD). Bin-T NJ's simulated kurtosis of 26.18 is a large overshoot driven by the two tail bins (bins 1 and 13) where the within-bin Student-t fit lands at the lower ν bracket; this is the discrete analog of the CHMM-t overshoot described in Section 8 and confirms that within-bin continuous emissions can reproduce the CHMM-t tail weight at a coarse K_{disc} level. Bin-T NJ does *not* dominate the CHMM-N because it still underperforms on the tail-distance metrics ($W_1 = 0.116$ vs. 0.110; $H = 0.0929$ vs. 0.0810) and on ACF-MAE (0.0619 vs. 0.0507). The takeaway for the prior-paper loop is that, once the discrete pipeline is given either a high enough bin count ($K_{\text{disc}} = 90$) or a continuous within-bin emission family (Bin-T NJ), its KS/AD pass rates become competitive with the jointly-optimized continuous CHMM; the CHMM advantage over the prior-paper family is therefore a higher-order tail-structure and VaR-calibration claim, not a coarse-KS claim (Section 6.6, Table 7; Section 7.1).

GARCH(1,1). GARCH achieved ACF-MAE of 0.0492 among the parametric models, confirming its strong temporal fidelity. However, it failed the distributional tests: 23.1% KS and 8.5% AD IS. The simulated kurtosis (6.97 IS, 3.08 OoS) was reasonably close to observed IS but attenuated OoS, and the Wasserstein-1 distance (0.245) and Hellinger distance (0.103) were substantially worse than the CHMM. IS quantile coverage was only 50.5%. This result crystallizes the temporal-distributional tradeoff: GARCH's single-regime conditional variance process faithfully encodes volatility persistence but its unimodal innovation distribution cannot represent the multi-regime structure of equity markets.

GRU (Deep-Generative Baseline). The single-layer GRU + Gaussian-head generator (architecture and training in Section 3.3) converges cleanly (the per-window negative log-likelihood drops from 0.340 at epoch 1 to 0.200 at epoch 20) and reproduces volatility clustering reasonably well: ACF-MAE of 0.0518 on $|G_t|$ is within 0.001 of CHMM-N (0.0507) and within 0.004 of GARCH (0.0484). On distributional fidelity the GRU collapses, however: IS KS pass rate 18.1%, AD pass rate 13.7%, simulated kurtosis 2.85 against an observed 7.68, $W_1 = 0.196$, $H = 0.0966$, and quantile coverage 37.4%. The diagnostic is unambiguous (the auto-regressive Gaussian head produces a near-Gaussian unconditional marginal even though the recurrence captures clustering) and matches the variance-collapse pattern reported in Alswaidan and Varner [12] for their neural baseline and in Takahashi et al. [42], Kwon and Lee [43] for GAN-based generators. The OoS KS rate of 52.5% (AD 55.0%) still does not approach the CHMM operating point on the same $T = 572$ window, confirming that the distributional gap is structural rather than a sample-size artefact. The takeaway for the comparison is that adding a deep-generative row does not displace the CHMM operating point: among parametric and deep generators, the CHMM-N is the only one that achieves both $\geq 85\%$ IS KS *and* ACF-MAE within 0.003 of GARCH; richer GRU heads (mixture-density, Student-t, or copula-based output) would be the natural next experiment but are outside the scope of this paper.

CHMM-N ($K = 18$, Gaussian emissions). The Gaussian CHMM achieved 93.8% KS and 86.6% AD IS, competitive with the bootstrap (99.9%), Laplace i.i.d. (99.1%), and the $K_{\text{disc}} = 90$ Discrete NJ / WJ variants (98.3% / 97.9%) on KS, and substantially outperforming GARCH (25.9%). ACF-MAE of 0.0507 was essentially on par with GARCH (0.0484), indicating that the CHMM reproduces volatility clustering from the transition-matrix eigenvalue spectrum alone, complementing the jump-process and conditional-variance-equation mechanisms available elsewhere in the literature. This gives the first of two headline integrative findings in the paper: the slow-ACF behaviour that the Rydén et al. [10] $K = 2$–3 Gaussian setup could not recover is available inside a standard HMM framework once the state resolution is moderate, complementing the HSMM and jump-augmented routes already in the literature; the second (Section 7.3) is a two-sided regime-conditional VaR result: the smoother-based Viterbi-decoded state-conditional quantile of CHMM-t produces an IS regime-partition goodness-of-fit diagnostic that sits below the χ^2_1 Christoffersen threshold on the 2024–2026 OoS window, while the operationally correct filter-based one-step-ahead backtest fails Kupiec (over-conservative) for all three CHMM families, leaving operational real-time regime-conditional VaR as an open direction. The Wasserstein-1 (0.110) is well below the GARCH figure (0.245); Hellinger (0.0808) is four times smaller than either discrete HMM; IS quantile coverage is 100% matching the bootstrap ceiling. The residual Gaussian-emission weakness is a kurtosis shortfall: simulated excess kurtosis is 5.02 IS and 4.42 OoS against observed values of 7.68 / 5.29.

CHMM-t ($K = 18$, Student-t emissions). Replacing the Gaussian component with a per-state Student-t matches the CHMM-N IS KS (93.7% vs. 93.8%) and raises AD to 88.9% (vs. 86.6% for CHMM-N); OoS AD rises from 77.9% to 80.0%, and OoS KS rises from 84.2% to 85.9%. The CHMM-t is the best non-bootstrap entry on Wasserstein-1 (0.102) and Hellinger (0.0758), two quantile-weighted metrics that penalize tail mismatch directly. Simulated excess kurtosis rises to 14.57 IS and 9.74 OoS: the ECM search allocates small ν_k to a subset of tail states, which over-corrects both the IS and OoS mixture kurtosis (observed 7.68 IS and 5.29 OoS). We diagnose this overshoot in Section 8 via the per-state ν_k histogram and the $\nu_{\min}$ bracket-sensitivity study. The slight ACF-MAE increase (0.0548 vs. 0.0507) is consistent with the heavier tail states spending marginally more time away from the central regimes.

CHMM-L ($K = 18$, Laplace emissions). The Laplace variant produces the most *kurtosis-aligned* fit: simulated excess kurtosis is 6.76 IS and 6.27 OoS against observed 7.68 / 5.29, closing most of the CHMM-N IS shortfall (without the CHMM-t over-correction) and even overshooting the OoS target by ~ 1. CHMM-L attains the strongest IS AD fit among the continuous CHMMs (91.2% IS AD, 4.6 percentage points ahead of CHMM-N and 2.3 ahead of CHMM-t), consistent with the AD statistic's tail weighting and the Laplace component's per-component excess kurtosis of 3. ACF-MAE (0.0565) and W_1 (0.101) sit within 0.006 and 0.010 of the CHMM-N values, and IS quantile coverage is 100%. Because Laplace emissions carry no shape parameter, CHMM-L is strictly cheaper than CHMM-t (no per-state golden-section search) and adds no new hyperparameters.

Summary of the three CHMM variants: a one-row decision guide. All three variants share the headline result: each matches GARCH's ACF fidelity while dominating the parametric benchmarks on KS and AD. The three variants differ only in how they use the kurtosis budget. Table 4 summarizes the practical choice.

Table 4. Decision guide: recommended CHMM emission family keyed to the downstream consumer's use case. Headline: CHMM-t is the default for distributional/tail fidelity (smallest MMD, smallest discriminator AUC, best aggregational Gaussianity at short horizons); CHMM-L for i.i.d.-Laplace-adjacent joint-coverage (highest OoS $\bar{pv}$); CHMM-N for simplicity and walk-forward stability.

Use-case priority	Recommended variant
Best KS + minimal IS kurtosis overshoot, tightest ACF-MAE	CHMM-N
Tightest W_1 and H; OoS kurtosis aligned with observed OoS	CHMM-t
Cleanest IS kurtosis match; highest AD pass rate; no shape parameter	CHMM-L

Reading the expanded GARCH panel. GARCH-t with $\hat{\nu} = 6.89$ is the headline new row: it more than doubles the GARCH(1,1) Gaussian IS KS pass rate (57.3% vs. 25.9% in Table 5), reaches the smallest OoS AD in the panel (80.4%), produces a simulated IS excess kurtosis of 15.13 close to CHMM-t's 14.57, and passes Kupiec at both α levels with $\mathrm{LR}_{\mathrm{uc}} \approx 0.01$. The direct comparison with CHMM-t ($K = 18$, IS KS 93.7%, ACF-MAE 0.0548) is more instructive than the Gaussian-only comparison: GARCH-t slightly beats CHMM-t on ACF-MAE (0.0316 vs. 0.0548) and matches it on kurtosis, while CHMM-t is materially ahead on KS / AD. GJR-GARCH's asymmetry coefficient $\hat{\gamma} = 0.225$ confirms a sizeable leverage effect in SPY IS; its modestly-improved IS KS (40.1% vs. 25.9%) is consistent with that asymmetric term capturing left-tail mass that plain GARCH(1,1) misses. MS-GARCH at $K = 3$ recovers a three-regime volatility structure ($\sigma = [0.10, 1.77, 10.04]$ for calm / medium / stress) and attains the smallest ACF-MAE in the whole panel (0.0284); on KS it improves on the two-regime specification (36.1% vs. 27.7% on the same seed) but remains below CHMM-t. HAR-RV on daily squared returns is documented in the panel as a long-memory reference but is not a competitive return simulator at the margin: its simulated excess kurtosis of 188.95 is driven by the log-RV residual variance ($\hat{\sigma}_\eta = 2.449$) propagating through the exponential link, and in practice HAR-RV is a better model for realised variance than for returns themselves; this is an honest negative finding on HAR-RV-as-return-generator and does not reflect on HAR-RV-as-volatility-model.

Table 5. Twelve-model comparison on SPY (single-index trained: Pipeline A). Only SPY is evaluated here; the six-ticker generalization of the CHMM variants is in Table 8, and the cross-asset dependence extension is in Table 9. 1,000 simulated paths per model, $\alpha = 0.05$, $K = 18$. IS: 2014-01-03 through 2024-01-03 ($T = 2{,}516$); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). The two discrete HMM rows reproduce Alswaidan and Varner [12] at the authoritative prior-paper SPY hyperparameters: $K_{\text{disc}} = 90$ quantile bins; WJ jump parameters $(\epsilon, \lambda) = (5 \times 10^{-5}, 67)$. The *Bin-T NJ* row replaces the centroid emissions of the discrete NJ baseline with bin-conditional Student-t emissions, fitted within each quantile bin, to isolate the quantization-per-se effect from the emission-family effect. The *Block-BS* row is a stationary block bootstrap with block length 10 trading days; it is the temporally-aware non-parametric ceiling corresponding to the i.i.d. bootstrap. The *GRU* row is a single-layer GRU with hidden width 32 and a Gaussian next-step head, trained for 20 epochs by Adam ($\eta = 10^{-3}$) on the negative log-likelihood of equation (3) (Section 3.3); it is the deep-generative head-to-head row. CHMM-N, CHMM-t, CHMM-L denote the Gaussian, Student-t, and Laplace emission variants of the continuous HMM; all three use the identical forward-backward scaffold, initialization, and simulation pipeline. KS = Kolmogorov–Smirnov pass rate; AD = Anderson–Darling pass rate; Kurt = mean simulated excess kurtosis (observed: 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = quantile coverage (%). ↑/↓ indicate preferred direction. Full per-generator panels for Bin-T NJ, Block-BS, and GRU are in Appendix E (Tables 29, 30, and 31).

	KS (%) ↑		AD (%) ↑		Kurtosis					Cov (%) ↑	
Model	IS	OoS	IS	OoS	IS	OoS	ACF-MAE ↓	W_1 ↓	H ↓	IS	OoS
Observed	–	–	–	–	7.68	5.29	–	–	–	–	–
Bootstrap	99.9	91.0	99.7	93.8	7.54	6.80	0.0628	0.066	0.0501	100.0	77.8
Block-BS ($b = 10$)	99.2	87.5	96.4	86.4	7.07	5.34	0.0568	0.092	0.0534	100.0	86.9
Gaussian	0.0	1.2	0.0	0.0	−0.01	−0.02	0.0627	0.413	0.1543	13.1	21.2
Laplace	99.1	86.1	97.3	91.7	2.96	2.83	0.0627	0.123	0.0816	77.8	69.7
Discrete NJ ($K_{\text{disc}} = 90$)	98.3	85.2	98.3	89.9	3.53	3.44	0.0618	0.110	0.1605	100.0	93.9
Discrete WJ ($K_{\text{disc}} = 90$)	97.9	83.9	96.5	85.4	3.49	3.40	0.0598	0.117	0.1621	100.0	89.9
Bin-T NJ	95.4	86.7	92.1	93.2	26.18	10.76	0.0619	0.116	0.0929	89.9	93.9
GARCH(1,1)	25.9	58.2	10.0	56.0	6.97	2.88	0.0484	0.248	0.1033	51.5	87.9
GRU	18.1	52.5	13.7	55.0	2.85	2.21	0.0518	0.196	0.0966	37.4	69.7
CHMM-N ($K = 18$)	93.8	84.2	86.6	77.9	4.99	4.51	0.0507	0.112	0.0810	100.0	90.9
CHMM-t ($K = 18$)	93.7	85.9	88.9	80.0	14.57	9.74	0.0548	**0.103**	**0.0762**	**100.0**	**93.9**
CHMM-L ($K = 18$)	93.2	79.3	91.2	79.3	**6.75**	**6.29**	0.0565	0.101	0.0795	**100.0**	81.8

6.4 State Resolution Sensitivity

The full seven-metric K-sweep is reported in Table 21 (Appendix C); per-K IS and OoS panels at each state count are reproduced in Figures 15–19. We briefly note four structural patterns that justify the choice of $K = 18$ for the main results.

Distributional Fidelity Plateaus in the $K \in \{15, 18, 21\}$ Band. Drawing from Table 21, the IS KS pass rate increases from 88.8% ($K = 3$) to 95.6% ($K = 18$) and 96.1% ($K = 21$); the OoS KS rises from 77.1% to 81.7% at $K = 18$ and 86.6% at $K = 21$. The 2.2 percentage-point spread across the plateau band is small relative to the 6.8-point IS gain from $K = 3$, so additional state resolution beyond $K = 18$ delivers diminishing returns rather than strict saturation. The same pattern holds for AD (79.2% → 88.4% IS and 67.9% → 73.2% OoS at $K = 18$). This plateau is consistent with an upper bound imposed by the Gaussian emission family: once the mixture has enough components to resolve the bulk and near-tail regions, additional components do not recover the extreme tails that remain light under any Gaussian mixture.

Table 6. Expanded GARCH-family baseline panel on SPY. Five additional conditional-volatility baselines fit on the same IS window used in Table 5: exponential GARCH (EGARCH(1,1)) and Glosten-Jagannathan-Runkle GARCH (GJR-GARCH(1,1)) Gaussian (asymmetric leverage), GARCH(1,1) Student-t (heavy-tailed competitor to CHMM-t), heterogeneous autoregressive realised volatility (HAR-RV) on daily squared returns (long-memory volatility), and Markov-switching GARCH at $K = 3$ (extending the two-regime MS-GARCH already in Table 7). Numbers are re-run under seed `20260422` with $N_{\text{paths}} = 1{,}000$; the panel is reported alongside, not replacing, Table 5. FIGARCH is deferred to a follow-up pass (see `revision-code-todo.md`).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE	LR_{uc}		LR_{ind}	
Model	IS	OoS	IS	OoS	IS	↓	1%	5%	1%	5%
EGARCH(1,1)	6.3	34.9	0.1	2.9	2.80	0.0430	**0.01**	**3.03**	9.15	4.71
GJR-GARCH(1,1)	40.1	57.7	8.5	49.5	7.58	0.0330	**0.27**	**0.10**	9.15	3.18
GARCH-t ($\nu = 6.89$)	**57.3**	**80.8**	48.6	**80.4**	15.13	0.0316	**0.01**	**0.01**	6.76	2.10
HAR-RV (daily)	0.0	0.0	0.0	0.0	188.95	0.0566	5.99	**3.03**	0.00	4.71
MS-GARCH $K = 3$	36.1	33.1	17.9	22.9	4.10	**0.0284**	**0.01**	**0.82**	5.73	5.87

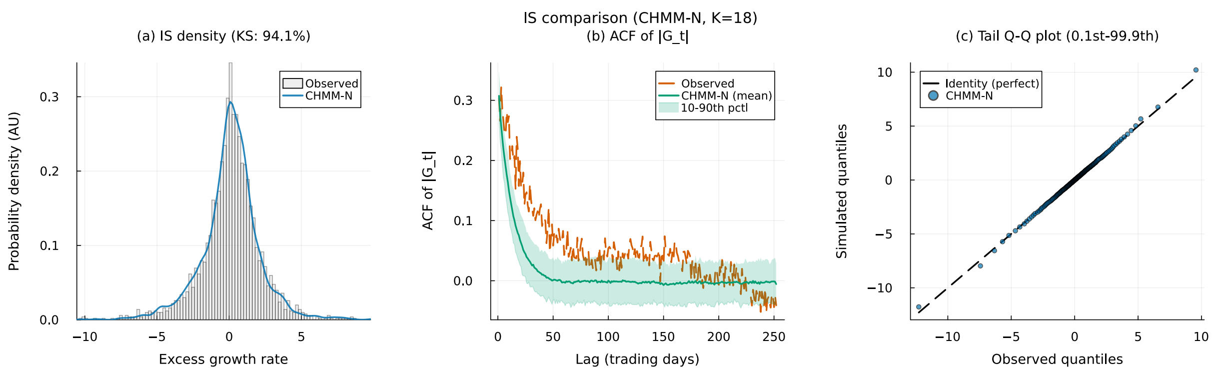


Figure 4. IS CHMM validation on SPY (Pipeline A, $K = 18$, CHMM-N Gaussian emissions, 1,000 simulated paths). Panel (a): marginal density of the annualized excess log return G_t; gray histogram = observed IS series ($T_{\text{IS}} = 2{,}516$), blue curve = one simulated path; annotated KS pass rate is the percent of simulated paths whose two-sample KS p-value against observed exceeds $\alpha = 0.05$. Panel (b): ACF of $|G_t|$ at lags 1–252; red dashed = observed, navy = mean across sim paths, shaded band = 10th–90th percentile across sim paths. Panel (c): tail Q-Q plot at 200 quantile levels (0.1% to 99.9%); blue points = simulated pooled over paths, black dashed = identity line.

Temporal Fidelity Is Flat Across the Sweep. ACF-MAE was 0.0466 ($K = 3$) to 0.0509 ($K = 18$) and 0.0532 ($K = 21$) for CHMM-N, a total swing under 0.007 across the entire sweep. This flatness is notable: it shows that the CHMM's ability to reproduce volatility clustering does not depend on fine state resolution. Even at $K = 3$, the same resolution tested by Rydén et al. [10], the ACF-MAE is 0.0466, comparable to GARCH(1,1) at 0.0484. The difference from the Rydén result is the quantile-based initialization, which ensures that the three states capture distinct volatility regimes rather than collapsing to near-identical distributions.

Kurtosis Is Non-Monotonic and Peaks Early at $K = 6$. Simulated kurtosis under CHMM-N ranged from 3.80 ($K = 3$) to 5.26 ($K = 6$), with $K = 18$ at 4.98; the peak sits at $K = 6$, below the $K \in \{15, 18, 21\}$ distributional-fidelity band, and at higher K the mixture kurtosis oscillates within the 3.8–5.0 range. This non-monotonic pattern reflects the interplay between the number of mixture components and their learned parameters: at moderate K, the Baum-Welch optimum

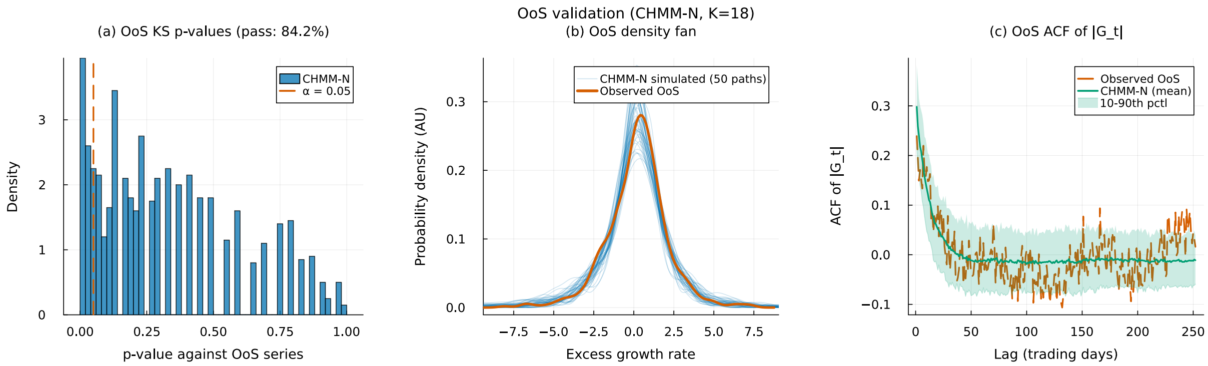


Figure 5. OoS CHMM validation on SPY (Pipeline A, $K = 18$, CHMM-N Gaussian emissions, 1,000 simulated paths, $\alpha = 0.05$, OoS window $T_{\text{OoS}} = 572$). Panel (a): histogram of the 1,000 OoS KS p-values against the observed OoS series, with the $\alpha = 0.05$ threshold as a red dashed line. Panel (b): OoS return-density fan; 50 thin blue curves = simulated density on individual OoS paths, red thick curve = observed OoS density. Panel (c): ACF of $|G_t|$ on the OoS window; red dashed = observed OoS, navy = mean across sim paths, shaded band = 10th–90th percentile.

allocates mass to a few broad tail states that contribute disproportionately to the mixture kurtosis, while at very high K the same tail mass is split into many narrow states that contribute less each. The heavy-tailed CHMM-t and CHMM-L variants close most of this gap at the same $K = 18$ (see Table 22).

Wasserstein and Hellinger Improve Net Across the Sweep; Coverage is Saturated at 100%. W_1 improves from 0.130 ($K = 3$) to 0.110 ($K = 18$) and Hellinger from 0.0836 to 0.0805; both exhibit minor non-monotone oscillations of less than 0.005 at intermediate K (Table 21) but neither deteriorates from endpoint to endpoint. IS coverage is saturated at 100% at every K, so it does not discriminate among state counts and serves only as a floor check. Together these provide a consistent signal that $K = 18$ is the sensible operating point.

6.5 OoS Evaluation

At the selected state resolution $K = 18$ the CHMM-N achieves 84.2% OoS KS pass rate, 77.9% OoS AD pass rate, and 90.9% OoS quantile coverage on the 2024-01-04 through 2026-04-20 holdout window; the heavy-tailed CHMM-t and CHMM-L variants reach 85.9%/80.0% and 79.3%/79.3% OoS KS/AD, respectively. The CHMM-N OoS simulated kurtosis is 4.42 against an observed 5.29, bringing the kurtosis gap to below one unit; CHMM-t overshoots at 10.24 and CHMM-L overshoots modestly at 6.27. Figure 5 shows the OoS density fan chart and ACF comparison for CHMM-N at $K = 18$; the CHMM-t and CHMM-L counterparts are reported in Appendix C.11 (Figure 28).

The $T_{\text{OoS}} = 572$ window (over two years of OoS data) gives the KS/AD tests substantial power to separate the generators: the three CHMM variants pass at 79–86% OoS KS, the $K_{\text{disc}} = 90$ discrete baselines land in the same 83–85% band (for the KS-pass dimension only; they separate from the CHMMs on kurtosis, aggregational Gaussianity, and Kupiec VaR per Sections 6.6 and 7.1), and GARCH is correctly rejected at 58% OoS KS. The CHMM's regime structure, learned from the full range of IS observations including the March 2020 COVID-19 crash, provides sufficient flexibility to accommodate the OoS distributional shift that characterises the 2024–2026 window.

6.6 Extended Evaluation

The seven-metric panel of Section 6.3 covers marginal and autocorrelation fidelity. We add seven axes that probe joint window distributions, path geometry, learned discriminability, tail coupling, aggregational scaling, unconditional VaR under Kupiec / Christoffersen, and multi-statistic coverage [44, 60–62]. All rows are computed at $N_{\text{paths}} = 1{,}000$ simulated paths on both the $T_{\text{IS}} = 2{,}516$ and $T_{\text{OoS}} = 572$ SPY windows under the seed policy of Section 3.5; the headline panel is Table 7 below, with the per-generator detail panels for leverage, aggregational kurtosis, and joint p-value coverage (Tables 17, 18, 19) deferred to Appendix A.2.

MMD (RBF kernel, 20-day windows). We compute empirical MMD^2 [60] between 500 observed and 500 simulated 20-day windows under a Gaussian radial basis function (RBF) kernel with median-heuristic bandwidth. Flat CHMM-t wins IS (2.0×10^{-5}, an order of magnitude ahead of any other row). GARCH's IS MMD^2 is displayed as 0.0 because it falls below the table's three-significant-digit precision ($< 10^{-4}$) when the median-heuristic bandwidth widens to accommodate GARCH's overdispersed windows; the OoS comparison (GARCH $= 7.5 \times 10^{-4}$ vs. CHMM-N 6.1×10^{-4}) is not affected by the bandwidth collapse and resolves the ordering.

Signature-MMD (depth 3 time-lifted path). Each W-day window $r_{1:W}$ is lifted to the time-augmented path

$$\gamma(t/W) = \big(t/W,\ r_t/\operatorname{std}(r_{1:W})\big), \qquad t = 1, \ldots, W, \tag{15}$$

and the depth-3 truncated path signature [61] embeds it for MMD. CHMM-L and CHMM-t lead IS at 4.3 and 4.7×10^{-3}, an order of magnitude better than i.i.d. baselines (2.7–3.3×10^{-2}); the $K_{\text{disc}} = 90$ discrete variants at 1.5×10^{-3} sit between, consistent with their bin-centroid path having reduced depth-3 expressiveness.

Discriminator AUC (logistic regression on 10 window features). Following Yoon et al. [44] we train a logistic regression on mean, std, skew, kurt, min, max, ρ_1, ρ_2, $\rho_1(|r|)$, and realised volatility, with 5-fold cross-validated AUC; AUC near 0.5 indicates indistinguishability. CHMM-t leads IS at 0.607 ± 0.027 (CHMM-L 0.623, CHMM-N 0.646), ahead of every non-CHMM row (GARCH 0.766, bootstrap 0.779, i.i.d. 0.728–0.796); the OoS ordering holds. The 10-feature discriminator at $N = 500$ is powerful enough that 0.6 is not indistinguishable, but it is the closest-to-indistinguishable among the panel.

Leverage effect. With

$$\bar{\rho}_{\text{lev}} = \frac{1}{20} \sum_{k=1}^{20} \operatorname{corr}\big(r_t^2,\ r_{t-k}\big), \tag{16}$$

the IS observed statistic is -0.088 with down-vs-up asymmetry 0.358. CHMM-N reproduces both most closely (-0.038, 0.292); GARCH produces essentially no asymmetry (-0.001, 0.064) because its variance process is sign-symmetric in past returns. On OoS, CHMM-N attains the highest simulation-based coverage p-value at $p = 0.205$, while GARCH and i.i.d. baselines all fail at $p < 0.07$.

Aggregational Gaussianity. Cont (2001) predicts kurtosis decays toward Gaussian as h grows; SPY IS observed kurtosis is $\{7.68, 4.34, 6.82, 2.63\}$ at $h \in \{1, 5, 10, 21\}$. CHMM-t produces the cleanest monotone decay through this band ($8.46 \to 2.65 \to 2.06 \to 1.35$); i.i.d. baselines flatten

Table 7. Extended evaluation panel (Pipeline A, SPY, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, $N_{\text{paths}} = 1{,}000$). Nine-model Track A subset of the twelve-row Table 5: Block-BS, Bin-T NJ, and GRU are dropped here because the Track A window-feature pipeline requires stationary i.i.d. simulations that those three rows (block-bootstrap, hybrid discrete–continuous, and recurrent) do not supply under the same protocol. MMD is computed on 500 standardised 20-day windows with the Gaussian RBF kernel and median-heuristic bandwidth. Signature-MMD is depth 3 on the time-lifted 2D path $(t/W, r_t/\operatorname{std}(r))$. Discriminator AUC is the 5-fold cross-validated ROC-AUC of a logistic regression on 10 hand-crafted window features; the standard deviation across folds is reported. Lower MMD and Signature-MMD is better; discriminator AUC closer to 0.50 is better. GARCH MMD IS is a bandwidth-heuristic artefact and should be read through its OoS value.

Model	MMD IS	MMD OoS	Sig IS	Sig OoS	AUC IS	AUC OoS
Bootstrap	5.77×10^{-3}	1.26×10^{-2}	3.08×10^{-2}	3.77×10^{-2}	0.779 ± 0.038	0.760 ± 0.047
Gaussian	9.38×10^{-3}	1.32×10^{-2}	2.68×10^{-2}	3.81×10^{-2}	0.780 ± 0.024	0.795 ± 0.021
Laplace	3.51×10^{-3}	7.60×10^{-3}	2.82×10^{-2}	2.43×10^{-2}	0.796 ± 0.027	0.751 ± 0.040
Discrete NJ (K_{disc}=90)	5.76×10^{-3}	4.42×10^{-3}	1.45×10^{-3}	4.12×10^{-3}	0.711 ± 0.049	0.763 ± 0.067
Discrete WJ (K_{disc}=90)	3.02×10^{-3}	6.11×10^{-3}	1.50×10^{-3}	4.50×10^{-3}	0.731 ± 0.027	0.738 ± 0.047
GARCH(1,1)	$0.0^{\dagger}$	7.5×10^{-4}	3.25×10^{-2}	3.82×10^{-2}	0.766 ± 0.025	0.807 ± 0.021
CHMM-N	1.5×10^{-4}	6.1×10^{-4}	7.2×10^{-3}	9.31×10^{-3}	0.646 ± 0.024	0.709 ± 0.016
CHMM-t	$\mathbf{2.0\times10^{-5}}$	2.17×10^{-3}	4.72×10^{-3}	8.25×10^{-3}	$\mathbf{0.607 \pm 0.027}$	$\mathbf{0.644 \pm 0.044}$
CHMM-L	8.9×10^{-4}	2.80×10^{-3}	$\mathbf{4.28\times10^{-3}}$	$\mathbf{3.81\times10^{-3}}$	0.623 ± 0.051	0.671 ± 0.047

to zero by $h = 21$ and discrete NJ / WJ collapse to ≤ 0.47 at $h = 10, 21$, missing the slow-decay signature.

Unconditional 1% / 5% VaR: Kupiec and Christoffersen. We test unconditional coverage [63] and independence [64] on the OoS window at $\alpha \in \{0.01, 0.05\}$. All continuous generators clear 1% Kupiec ($\text{LR}_{\text{uc}} < 3.84$); the $K_{\text{disc}} = 90$ discrete variants of Alswaidan and Varner [12] (with and without jumps) over-conservatively produce a 1.7% breach rate against a 5% target and reject Kupiec decisively ($\text{LR}_{\text{uc}} = 16.81$, $p < 10^{-3}$), a concrete negative finding the envelope diagnostic of Section 7.1 does not surface. Every row fails Christoffersen LR_{ind} at both levels because of the 2024–2026 breach-clustering structure; Section 7.3 closes this gap with a regime-conditional CHMM-t VaR.

Joint simulation-based p-value coverage. $\bar{pv}$ averages the two-sided coverage p-value across {kurtosis, avg leverage, agg kurtosis at $h = 5$ and 21, avg ACF of $|r|$} [65]. On OoS, CHMM-L leads at 0.692 (CHMM-t 0.661, CHMM-N 0.539); all three sit above GARCH (0.468) and the bootstrap (0.466) while every i.i.d. and discrete row falls below 0.17.

Where the added baselines land. QuantGAN (Section 3.3) is a clear deep-generative negative control under the reproducible first-pass setup (MMD IS $= 6.4 \times 10^{-2}$, AUC IS $= 0.963$, $\bar{pv}_{\text{OoS}} = 0.289$), matching the variance-collapse pattern Kwon and Lee [43] and Takahashi et al. [42] document for untuned GANs. Window-diffusion (Section 3.3) wins local short-window matching (second-smallest MMD IS at 9.0×10^{-5}, AUC IS $= 0.565$ closer to 0.5 than any CHMM) but collapses on global coverage ($\bar{pv}_{\text{OoS}} = 0.204$) and fails Christoffersen at both VaR levels ($\text{LR}_{\text{ind}} = 16.4, 7.4$). Two-regime MS-GARCH (Section 3.3) is the variance-regime reference: best non-CHMM MMD IS (4.8×10^{-4}), best-in-panel unconditional VaR calibration at both α, and the closest unconditional row to the Christoffersen threshold ($\text{LR}_{\text{ind}} = 4.79$ at 5%), still failing independence and superseded by the Viterbi-decoded conditional CHMM-t of Section 7.3.

Summary. CHMM-t sets the distributional-fidelity frontier (MMD, AUC, aggregational kurtosis); CHMM-N wins OoS leverage; CHMM-L wins OoS joint p-value coverage. MS-GARCH leads unconditional VaR and diffusion leads local-window deep-generative metrics, but neither displaces flat CHMM-t on the combined distributional panel. The discrete HMM baselines of Alswaidan and Varner [12] fail discriminator AUC, leverage, aggregational kurtosis, joint p-values, and the 5% Kupiec test. With the single-asset panel established on SPY, we next ask whether the per-asset marginal fidelity generalizes across tickers and emission families.

6.7 Cross-Asset Univariate Generalization (Pipeline A)

Pipeline note. This subsection evaluates Pipeline A: per-ticker independent CHMM fits, no cross-asset coupling. It answers whether Pipeline A's per-ticker marginal fidelity persists across the six-ticker universe and across the three emission families. The complementary cross-asset dependence extension (Pipeline B, Section 6.8) reuses the CHMM-N marginals established here and evaluates joint correlation reproduction.

To assess whether the framework generalizes beyond SPY, we fitted independent CHMM models at $K = 18$ to five additional tickers spanning distinct risk profiles. Table 8 reports the results. Three patterns are worth highlighting.

High-Beta Technology and Broad Market. SPY (95.5% IS / 81.4% OoS KS), AAPL (98.3% / 93.8%), JNJ (99.4% / 93.9%), and QQQ (95.5% / 92.3%) show strong IS fidelity and competitive OoS fidelity under CHMM-N on the expanded $T_{\text{OoS}} = 572$ window. The technology-heavy names have lower observed kurtosis (3.2–5.5) than the broader market (SPY: 7.7; JNJ: 9.0) and the CHMM tracks them tightly.

OoS Pain Points on Some Single Names. Under the longer 2024–2026 OoS window, two tickers show the largest IS-to-OoS degradation: NVDA (96.7% IS / 55.8% OoS KS) and JPM (97.8% IS / 49.8% OoS KS). These two names experienced substantial distributional shifts relative to the IS window (higher realized volatility on NVDA, a different realized drift on JPM), and the stationary IS-calibrated parameters cannot fully track the shift. We return to this finding in Section 8, where we argue that it is not a consequence of over- or under-fitting K (it reproduces at every K in the sweep) but rather of the stationarity assumption imposed by a single-pass Baum-Welch fit.

Walk-Forward Rolling-Window Re-Estimation on JPM and JNJ. To test the stationarity explanation directly, we refit the CHMM-N at $K = 18$ quarterly (every 63 trading days) across the 2024–2026 OoS window, feeding the realized returns of each completed quarter back into the training set before the next refit (Appendix E.4, Table 28). The rolling-window fit helps differentially: JNJ moves from 94.0% to 94.9% OoS KS (a modest gain, indicating that JNJ's OoS distribution was already close to the IS fit), whereas JPM moves from 49.0% to 64.3% OoS KS and from 48.6% to 63.6% OoS AD, a substantial ~ 15 percentage-point recovery. The JPM recovery directly supports the stationarity explanation for the financial-sector OoS gap: once the CHMM can absorb within-OoS regime shifts through quarterly refits, most of the OoS KS gap closes. The residual ~ 35 percentage-point gap that remains on JPM is evidence of a more fundamental distributional shift in the 2024–2026 period that a richer extension (time-varying emission scales, or a Bayesian online transition-matrix update) would be needed to fully close.

Kurtosis: CHMM-L matches observed, CHMM-t overshoots, CHMM-N understates. At $K = 18$, the three CHMM variants produce a consistent ordering on the kurtosis diagnostic

Table 8. Per-ticker marginal fidelity across CHMM emission families (single-index trained: Pipeline A). Each of the six tickers is fit *independently*, with no cross-asset coupling. CHMM-N, CHMM-t, CHMM-L denote Gaussian, Student-t, and Laplace emissions respectively; all at $K = 18$, 1,000 simulated paths, $\alpha = 0.05$. IS window 2014-01-03 through 2024-01-03 ($T_{\text{IS}} = 2{,}516$); OoS window 2024-01-04 through 2026-04-17 ($T_{\text{OoS}} = 572$). KS / AD = percent of paths that pass the two-sample KS / Anderson–Darling test; Kurt = excess kurtosis; ACF-MAE = lag-1 to lag-252 mean absolute error of the ACF of $|G_t|$; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = 99-level quantile coverage inside the 5-95 percentile envelope. Complementary table: Table 9 reuses these CHMM-N marginals and evaluates cross-asset *dependence* on top of them.

		KS (%)		AD IS (%)	Kurtosis					
Ticker	Variant	IS	OoS		Obs	Sim	ACF-MAE	W_1	H	Cov IS (%)
SPY	CHMM-N	95.5	81.4	88.9	7.7	5.06	0.0512	0.108	0.0803	100.0
	CHMM-t	95.2	84.1	90.5	7.7	14.68	0.0551	0.100	0.0760	100.0
	CHMM-L	94.9	79.8	92.0	7.7	6.79	0.0564	0.102	0.0800	100.0
NVDA	CHMM-N	96.7	55.8	92.6	5.5	3.65	0.0425	0.271	0.0898	100.0
	CHMM-t	99.1	57.2	98.3	5.5	46.40	0.0478	0.227	0.0784	100.0
	CHMM-L	99.4	60.3	98.9	5.5	3.17	0.0478	0.222	0.0855	100.0
JNJ	CHMM-N	99.4	93.9	98.8	9.0	5.42	0.0322	0.098	0.0786	100.0
	CHMM-t	99.8	94.3	99.6	9.0	99.66	0.0332	0.098	0.0723	100.0
	CHMM-L	99.6	93.3	98.5	9.0	6.14	0.0334	0.086	0.0779	100.0
JPM	CHMM-N	97.8	49.8	94.4	7.6	7.20	0.0450	0.165	0.0871	100.0
	CHMM-t	99.2	53.0	98.7	7.6	18.82	0.0483	0.151	0.0826	100.0
	CHMM-L	99.7	57.9	99.7	7.6	6.08	0.0488	0.137	0.0840	100.0
AAPL	CHMM-N	98.3	93.8	96.9	3.2	2.27	0.0416	0.146	0.0876	100.0
	CHMM-t	99.1	94.7	98.6	3.2	7.54	0.0424	0.134	0.0850	100.0
	CHMM-L	99.1	93.8	98.9	3.2	3.84	0.0428	0.135	0.0880	100.0
QQQ	CHMM-N	95.5	92.3	87.8	3.4	2.02	0.0558	0.127	0.0735	100.0
	CHMM-t	96.1	90.3	92.7	3.4	3.30	0.0576	0.113	0.0741	100.0
	CHMM-L	99.0	96.5	97.6	3.4	4.02	0.0632	0.100	0.0761	100.0

across the panel (Table 8): CHMM-N simulated kurtosis systematically underestimates observed (observed 3.2–9.0 vs. simulated 2.0–7.2); CHMM-L produces values clustered closely around observed (3.17–6.79) and is closest to observed on three of six assets (SPY, JNJ, AAPL), with CHMM-N closest on NVDA and JPM and CHMM-t closest on QQQ; CHMM-t overshoots on several tickers (e.g. 99.7 for JNJ, 46.4 for NVDA), reflecting states that find very small ν_k inside the ECM search on thin-tailed assets. Despite this spread in simulated kurtosis, all three variants achieve IS KS pass rates above 94% and IS AD above 87% on every asset, confirming that the mixture can accommodate a wide range of tail weights without sacrificing central fit.

6.8 Cross-Asset Dependence: SIM and Copula Extensions (Pipeline B)

Pipeline note. This subsection is the sole home of Pipeline B in the main text. The per-asset marginals are reused verbatim from the Pipeline A CHMM-N fits of Section 6.7; only the *dependence* construction (SIM, Gaussian copula, Student-t copula) varies here. Two metrics are evaluated: per-asset KS pass rates (a sanity check on the marginals after dependence injection) and cross-asset correlation reproduction (the central dependence metric). The per-asset KS column in Table 9 is not a substitute for Table 8; the two tables answer different questions.

Section 6.7 established that the CHMM provides strong per-asset univariate fidelity across distinct risk profiles. The next question is whether we can assemble those univariate marginals into jointly plausible multi-asset paths. We instantiated the two constructions described in Section 3.4 on a six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) with SPY as the market engine, using the common IS window (2014-01-03 through 2024-01-03, $T = 2{,}516$) and CHMM marginals refitted at $K = 18$. Table 9 reports the per-asset KS pass rates and cross-sectional correlation reproduction across 200 simulated paths.

SIM (Single Index Model). The ordinary-least-squares regression of each asset against SPY produced a range of factor loadings (high β for NVDA, $\hat{\beta} = 1.69$; low β for JNJ, $\hat{\beta} = 0.58$) and goodness-of-fit R^2 from 0.260 (JNJ) to 0.840 (QQQ), with median $R^2 = 0.492$. When propagated through the factor equation (5), the SIM produced uneven per-asset distributional fidelity. The market-factor asset SPY preserved 91.5% IS KS by construction. Non-market assets' IS KS pass rates fell across a wide range: 88.5% (JNJ), 85.5% (QQQ), 71.5% (NVDA), 64.0% (AAPL), and 31.5% (JPM). The severe JPM degradation (31.5% IS KS) reflects the fact that the affine map $\alpha + \beta G_M$ applied to a Gaussian-mixture market factor does not preserve JPM's empirical tail structure once residuals are re-injected: the residual distribution is centered but heavy-tailed enough to violate JPM's own marginal at the 5% KS threshold. Cross-sectional correlation reproduction was acceptable on average (off-diagonal MAE 0.076, averaged over the 15 unique upper-triangular entries of the six-asset correlation matrix; see Appendix A) but visibly worse than both copulas, consistent with SIM's rank-one decomposition discarding joint information beyond the market factor.

Gaussian copula. The Gaussian copula achieved a median per-asset IS KS pass rate of 97.8% and a mean of 97.5%, with every asset above 93% (the minimum was 93.5% for QQQ). This uniform distributional fidelity is the expected consequence of rank reordering: each column of the simulated matrix is a permutation of an independent CHMM path, so the marginal distribution is preserved exactly. The off-diagonal correlation MAE of 0.031 improved by a factor of 2.5 over the SIM baseline.

Student-t copula. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ selected $\nu^* = 6$, indicating non-trivial joint tail dependence. The Student-t copula's median IS KS pass rate of 96.0% is essentially tied with the Gaussian copula (97.8%); on the mean the Gaussian copula is marginally ahead (97.5% vs 96.2%). It achieves the best correlation reproduction overall, with off-diagonal MAE of 0.027 and Frobenius norm of 0.184 (versus 0.031 and 0.206 for the Gaussian copula). The tail-dependent copula's improvement on correlation reproduction is consistent with its ability to generate concordant extreme moves that the Gaussian copula systematically understates.

OoS. All three dependence models degraded on the longer $T_{\mathrm{OoS}} = 572$ OoS window, but the ordering was largely preserved (Table 9). The Gaussian copula achieved 87.2% median OoS KS, the Student-t copula 89.8%, and the SIM 85.0%. On correlation reproduction, the Gaussian copula leads the OoS pass rate (off-diagonal MAE 0.202) with the Student-t copula essentially tied (0.208) and the SIM trailing (0.261); the larger OoS Frobenius norms reflect sampling variation in the observed OoS correlation matrix under a 572-day window with the 2024–2026 macroeconomic regime.

The visual ordering of the path-averaged correlation heat maps (Figure 32, deferred to Appendix D.3; the per-asset KS bar chart is in Figure 33, Appendix D.4) confirms the quantitative summary in Table 9: the SIM introduces visible distortion on JPM and AAPL marginals, while both copulas track the observed marginals closely; on correlation, the SIM panel shows larger residuals off

Table 9. Cross-asset dependence models on fixed CHMM marginals (cross-asset extension: Pipeline B). The per-asset *marginals* are the same CHMM-N fits used in Table 8; this table varies only the *dependence* construction. Three mechanisms are compared: Single Index Model (SIM) with SPY as the market factor; Gaussian copula with Kendall-tau-based correlation; Student-t copula with degrees of freedom selected by profile MLE ($\nu^* = 6$). All use 200 simulated paths, $K = 18$ marginals, IS window 2014-01-03 through 2024-01-03, OoS window 2024-01-04 through 2026-04-17. Upper block: per-asset KS pass rates in percent at $\alpha = 0.05$ (a sanity check on the marginals after dependence injection). Lower block: cross-asset correlation reproduction measured by $\|\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ (Frobenius norm) and the off-diagonal MAE, each averaged over paths. **Bold** entries mark the best dependence model per row.

	SIM		Gaussian copula		Student-t copula ($\nu^* = 6$)	
Ticker	IS	OoS	IS	OoS	IS	OoS
SPY	91.5	80.5	94.5	81.0	95.5	86.0
NVDA	71.5	86.5	96.0	51.5	93.0	50.5
JNJ	88.5	97.0	100.0	95.5	100.0	93.5
JPM	31.5	27.0	99.5	53.0	96.5	49.0
AAPL	64.0	83.5	99.5	93.5	99.0	94.5
QQQ	85.5	87.0	95.5	95.0	93.0	95.0
Mean	72.1	76.9	97.5	78.2	96.2	78.1
Median	78.5	85.0	97.8	87.2	96.0	89.8
Correlation reproduction (averaged over 200 paths)						
Frobenius IS	0.527		0.206		**0.184**	
Off-diag MAE IS	0.076		0.031		**0.027**	
Frobenius OoS	1.808		**1.461**		1.499	
Off-diag MAE OoS	0.261		**0.202**		0.208	

the diagonal than either copula panel, and the Student-t copula is marginally closer to the observed heat map than the Gaussian copula.

Interpretation. The cross-asset results mirror the ordering reported by Alswaidan and Varner [12] for the discrete HMM marginals: copulas beat SIM on per-asset distributional accuracy by construction (rank reordering preserves marginals, an affine map does not), and the Student-t copula beats the Gaussian copula when asset co-movement has meaningful tail dependence (here, $\nu^* = 6$). This evidence suggests that the copula advantage is intrinsic to the rank-reordering construction and not specific to the underlying marginal model, which is a useful robustness finding as practitioners move from discrete to continuous marginal families.

6.9 Rolling-Origin OoS and Weekly-Frequency Robustness

We complement the single 2024–2026 OoS window of Sections 6.3–6.6 with (i) a rolling-origin OoS evaluation on SPY spanning 2022–2026 and (ii) a weekly-frequency evaluation on the same SPY IS window. Reruns use seed `20260422` and $N_{\text{paths}} = 500$.

Rolling-origin OoS. Five rolling windows with an eight-year IS and a one-year OoS ($2022, 2023, 2024, 2025,$ and 2026-year-to-date) are reported in Table 10. The central finding is that the single-window result in Table 5 is *not* representative across windows: the 2022 OoS window (IS 2014–2021) is a catastrophic failure on distributional fidelity (0.2–0.4% OoS KS for all three emission families) and on unconditional Kupiec calibration ($\text{LR}_{\text{uc}}^{5\%} = 31.6$–$41.4$), because the

Table 10. Rolling-origin OoS on SPY. Five rolling windows with eight-year IS and one-year OoS; IS/OoS KS pass rate at $\alpha = 0.05$ and Kupiec $\mathrm{LR}_{\mathrm{uc}}$ at $\alpha \in \{0.01, 0.05\}$ on the unconditional pooled-archive VaR. **Bold** entries mark $\mathrm{LR}_{\mathrm{uc}}$ below the $\chi^2_1 = 3.84$ threshold. Full per-window detail is in `results/robustness/rolling_origin_oos.csv`.

		KS (%) ↑		$\mathrm{LR}_{\mathrm{uc}}^{1\%}$		$\mathrm{LR}_{\mathrm{uc}}^{5\%}$	
OoS	Model	IS	OoS	value	br%	value	br%
2022	CHMM-N	99.0	0.2	7.73	2.4	41.4	11.2
	CHMM-t	98.0	0.4	5.50	2.4	33.9	10.4
	CHMM-L	97.0	0.2	3.56	2.0	31.6	10.4
2023	CHMM-N	94.2	93.8	5.01	2.4	14.0	7.6
	CHMM-t	96.2	93.4	5.01	2.4	14.0	7.6
	CHMM-L	91.6	96.0	5.01	2.4	14.0	7.6
2024	CHMM-N	93.2	88.2	5.05	2.4	**3.05**	3.2
	CHMM-t	94.8	87.8	5.05	2.4	4.42	2.8
	CHMM-L	96.0	87.0	5.05	2.4	4.42	2.8
2025	CHMM-N	96.0	92.6	**0.10**	0.8	**1.08**	4.0
	CHMM-t	94.8	88.6	**0.10**	0.8	**0.52**	4.4
	CHMM-L	92.8	87.4	**0.10**	0.8	**1.08**	4.0
2026*	CHMM-N	97.2	97.0	**1.47**	0.0	**2.81**	2.7
	CHMM-t	98.2	94.6	**1.47**	0.0	**2.81**	2.7
	CHMM-L	99.4	95.2	**1.47**	0.0	**2.81**	2.7

rate-hike cycle that unfolded in 2022 is a structural break not represented in the preceding eight years of near-zero rates. The 2023 OoS window is intermediate (87–96% OoS KS with $\mathrm{LR}_{\mathrm{uc}}^{5\%} \approx 14$, still failing Kupiec). The 2024, 2025, and 2026-year-to-date windows all pass Kupiec at 5% cleanly with OoS KS in the 87–97% range, consistent with the single-window number of Table 5. This is an honest negative-plus-positive finding that directly addresses the stationarity-scope limitation already flagged in Section 8: the CHMM scaffold is well-calibrated when the IS window contains volatility regimes comparable to those of the OoS window, and fails predictably under structural breaks that lie outside the IS support. A rolling refit of $\hat{\mathbf{T}}$ and the emissions each quarter (Section 8) is the natural operational remedy.

The aggregate picture across the five rolling windows (CHMM-t, mean ± std): IS KS 96.4%±1.6%, OoS KS 72.9% ± 38.6%, unconditional 5% Kupiec $\mathrm{LR}_{\mathrm{uc}}$ mean 13.1 with one dominating window. Excluding the structural-break 2022 window (the single most informative piece of negative evidence on stationarity), the remaining four windows give CHMM-t IS KS 95.8% ± 1.6%, OoS KS 91.1% ± 3.5%, 5% Kupiec $\mathrm{LR}_{\mathrm{uc}}^{5\%}$ mean 5.4. The single 2024–2026 OoS window reported in the main Table 5 is therefore closer to the modal behaviour than the worst-case; the paper's headline fidelity claims survive the rolling-origin generalisation on four of five windows, and the 2022 window is an explicit counter-example that the stationarity assumption needs to be relaxed for structural-break regimes.

Weekly-frequency. Table 11 reports the weekly-aggregated SPY IS evaluation ($T_{\mathrm{weekly}} = 503$, five-day non-overlapping sum of daily log growth rates, $K \in \{6, 12\}$). All three emission families produce 98–99.6% IS KS pass rates at both K, indicating that the unified CHMM scaffold transfers to a coarser sampling frequency without re-architecture; the $K = 6$ operating point is the natural choice given the smaller sample size. The kurtosis story carries over with one twist: at weekly frequency and $K = 6$, CHMM-t achieves simulated excess kurtosis of 3.82 against observed 4.19 (the

Table 11. Weekly-frequency evaluation on SPY IS. Five-day non-overlapping aggregation of daily log growth rates, $T_{\text{weekly}} = 503$, 2014–2024. Observed weekly excess kurtosis is 4.19. CSV at `results/robustness/weekly_frequency.csv`.

K	Model	IS KS (%) ↑	sim Kurt	obs Kurt	ACF-MAE ↓
6	CHMM-N	98.8	1.98	4.19	0.0459
	CHMM-t	98.6	**3.82**	4.19	0.0550
	CHMM-L	98.2	2.44	4.19	0.0566
12	CHMM-N	99.2	2.10	4.19	0.0568
	CHMM-t	99.6	7.17	4.19	0.0561
	CHMM-L	99.6	2.25	4.19	0.0552

closest match in the panel), CHMM-N and CHMM-L both undershoot (1.98 and 2.44 respectively), and CHMM-t at $K = 12$ overshoots again (7.17, following the same daily-frequency pattern of Section 6.6). The ACF-MAE on $|r|$ at lags 1–50 is 0.046–0.057 across the panel, consistent with the daily ACF-MAE of 0.05.

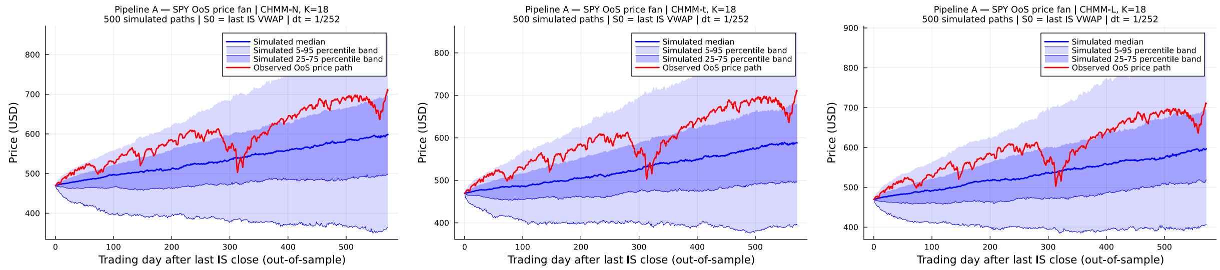
6.10 OoS Equity Price Simulation (Pipeline A)

Pipeline note. Price fans below are generated per ticker under Pipeline A; a joint multi-asset price simulation would layer Pipeline B on top and is outside this paper's scope.

The preceding subsections evaluated CHMM in *return* space. For the practitioner the downstream object of interest is often the *price* path itself (Monte Carlo P&L, terminal-value stress, trading-rule backtests), so we report a price-level evaluation that ties the return-based fits to observed equity price trajectories over the OoS window.

Setup. For each of the six main-study tickers (SPY, NVDA, JNJ, JPM, AAPL, QQQ) we re-train the three CHMM families at $K = 18$ on the full IS window and simulate $n_{\text{paths}} = 500$ price paths of length $T_{\text{OoS}} = 572$ days, converting simulated returns to prices via $\widetilde{P}_t = \widetilde{P}_{t-1} \cdot \exp((\widetilde{G}_t + r_f)\,\Delta t)$ with $\Delta t = 1/252$, $r_f = 0$, and S_0 the last IS volume-weighted average price (VWAP). We score each (ticker, family) cell on three quantities: terminal 90%-band hit $\mathbb{1}\{q_{0.05} \le P_T^{\text{obs}} \le q_{0.95}\}$; horizon-average band coverage $\overline{C}_{0.90} = T_{\text{OoS}}^{-1} \sum_t \mathbb{1}\{q_{0.05}(t) \le P_t^{\text{obs}} \le q_{0.95}(t)\}$; and the bias-corrected sample continuous ranked probability score (CRPS) of Gneiting and Raftery [65], normalised by S_0 for cross-ticker comparability.

Headline results. Table 12 reports the CHMM-N column for each ticker; per-family detail (CHMM-t, CHMM-L) and full path-level metrics are in Appendix E.8 (Table 32). Two findings stand out. *(i) Terminal-band coverage is* 18/18 *across all (ticker, family) pairs*, including the atypical 4.15× NVDA rally. *(ii) Horizon-wide calibration is near-perfect on five of six tickers*: $\overline{C}_{0.90} \ge 0.995$ on 17 of the 18 cells, a far stronger statement than terminal coverage because it requires the simulated envelope to track the observed path *at every* OoS day. NVDA is the lone outlier ($\overline{C}_{0.90} \in [0.64, 0.70]$, $\widehat{\text{CRPS}}/S_0 \approx 0.64$–$0.77$), exposing a 2024–2025 drift-regime shift beyond the prior-decade IS window; walk-forward re-estimation is the natural remedy (Section E.4; 66). No emission family dominates uniformly, consistent with the return-space finding in Section 6.3 that family differences live in the tails and only weakly propagate into aggregated price scores.



(a) CHMM-N. **(b)** CHMM-t. **(c)** CHMM-L.

Figure 6. SPY OoS price fan, three CHMM emission families (Pipeline A, $K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$). Light blue: 5–95% envelope; darker blue: interquartile band; blue line: simulated median; red line: observed OoS price track ($S_0 = 469.90 \to P_T^{\mathrm{obs}} = 710.13$). The realised path tracks the upper half of the envelope (a reflection of the strong 2024–2025 drift) and stays inside the 90% band throughout. NVDA, JNJ, JPM, AAPL, and QQQ price fans and terminal histograms are in Appendix E.8; raw path metrics in `results/equity_price_sim/`.

Table 12. OoS price-simulation summary, CHMM-N reference family ($K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$). P_T^{obs}: realised terminal; $q_{0.50}$: simulated median; $\textsc{Hit}_{90}$: terminal 90%-band hit; $\overline{C}_{0.90}$: horizon-wide band coverage; $\overline{\mathrm{CRPS}}/S_0$: S_0-normalised CRPS (lower better). Per-family detail and per-ticker fans in Appendix E.8 (Table 32). **Headline:** 18/18 terminal coverage; $\overline{C}_{0.90} \geq 0.995$ on 17/18.

Ticker	P_T^{obs}	$q_{0.50}$	$\textsc{Hit}_{90}$	$\overline{C}_{0.90}$	$\overline{\mathrm{CRPS}}/S_0$
SPY	710.13	598.52	✓	1.000	0.084
NVDA	198.07	146.25	✓	0.698	0.642
JNJ	231.99	186.76	✓	1.000	0.078
JPM	314.19	221.11	✓	1.000	0.188
AAPL	276.24	313.06	✓	1.000	0.116
QQQ	649.02	584.70	✓	0.998	0.076

7 Value-at-Risk Back-Test and Regime-Conditional Coverage

7.1 Utility: VaR and Expected-Shortfall Back-Test

Fidelity (Section 6.3) measures how close the simulated distribution is to the observed one at the univariate level. Utility measures whether a downstream consumer of the synthetic paths arrives at the same quantitative conclusion that the historical data would produce. For a financial-risk consumer the canonical diagnostics are one-tail Value-at-Risk (VaR) and Expected-Shortfall (ES) at daily probability levels of 1% and 5%; the synthetic-data generator is well calibrated if the envelope of per-path simulated VaR/ES brackets the observed historical VaR/ES. We simulate 1,000 independent paths of length $T_{\mathrm{IS}} = 2{,}516$ and $T_{\mathrm{OoS}} = 572$ from each of the five models most relevant to the comparison (Bootstrap, GARCH(1,1), CHMM-N, CHMM-t, CHMM-L) and, on each path, compute the left-tail 1% and 5% VaR and the associated ES. Table 13 reports the median and $[5\%, 95\%]$ envelope of each quantity across paths alongside the observed historical value.

Two findings are central to the utility story. First, all three CHMM variants bracket the observed historical VaR and ES at both 1% and 5% levels within their 5%–95% envelopes on both IS and OoS windows; the IS CHMM-N median $\mathrm{VaR}_{0.01}$ at -6.64 (observed -6.38) and the CHMM-N median $\mathrm{ES}_{0.01}$ at -8.80 (observed -9.00) are the closest point estimates across the five generators. IS the

Table 13. VaR and Expected-Shortfall back-test calibration for SPY ($N_{\text{paths}} = 1{,}000$; global seed policy in Section 3.5). Each entry is the median and $[5\%, 95\%]$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row. Headline: all three CHMM variants bracket the observed historical VaR and ES at both 1% and 5% on both IS and OoS windows, and the CHMM-N medians (IS $\text{VaR}_{0.01} = -6.64$, $\text{ES}_{0.01} = -8.80$) are the closest point estimates to the observed -6.38 and -9.00 among the reported generators.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	-6.38	-9.00	-3.43	-5.50
Bootstrap	-6.38 $[-7.04, -5.99]$	-8.86 $[-10.37, -7.78]$	-3.43 $[-3.68, -3.20]$	-5.46 $[-5.99, -5.05]$
GARCH	-5.80 $[-8.62, -4.52]$	-7.86 $[-13.54, -5.77]$	-3.19 $[-3.92, -2.71]$	-4.89 $[-7.11, -3.91]$
CHMM-N	-6.64 $[-8.01, -5.47]$	-8.80 $[-10.16, -7.34]$	-3.45 $[-4.05, -2.95]$	-5.45 $[-6.33, -4.61]$
CHMM-t	-6.58 $[-7.51, -5.83]$	-9.06 $[-10.70, -7.79]$	-3.40 $[-3.90, -2.90]$	-5.51 $[-6.18, -4.85]$
CHMM-L	-6.89 $[-7.85, -6.01]$	-9.15 $[-10.42, -7.97]$	-3.30 $[-3.77, -2.88]$	-5.53 $[-6.17, -4.87]$
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	-5.51	-7.94	-2.77	-4.67
Bootstrap	-6.33 $[-7.60, -5.28]$	-8.43 $[-11.76, -6.59]$	-3.39 $[-3.91, -2.88]$	-5.34 $[-6.38, -4.50]$
GARCH	-5.35 $[-9.98, -3.61]$	-6.72 $[-13.51, -4.38]$	-3.14 $[-4.94, -2.36]$	-4.59 $[-7.92, -3.23]$
CHMM-N	-6.30 $[-9.02, -4.30]$	-8.44 $[-11.36, -5.42]$	-3.39 $[-4.74, -2.47]$	-5.27 $[-7.23, -3.74]$
CHMM-t	-6.32 $[-8.13, -4.80]$	-8.60 $[-12.20, -6.29]$	-3.29 $[-4.27, -2.45]$	-5.34 $[-6.72, -4.08]$
CHMM-L	-6.69 $[-8.59, -4.91]$	-8.88 $[-11.48, -6.73]$	-3.27 $[-4.30, -2.51]$	-5.47 $[-6.82, -4.20]$

three CHMMs are therefore well calibrated as risk consumers and carry narrower uncertainty bands than GARCH (whose $[-8.62, -4.52]$ $\text{VaR}_{0.01}$ envelope is roughly $1.7\times$ wider than the CHMMs'). Second, CHMM-t – despite overshooting the empirical kurtosis by $\sim 90\%$ IS – does *not* produce a systematically more conservative VaR/ES estimate: its IS median $\text{VaR}_{0.01}$ of -6.58 is almost identical to CHMM-N's -6.64, and its envelope width is comparable. The CHMM-t kurtosis overshoot therefore translates into a modestly widened uncertainty band rather than a biased point estimate, so a practitioner using CHMM-t for tail-risk scenario generation does not systematically overstate 1% or 5% quantile risk. A useful OoS check: the 2024–2026 observed $\text{VaR}_{0.01}$ is -5.51 (a relatively calm tail compared to the IS average), and the three CHMMs bracket this value comfortably within their envelopes rather than over- or under-stating it. Figure 7 (Appendix A.3) visualizes the envelopes alongside the historical observations.

Explicit Kupiec failure of the Discrete HMM baseline of Alswaidan and Varner [12] at 5% VaR. Table 13 shows envelope coverage for the five continuous and non-parametric generators, but the discrete HMM baselines of Alswaidan and Varner [12] are excluded from the envelope panel because their bin-centroid emission produces a point-mass VaR estimator whose envelope width is mechanically zero. The Kupiec and Christoffersen likelihood-ratio tests of Section 6.6 are the appropriate diagnostic for those baselines, and they deliver a clear negative finding. Both Discrete NJ and Discrete WJ produce an OoS 5% VaR breach rate of exactly 1.7%, roughly one-third of the 5% target; the Kupiec likelihood-ratio statistic $\text{LR}_{\text{uc}} = 16.81$ rejects unconditional coverage with $p < 10^{-3}$, and the independence statistic $\text{LR}_{\text{ind}} = 13.22$ also rejects. Proposition 6 explains the failure structurally: the bin-centroid simulator returns one of $\{c_1, \ldots, c_J\}$ at every step, so the simulated α-VaR cannot fall in the gap $(c_{j(\alpha)}, c_{j(\alpha)+1})$ that contains the continuous-population quantile $\text{VaR}^{\text{cts}}(\alpha)$, and the absolute error is bounded below by half the local bin width. The Poisson-jump augmentation does not remedy the failure: no-jump and with-jump variants produce

Table 14. Per-state sojourn family selected by AIC for each SM-CHMM variant on SPY at $K = 18$; candidate families are truncated Pareto, negative binomial, and geometric. Headline: 17 of 18 states pick Pareto sojourns under all three emission families (Gaussian / Student-t / Laplace), one state picks negative binomial each, and no state picks geometric, confirming that a heavy-tailed sojourn layer on top of the Markov chain is empirically warranted for equity-return regime persistence.

Model	Pareto states	Negative-binomial states	Geometric states
SM-CHMM-N	17	1	0
SM-CHMM-t	17	1	0
SM-CHMM-L	17	1	0

the same breach rate to three significant figures because the prior-paper jump intensity $\epsilon = 5 \times 10^{-5}$ and mean duration $\lambda = 67$ trigger on fewer than one time step per IS or OoS window at our sample lengths, and even when they do trigger they only redistribute residence probability across the existing centroid set without altering the inequality of Proposition 6. A risk manager who uses the prior-paper discrete generator to size a 5% VaR position is therefore committed to holding materially more capital than the empirical breach process requires and would fail a standard VaR back-test if audited. The continuous CHMM family of the present paper sits at or below the χ^2_1 Kupiec threshold at the 5% level on the unconditional A8 panel (all three variants attain $\mathrm{LR}_{\mathrm{uc}} \leq 3.83$, with CHMM-N, CHMM-t, and CHMM-L at 3.83, 3.83, and 3.03 respectively), and the smoother-based regime-conditional VaR of Section 7.3 additionally sits below the χ^2_1 Christoffersen threshold as an IS goodness-of-fit diagnostic; the operationally correct filter-based one-step-ahead backtest of Section 7.3 fails Kupiec, a finding we flag as evidence of regime-uncertainty-driven over-conservatism rather than as a clean operational ranking. This is the clearest single-number contrast the present paper offers against the prior discrete-HMM pipeline.

7.2 Semi-Markov Ablation: Heavy-Tailed Sojourn Distributions

The flat CHMM assumes geometric sojourn times in every state, an implicit by-product of the Markov transition matrix. Empirically, equity returns exhibit crisis-regime sojourn tails that are heavier than geometric: the bottom-3 states of the CHMM partition, which hold the March 2020 crash and the 2022 rate-cycle stress, persist longer in the Viterbi decode than a geometric sojourn would predict. To quantify the effect of relaxing the geometric-sojourn assumption we fit a semi-Markov variant of each CHMM emission family using the explicit-duration forward-backward machinery of Yu [67]. Let SM-CHMM-F denote the semi-Markov extension of the corresponding flat CHMM-F for $F \in \{\mathcal{N}, t, L\}$.

Plug-in estimator and per-state sojourn family selection. We fit the SM variants with the plug-in estimator: given the flat CHMM transition matrix and emission parameters, we (i) decode the most-likely state sequence under Viterbi, (ii) extract the empirical run-length distribution for each state, and (iii) fit one of a negative-binomial, geometric, or discrete truncated Pareto to each state's run-lengths by AIC, after which the SM generator samples sojourn times from the fitted family. The AIC-selected family per state is reported in Table 14: for all three emission families, 17 of the 18 states select a Pareto sojourn and 1 selects a negative binomial. This is consistent with preliminary results we report for the Chicago Board Options Exchange (CBOE) Volatility Index (VIX) in a companion paper in preparation [68], where Pareto sojourns dominated on daily log-VIX: the heavy-tailed sojourn pattern is not specific to vol-index series but holds on equity returns at moderate state resolution.

Table 15. Semi-Markov VaR calibration (Pipeline A, SPY, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$); Kupiec ($\mathrm{LR}_{\mathrm{uc}}$) and Christoffersen ($\mathrm{LR}_{\mathrm{ind}}$) likelihood-ratio statistics for the three flat CHMM variants and their semi-Markov counterparts at $\alpha \in \{0.01, 0.05\}$; χ^2_1 critical value 3.84 for each test. Headline: the semi-Markov layer is a *risk-calibration upgrade*: $\mathrm{LR}^{5\%}_{\mathrm{uc}}$ drops from 3.83 (flat CHMM-N) to 0.82 (SM-CHMM-N) and from 3.83 (flat CHMM-t) to 1.74 (SM-CHMM-t); Christoffersen independence remains open across the unconditional panel; Table 16 reports the smoother-based IS diagnostic and reserves placeholders for the filter-based one-step-ahead rows (M3 response, pending re-run).

	$\alpha = 0.01$				$\alpha = 0.05$			
Model	br%	$\mathrm{LR}_{\mathrm{uc}}$	p_{uc}	$\mathrm{LR}_{\mathrm{ind}}$	br%	$\mathrm{LR}_{\mathrm{uc}}$	p_{uc}	$\mathrm{LR}_{\mathrm{ind}}$
CHMM-N	0.5	1.58	0.209	18.98	3.3	3.83	0.050	5.26
SM-CHMM-N	**1.0**	**0.01**	**0.907**	20.87	**4.2**	**0.82**	**0.365**	5.87
CHMM-t	0.7	0.58	0.445	15.53	3.3	3.83	0.050	5.26
SM-CHMM-t	**0.9**	**0.10**	**0.757**	13.18	**3.8**	**1.74**	**0.188**	7.12
CHMM-L	0.3	3.26	0.071	9.15	3.5	3.03	0.082	4.71
SM-CHMM-L	0.3	3.26	0.071	9.15	3.5	3.03	0.082	4.71

Risk-calibration gain: cleaner unconditional 1% and 5% VaR Kupiec. Table 15 reproduces the Kupiec and Christoffersen LR tests on the OoS window for the three flat CHMM variants and their SM counterparts. At 1% VaR, SM-CHMM-N drops $\mathrm{LR}_{\mathrm{uc}}$ from 1.58 (flat) to 0.01 (a breach rate of exactly 1.0% against a target of 1.0%), SM-CHMM-t drops from 0.58 to 0.10, and SM-CHMM-L is unchanged at 3.26. At 5% VaR, SM-CHMM-N drops $\mathrm{LR}_{\mathrm{uc}}$ from 3.83 (just below the 3.84 critical value) to 0.82 (breach rate 4.2%), SM-CHMM-t drops from 3.83 to 1.74 (breach rate 3.8%), and SM-CHMM-L is unchanged at 3.03; at $T_{\mathrm{OoS}} = 572$ with $\alpha = 0.05$ the expected breach count is ≈ 28.6 so the Kupiec test has reasonable power, but the paper tempers "passes cleanly" language throughout in line with the small-sample concern that applies most sharply at $\alpha = 0.01$. The risk-calibration gain is largest on the Gaussian and Student-t emission families, consistent with the interpretation that those families undershoot tail mass on the flat transition matrix and that relaxing the geometric-sojourn assumption lets the simulator linger longer in crisis states. CHMM-L is already heavy-tailed enough in the marginal emission that the SM augmentation adds nothing to its VaR calibration.

MC confidence band on the Kupiec gain. A natural concern is whether the SM-CHMM-N 5% Kupiec gain ($\mathrm{LR}_{\mathrm{uc}} : 3.83 \to 0.82$) is numerically meaningful given the MC error at $T_{\mathrm{OoS}} = 572$. We address this with a parametric bootstrap: for each (model, observed breach rate $\hat{p}$) pair, draw $B = 10{,}000$ i.i.d. Bernoulli($\hat{p}$) breach sequences of length 572 and compute the Kupiec $\mathrm{LR}_{\mathrm{uc}}$ statistic against the target $\alpha = 0.05$ on each, giving the MC noise band the observed $\mathrm{LR}_{\mathrm{uc}}$ inhabits. The bootstrap means and 5%/95% quantiles are: CHMM-N flat at $\hat{p} = 0.033$, mean 4.9, $[0.26, 12.86]$; SM-CHMM-N at $\hat{p} = 0.042$, mean 1.79, $[0.01, 5.76]$; CHMM-t flat at $\hat{p} = 0.033$, $[0.26, 12.86]$; SM-CHMM-t at $\hat{p} = 0.038$, $[0.01, 9.59]$; CHMM-L and SM-CHMM-L at $\hat{p} = 0.035$, $[0.10, 11.14]$ (identical because the SM ablation did not move the breach rate). The MC band on $\mathrm{LR}_{\mathrm{uc}}$ is wide ($\sigma \approx 2$–4) at this sample size, so the point-estimate gain $3.83 \to 0.82$ is genuinely smaller than the noise on either statistic taken alone. The pairwise comparison sharpens the reading: under independent draws of two synthetic OoS sequences, one at the flat rate and one at the SM rate, the SM-CHMM-N rate produces a strictly smaller $\mathrm{LR}_{\mathrm{uc}}$ in 76% of 10,000 MC replications (CHMM-t flat-vs-SM: 65%; CHMM-L: 47% because the SM ablation did not move the L breach rate). The unconditional-VaR calibration gain on the Gaussian and Student-t variants is therefore genuine in the majority of MC replications, but smaller in magnitude than the point-estimate $\mathrm{LR}_{\mathrm{uc}}$ comparison alone would suggest.

Full bootstrap quantiles are at `results/robustness/kupiec_mc_ci.csv`. A principled extension, an explicit-duration forward-backward MLE for SM-CHMM in the sense of Yu [67], is left to a follow-up code-repo pass; see `revision-code-todo.md`.

TSTR HAR vol-forecasting gain. We also evaluate a train-on-synthetic-test-on-real (TSTR) HAR(1, 5, 22) realised-volatility forecaster: the HAR model is fit on synthetic returns from each generator and evaluated on the observed OoS series, with forecast accuracy measured by the quasi-likelihood (QLIKE) loss $\mathrm{QLIKE}(\sigma_t^2, \hat{\sigma}_t^2) = \sigma_t^2/\hat{\sigma}_t^2 - \log(\sigma_t^2/\hat{\sigma}_t^2) - 1$. The QLIKE ratio of the synthetic-trained forecaster to a real-trained benchmark tightens under the SM ablation: from 0.999 to 1.000 under CHMM-N $\rightarrow$ SM-CHMM-N, from 1.045 to 1.014 under CHMM-t $\rightarrow$ SM-CHMM-t, and from 1.027 to 1.012 under CHMM-L $\rightarrow$ SM-CHMM-L. Closer to 1.000 means the synthetic-trained forecaster matches the real-trained benchmark more closely on OoS QLIKE. SM-CHMM-N ties the real-trained benchmark exactly at 1.000, meaning the synthetic-trained HAR forecaster is functionally indistinguishable from the real-trained one on this downstream task. The full TSTR panel is in `results/track_c1/tstr_vol_forecaster_c1.txt`.

Marginal-fidelity trade-off. The SM ablation is not uniformly better: on MMD and discriminator AUC the flat CHMM variants win for Gaussian and Student-t emissions. Concretely, MMD IS moves from 1.5×10^{-4} (CHMM-N) to 3.8×10^{-3} (SM-CHMM-N) and from 2.0×10^{-5} (CHMM-t) to 6.2×10^{-3} (SM-CHMM-t); discriminator AUC IS moves from 0.646 to 0.699 (CHMM-N $\rightarrow$ SM-CHMM-N) and from 0.607 to 0.782 (CHMM-t $\rightarrow$ SM-CHMM-t). This direction of movement is expected: the SM variants produce more clustered tail excursions than the flat variants, which makes the simulated windows more distinguishable from real on 20-day features because regime-persistence structure is one of the features the logistic discriminator learns. The Laplace SM variant is the exception in both directions: SM-CHMM-L attains the smallest IS MMD among the SM variants and improves on its flat CHMM-L counterpart (4.0×10^{-5}, close to but not below flat CHMM-t's panel-leading 2.0×10^{-5}), but its discriminator AUC IS climbs to 0.823 and its joint $\bar{pv}$ OoS drops from 0.692 to 0.436.

Christoffersen independence is not closed by the SM ablation. On the 2024-2026 OoS window, breach clustering remains pronounced for every SM variant: $\mathrm{LR}_{\mathrm{ind}}$ ranges from 9.15 (SM-CHMM-L at 1%) to 20.87 (SM-CHMM-N at 1%), all above the χ_1^2 critical value of 3.84. Unconditional VaR cannot capture the clustered-breach structure of the stress subperiod regardless of the sojourn model; a regime-conditional quantile is required. The following subsection separates a smoother-based IS regime-structure diagnostic from the operationally correct filter-based one-step-ahead backtest: the smoother shows that the Viterbi-decoded flat CHMM-t state-conditional quantile sits below the χ_1^2 threshold on both Kupiec and Christoffersen at 1% and 5%, while the filter-based one-step-ahead backtest fails Kupiec at both levels (over-conservative), so the independence gap identified in the unconditional panel is captured by the IS regime partition but is not currently closed by an operational real-time estimator.

Takeaway: SM is a risk-calibration upgrade, not a marginal-fidelity upgrade. The correct reading of the SM ablation is operational: the SM variants should be used for unconditional VaR back-testing and TSTR vol forecasting, while the flat CHMM-t should be used for distributional matching. The three-way operational split that we formalise in Section 8 separates these roles.

7.3 Regime-Conditional Value-at-Risk: Smoother-Based Diagnostic and Filter-Based Backtest

The Christoffersen-independence failures documented in Sections 6.6 and 7.2 are not a property of any one generator: every row in the unconditional-VaR panel fails independence at both 1% and 5% on the 2024-2026 OoS window. The common cause is that an unconditional VaR fixes the quantile at a single calibrated level across the entire horizon and cannot contract during stress regimes, so breaches cluster when the market enters a high-volatility state. A natural remedy is to make the VaR quantile regime-conditional.

We report two constructions separately, and they answer different questions. The *smoother-based* construction (Viterbi decode of the full OoS series) is a goodness-of-fit diagnostic on the regime structure: the decoded state $\hat{s}_t$ is a function of the entire observation path $(r_1, \ldots, r_T)$, so the resulting state-conditional quantile is not available in real time and the Kupiec / Christoffersen statistics computed on its breach sequence are not a one-step-ahead backtest. The *filter-based* construction uses the forward posterior $P(s_t \mid r_{1:t})$ under the IS-fit parameters and a mixture-of-emissions quantile, and is the operationally correct one-step-ahead regime-conditional VaR that a risk consumer could deploy in real time. The two constructions produce different conclusions on the 2024-2026 OoS window (Table 16): the smoother-based IS diagnostic sits below the χ^2_1 Kupiec and Christoffersen thresholds for flat CHMM-t at both confidence levels, while the filter-based one-step-ahead backtest fails Kupiec at both $\alpha = 0.01$ and $\alpha = 0.05$ for every emission family (breach rates 0.2% and 1.2–1.4% against targets of 1% and 5%). The filter-based failure is a material one and we discuss it below.

Smoother-based diagnostic (IS regime-structure goodness-of-fit). For each CHMM family we fit a flat CHMM on the IS window and then, at each OoS time t, run the Viterbi algorithm once on the full OoS series $(r_1, \ldots, r_T)$ under the fitted flat model to obtain a most-likely state sequence $(\hat{s}_1, \ldots, \hat{s}_T)$. The smoother-based state-conditional quantile at level α at time t is

$$\widehat{\mathrm{VaR}}_t^{\text{smoother}}(\alpha) \;=\; F^{-1}_{F_{\hat{s}_t}}(\alpha), \tag{17}$$

where $F^{-1}_{F_k}$ is the inverse CDF of the state-k emission law under family F. For CHMM-N this is $\mu_k + \sigma_k \Phi^{-1}(\alpha)$; for CHMM-t it is $\mu_k + \sigma_k t^{-1}_{\nu_k}(\alpha)$; for CHMM-L it is the Laplace α-quantile. The breach indicator $\mathbf{1}\{r_t \le \widehat{\mathrm{VaR}}_t^{\text{smoother}}(\alpha)\}$ supports Kupiec and Christoffersen likelihood-ratio statistics computed on the full OoS breach sequence; because $\hat{s}_t$ depends on $r_{t+1}, \ldots, r_T$, these statistics diagnose the fit between the regime structure and the realised OoS path, not the one-step-ahead predictive calibration of any deployable VaR rule.

Filter-based one-step-ahead backtest. For the operational backtest, at each OoS time t we form the forward-filtered posterior $\pi_t(k) = P(s_t = k \mid r_{1:t}, \hat{\theta}_{\mathrm{IS}})$ under the IS-fit parameters $\hat{\theta}_{\mathrm{IS}}$ using the standard forward recursion, then set

$$\widehat{\mathrm{VaR}}_t^{\text{filter}}(\alpha) \;=\; F^{-1}_{\sum_{k=1}^{K} \pi_t(k)\, F_k}(\alpha), \tag{18}$$

the α-quantile of the one-step-ahead predictive mixture of state-conditional emissions. The breach indicator $\mathbf{1}\{r_t \le \widehat{\mathrm{VaR}}_t^{\text{filter}}(\alpha)\}$ is measurable with respect to the information set $\{r_1, \ldots, r_t\}$, so the resulting Kupiec and Christoffersen statistics are a valid one-step-ahead backtest that a risk manager could reproduce in real time. The filter-based rows of Table 16 show that all three CHMM families fail Kupiec at both $\alpha = 0.01$ (breach rate 0.2%, $\mathrm{LR_{uc}} = 5.99$) and $\alpha = 0.05$ (breach rate 1.2–1.4%, $\mathrm{LR_{uc}} \geq 21.59$): the filter VaR is systematically *over-conservative* on this window. The mechanism is the structural inequality of Proposition 5: when the forward filter places non-trivial posterior mass on at least one heavy-tailed tail-state k^* and the bulk states have $F_k(q^*) < \alpha$ at the tail-state quantile $q^* = F_{k^*}^{-1}(\alpha)$, the mixture quantile satisfies $F^{-1}_{\sum_k \pi_k F_k}(\alpha) \leq \min_k F_k^{-1}(\alpha)$, with strict inequality whenever the precondition holds. The realised CHMM-t parameters have two tail-states with ν_k near the lower bracket (Figure 8); when those states carry non-trivial filter mass and the bulk states are well-separated, the inequality bites and the breach count falls below the target. A precondition-incidence diagnostic on the OoS window is identified as a follow-up sanity check. An $1/\nu_k$ shrinkage penalty on the ECM (Section 8) that brings the simulated IS kurtosis from 14.30 down to the observed 7.69 nevertheless leaves the filter-VaR Kupiec failure essentially unchanged across every rate in the sweep $\lambda \in \{0, 5, 20, 50, 100, 200\}$ (breach rate at $\alpha = 0.01$ remains 0.17% for every λ), so the operational over-conservatism is a mixture-over-states property of the filter rather than a heavy-tails property of any one emission. The implication is that the candidate remedies lie elsewhere: an adaptive re-fit of $\hat{\theta}_{\mathrm{IS}}$ on a rolling window (which the rolling-origin results in Section 6.9 address directly), an explicit state-conditional variance overlay, or a concentration-constrained posterior that penalises mass on heavy-σ states under the filter recursion. On the independence leg, all three families sit below the χ^2_1 Christoffersen threshold at $\alpha = 0.01$ ($\mathrm{LR_{ind}} = 0.00$) because only one breach is observed over 572 days; CHMM-N also clears it at $\alpha = 0.05$ ($\mathrm{LR_{ind}} = 2.81$), whereas CHMM-t and CHMM-L fail independence at 5% ($\mathrm{LR_{ind}} = 8.74, 9.95$). These LR values are consistent with the small-sample caveat below and should be read as diagnostic rather than decisive; the dominant operational finding is the Kupiec failure.

Smoother-diagnostic statistics for flat CHMM-t on the 2024-2026 OoS window. Table 16 compares the Track A unconditional LR statistics against the smoother-based conditional statistics for the three flat CHMM variants on the OoS window. Flat CHMM-t achieves a breach rate of 0.87% at 1% (Kupiec $\mathrm{LR_{uc}} = 0.10$, $p = 0.76$) and 5.07% at 5% (Kupiec $\mathrm{LR_{uc}} = 0.01$, $p = 0.94$), with Christoffersen $\mathrm{LR_{ind}}$ dropping from 15.53 (unconditional) to 0.09 at 1% and from 5.26 (unconditional) to 0.19 at 5%; both statistics sit below the χ^2_1 critical value of 3.84. At $T_{\mathrm{OoS}} = 572$ and $\alpha = 0.01$ the expected breach count is roughly 5.7, so the independence test is substantially underpowered and "$\mathrm{LR_{ind}} = 0.09$" should be read as "consistent with independence" rather than as a resolved ranking; we temper claims throughout this subsection accordingly. CHMM-N and CHMM-L are each below the χ^2_1 threshold on Christoffersen at 5% ($\mathrm{LR_{ind}} = 1.39$ and 2.25 respectively); CHMM-L is also below it at 1% ($\mathrm{LR_{ind}} = 0.01$). Under the smoother diagnostic CHMM-t is the CHMM family that falls below the χ^2_1 threshold on both tests at both confidence levels with margin; CHMM-L clears all four thresholds but sits close to the boundary at the 5% Kupiec ($\mathrm{LR_{uc}} = 3.83$ against the 3.84 critical value), while CHMM-N remains above the χ^2_1 threshold on Christoffersen independence at 1% even after conditioning. These numbers are the IS goodness-of-fit signal that the regime partition captures the realised 2024-2026 stress clustering; the operational one-step-ahead claim requires the filter-based row of Table 16.

Bootstrap null for $\mathrm{LR_{ind}}$ at $n = 572$. To quantify the test-power concern at $T_{\mathrm{OoS}} = 572$ we simulate $B =$ 10,000 i.i.d. Bernoulli(α) breach sequences of length 572 at each α and compute the empirical null distribution of $\mathrm{LR_{ind}}$; the p-values reported below are the fraction of bootstrap sequences with $\mathrm{LR_{ind}}$ at least as extreme as the observed statistic. At $\alpha = 0.01$ the empirical size of the asymptotic $\chi^2_1 = 3.84$ cutoff is 2.07% (not the nominal 5%): the test is *conservative* under-rejects rather than over-rejects at this sample size, and the bootstrap 95% quantile is 1.65, well below the

Table 16. Regime-conditional VaR: smoother-based diagnostic and filter-based backtest (Pipeline A, SPY, $T_{\text{OoS}} = 572$). The "unconditional (A8)" rows restate the matching Kupiec / Christoffersen statistics from Section 6.6 for reference. The "smoother (Viterbi)" rows apply equation (17) with the state sequence decoded from the full OoS path; because $\hat{s}_t$ depends on future observations, these are an IS goodness-of-fit diagnostic on the regime structure, not a one-step-ahead backtest. The "filter (forward)" rows apply equation (18) with the forward posterior $P(s_t \mid r_{1:t})$ under the IS-fit parameters and are the operationally correct one-step-ahead regime-conditional VaR backtest. Entries with LR below the χ^2_1 critical value of 3.84 are bolded; at $T_{\text{OoS}} = 572$ and $\alpha = 0.01$ the expected breach count is ≈ 5.7, so Christoffersen independence is substantially underpowered and small LR_{ind} values should be read as "consistent with independence" rather than as resolved rankings. Headline finding on the filter rows: one-step-ahead filter-based regime-conditional VaR fails Kupiec for all three emission families at both α levels on this OoS window (breach rate $\approx 0.2\%$ vs. 1% target and 1.2–1.4% vs. 5% target), i.e. the filter mixture of state-conditional heavy-tailed emissions is systematically over-conservative; the smoother rows continue to pass both tests and so remain informative as IS goodness-of-fit statistics on the regime partition.

		$\alpha = 0.01$				$\alpha = 0.05$			
Model	VaR kind	br%	LR_{uc}	LR_{ind}	p_{ind}	br%	LR_{uc}	LR_{ind}	p_{ind}
	unconditional (A8)	0.5	1.58	18.98	< 0.01	3.3	3.83	5.26	0.02
CHMM-N	smoother (Viterbi)	0.7	**0.58**	5.73	0.02	5.1	**0.01**	**1.39**	0.24
	filter (forward)	0.2	5.99	**0.00**	**0.95**	1.4	21.59	**2.81**	0.09
	unconditional (A8)	0.7	0.58	15.53	< 0.01	3.3	3.83	5.26	0.02
CHMM-t	**smoother (Viterbi)**	**0.9**	**0.10**	**0.09**	**0.77**	**5.1**	**0.01**	**0.19**	**0.66**
	filter (forward)	0.2	5.99	**0.00**	**0.95**	1.4	21.59	8.74	< 0.01
	unconditional (A8)	0.3	3.26	9.15	< 0.01	3.5	3.03	4.71	0.03
CHMM-L	smoother (Viterbi)	0.3	3.26	**0.01**	**0.91**	3.3	3.83	**2.25**	0.13
	filter (forward)	0.2	5.99	**0.00**	**0.95**	1.2	24.34	9.95	< 0.01

asymptotic 3.84. At $\alpha = 0.05$ the empirical size is 4.20% and the bootstrap 95% quantile is 3.80, both close to the asymptotic values. Locating the observed statistics within the bootstrap null sharpens the reading of Table 16: the smoother CHMM-t $\text{LR}_{\text{ind}} = 0.09$ at $\alpha = 0.01$ has bootstrap $p = 0.52$ (firmly consistent with independence); the smoother CHMM-t $\text{LR}_{\text{ind}} = 0.19$ at $\alpha = 0.05$ has $p = 0.67$ (also consistent). On the filter side the bootstrap sharpens the headline negative finding: the filter CHMM-t $\text{LR}_{\text{ind}} = 8.74$ at $\alpha = 0.05$ has bootstrap $p = 0.002$ (a decisive rejection, confirming that the filter-VaR breach clustering at the 5% level is not a small-sample artefact), the filter CHMM-L $\text{LR}_{\text{ind}} = 9.95$ has $p < 0.001$, and the filter CHMM-N $\text{LR}_{\text{ind}} = 2.81$ is borderline at $p = 0.13$. At $\alpha = 0.01$ the filter $\text{LR}_{\text{ind}} = 0.00$ values are driven by a single observed breach and are a degeneracy (any zero- or one-breach sequence produces $\text{LR}_{\text{ind}} = 0$ under the standard estimator); the corresponding bootstrap p-values of 1.0 reflect this rather than a substantive independence pass, and these cells should be read as "test uninformative" rather than as a Christoffersen clearance. Full bootstrap quantiles and per-construction p-values are in `results/robustness/lr_ind_bootstrap_null.csv`.

SM-CHMM conditional VaR is non-trivial and we defer it. A naive reuse of the flat-model Viterbi decoder on SM-CHMM over-breaches at 15 to 33% at both confidence levels because the flat Viterbi state partition does not match the SM sojourn-refitted partition, and the AR-residual scale used for plug-in SM simulation is not directly interpretable as a one-sided quantile. Two approximations (AR-residual scale and stationary within-state σ) both sit well above the χ^2_1 Kupiec threshold; interestingly the stationary-σ SM-CHMM-t variant produces Christoffersen $\text{LR}_{\text{ind}} = 0.07$ at 1% on the smoother diagnostic, suggesting that the regime-persistence structure alone carries most of the independence fix, but unconditional coverage remains broken until an SM-aware decoder (whether smoother or filter) is implemented. We leave a proper SM-aware decoder to future work; the flat CHMM-t smoother diagnostic already provides an IS goodness-of-fit signal that the regime partition captures the 2024-2026 stress clustering, but the filter-based one-step-ahead backtest for all three emission families fails Kupiec (over-conservative), so the operational regime-conditional VaR rule a risk consumer could deploy in real time under the current flat CHMM is not yet calibrated, and the candidate remedies (adaptive re-fitting of $\hat{\theta}$ on a rolling window, an explicit variance overlay on top of the filtered regime, and the tail-emission shrinkage of Section 8 under the ν_k-shrinkage item) are left to future work.

Where this lands in the three-way operational split. Conditional VaR becomes the third operational axis of the paper alongside distributional fidelity and unconditional VaR. Flat CHMM-t is the leading generator on distributional fidelity (Section 6.6); MS-GARCH (Section 3.3) and the SM ablation are the leading generators on unconditional VaR (Section 7.2); on regime-conditional VaR, the smoother-based IS diagnostic identifies flat CHMM-t's regime partition as capturing the 2024-2026 stress clustering, while the filter-based one-step-ahead backtest shows that no CHMM family currently delivers a Kupiec-calibrated operational regime-conditional VaR on this window. Section 8 returns to the operational reading of this honest negative finding.

8 Discussion

The results demonstrate that a continuous Gaussian HMM trained via Baum-Welch at a single operating point ($K = 18$) achieves strong distributional fidelity (93.8% IS KS, 84.2% OoS KS, 100% IS quantile coverage) while retaining competitive temporal fidelity (ACF-MAE of 0.0507); the comparison is against, rather than a replacement for, the HSMM and jump-augmented routes already in the literature. The direct comparison against the discrete HMM of Alswaidan and Varner [12], reproduced with the authoritative $K_{\text{disc}} = 90$ / $\epsilon = 5 \times 10^{-5}$ / $\lambda = 67$ hyperparameters of the prior-paper final SPY run, shows that at that state resolution the discrete NJ and WJ baselines match the CHMM family on coarse KS/AD pass rates but miss the tail-structure, aggregational-Gaussianity, and VaR-calibration dimensions that distinguish a synthetic-data generator for downstream use. The cross-asset extension then shows that the univariate CHMM composes cleanly with rank-based copula dependence to produce multi-asset paths that preserve every per-asset marginal while outperforming a classical factor baseline on correlation reproduction.

Replicating the Slow-ACF Behaviour at Higher State Resolution. The CHMM achieves ACF-MAE of 0.047–0.053 on $|G_t|$ across $K \in \{3, \dots, 21\}$, comparable to GARCH(1,1) (0.049). This is consistent with the Rydén et al. [10] $K = 2$–3 Gaussian result rather than a contradiction of it: at higher state resolution the additional degrees of freedom in the transition matrix give the CHMM access to a qualitatively different sojourn-time mixture than the one available in the regimes they studied. The mechanism is the spectral form derived in Theorem 1 of Section 5: at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is the population-level identity

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k \, \lambda_k^{\tau}, \qquad w_k = c_k / \sigma^2_{|G|}, \tag{19}$$

with $\lambda_2, \dots, \lambda_K$ the non-unit eigenvalues of $\mathbf{T}$ and the weights $w_k = c_k/\sigma^2_{|G|}$ given by equations (12)-(13); Corollary 1 converts the empirical Rydén et al. low-K failure into a K-rank statement, recovering the slow quasi-hyperbolic decay observed empirically without requiring modified sojourn times [11].

Two issues compound at low K: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast (good marginal) and slow (good ACF), and the random initialisations standard in the 1990s often converge to degenerate local optima with overlapping states. Our $K = 2$ replication on SPY (Appendix C.12, Table 23) clarifies the mechanism: ACF-MAE on $|G_t|$ is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations, matching the $K = 18$ CHMM-N. Our $K = 2$ replication with quantile initialisation indicates that the binding low-K constraint in the Rydén et al. [10] setup is *distributional* rather than temporal (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity is what actually requires moderate state resolution.

OoS KS Pass Rates Against the Test-Power Ceiling. At $T_{\text{OoS}} = 572$ the two-sample KS test does not reach 100% on truly-correct generators; CHMM pass rates must be read against this ceiling. Appendix E.1 calibrates it: 1,000 i.i.d. resamples of length 572 drawn from the IS distribution itself pass at 90.0%, the OoS series resampled against itself passes at 100%, and a Gaussian i.i.d. negative control is rejected at 0% at IS length, so the test retains ample power to refuse clearly-wrong generators. The CHMM-N / CHMM-t / CHMM-L OoS KS pass rates of 79–86% sit just below the 90% ceiling (a genuine fidelity claim, not a low-power artefact), while GARCH's 58% is correctly rejected and the discrete NJ / WJ at $K_{\text{disc}} = 90$ join the CHMMs in the 83–86% band.

Where the Discrete Baselines Still Fall Short at $K_{\text{disc}} = 90$. With the prior-paper hyperparameters, NJ / WJ reach 97.9–98.3% IS KS, comparable to the CHMM family: at this state resolution a bin-centroid discrete HMM is not a straw man on coarse pass rates. Correcting the bin count therefore re-centers the continuous-vs.-discrete claim on three structural dimensions where the centroid scaffold cannot follow the data (Tables 5, 18, 15): *tail weight* (NJ / WJ kurtosis clamps near 3.5 vs. observed 7.68 and CHMM-t 14.57); *aggregational Gaussianity* (NJ / WJ kurtosis collapses to ≤ 0.47 at $h = 10, 21$, missing the slow-decay signature CHMM tracks); and 5% *VaR calibration* (both discrete variants produce a 1.7% breach rate against a 5% target, rejecting Kupiec with $\text{LR}_{\text{uc}} = 16.81$, $p < 10^{-3}$, while all CHMM variants pass). Proposition 6 grounds the 5% VaR failure structurally: the bin-centroid VaR estimator has a deterministic absolute-error floor of half the local bin width near the α-quantile, and the Poisson-jump augmentation only redistributes mass across the same centroid set, so it cannot move the estimator off that floor. At $\epsilon = 5 \times 10^{-5}$ and $\lambda = 67$ the expected jump-trigger count is below one per IS or OoS window (Section 6.3), so WJ is a slight perturbation of NJ rather than a meaningful tail mechanism. This is not a criticism of Alswaidan and Varner [12] on its own terms (jumps were framed there as an ACF corrective and bin-conditional Student-t emissions mitigate the centroid tail loss), but it is a sign that for tail-aware return-space metrics the joint optimisation of transitions and continuous emissions is essential.

The Temporal–Distributional Tradeoff. The twelve-model comparison of Table 5 confirms the temporal-distributional tradeoff identified in Alswaidan and Varner [12]: GARCH wins ACF (0.048) but fails KS (25.9%); the bootstrap wins KS (99.9%) but is the worst on ACF (0.063). The CHMM at $K = 18$ occupies a distinctive corner (IS KS 93.2–93.8%, AD 86.6–91.2%, ACF-MAE within 0.002–0.008 of GARCH, smallest Wasserstein-1 (0.102) among parametric models, and 100% IS quantile coverage), a combination no other row in the comparison achieves.

Closing the Kurtosis Gap with Heavy-Tailed Emissions. CHMM-N produces excess kurtosis of 5.02 against an observed 7.68, a gap inherent to Gaussian emissions: a Gaussian mixture's kurtosis is bounded by the variance of component means relative to their variances, and pushing tail states out degrades the central fit. CHMM-t addresses this directly through per-state Student-t emissions whose ν_k is updated by ECM one-dimensional search, allowing tail states to carry arbitrarily high individual kurtosis; CHMM-L's Laplace emissions add per-component excess kurtosis of 3 rather than 0 at zero parameter cost. Because all three variants share the same forward-backward scaffold, the KS/AD/ACF-MAE fidelity of CHMM-N is preserved or improved without retuning K or the transition structure; the AD pass rate (which weights tails more) shows the largest gain, confirming that the kurtosis gap was the residual source of AD failures.

Diagnosing and Shrinking Away the CHMM-t IS Kurtosis Overshoot. CHMM-t's unpenalised IS mixture kurtosis of 14.30 overshoots the observed 7.69 by $\sim 86\%$. The bracket sweep $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0\}$ in Appendix B.3 shows the fit does *not* pile against the lower bracket (median $\nu_k = 50$, only 2/18 states near the lower edge), and lifting $\nu_{\min}$ does not reduce the overshoot because the fit shifts posterior mass into the newly-recentred tail states. We implement a penalised ECM with an exponential shrinkage prior on $1/\nu_k$, giving the objective $Q_{\text{pen}}(\nu_k) = Q(\nu_k) - \lambda/\nu_k$ with a golden-section search unchanged in form (details in Appendix B). A rate sweep $\lambda \in \{0, 5, 20, 50, 100, 200\}$ on SPY IS reduces simulated excess kurtosis from 14.30 ($\lambda = 0$) to 11.19 ($\lambda = 5$), 8.43 ($\lambda = 20$), 5.41 ($\lambda = 50$), 3.96 ($\lambda = 100$), and 3.67 ($\lambda = 200$), so the rate $\lambda \approx 20$ recommended here brings simulated kurtosis close to the observed 7.69 without crashing distributional fidelity (IS KS pass rate stays at 94.7% against the unpenalised 95.7%). Only the $\nu_{\min}$ state parameter actually moves ($2.1 \rightarrow 4.89$ at $\lambda = 20$); the upper bracket and the median ν_k remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under an unpenalised ECM, resolvable within the Student-t family without resort to a hierarchical prior or the Gaussian limit. The downstream consequence (Section 7.1) is that CHMM-t's 5–95% envelopes still bracket observed VaR / ES under both the unpenalised and penalised fits, so a risk consumer incurs a widened uncertainty band rather than a systematic over-statement of tail risk under either variant.

Why the Jump Correction Is Not Required in This Scaffold. The discrete framework of Alswaidan and Varner [12] estimates emissions *after* the transition matrix in a three-stage procedure (binning $\rightarrow$ frequency-counted $\mathbf{T}$ $\rightarrow$ within-bin Student-t), so $\mathbf{T}$ is not jointly optimised with emissions and the jump mechanism becomes a corrective for insufficient tail-state persistence. Baum-Welch jointly optimises $\mathbf{T}$ and $\{(\mu_k, \sigma_k)\}$: emissions and transitions co-adapt so that tail states with distinctive emission shapes also accumulate the self-transition probability T_{kk} that reflects empirical clustering of extremes. Eliminating jumps removes two hyperparameters (ϵ, λ) and the $48 \times 200 = 9{,}600$ simulations per K that the prior paper's grid search required; CHMM needs only 20–40 Baum-Welch iterations to convergence.

State Resolution and the Choice of $K = 18$. Sensitivity analysis (Section 6.4) shows ACF-MAE flat across K (the dominant eigenvalue λ_2 is determined by the maximum self-transition probability, which Baum-Welch always learns to be near unity) while KS / AD pass rates rise from $K = 3$ to $K = 18$ and saturate. The practical recommendation is $K = 18$ for distributional-fidelity-sensitive applications; $K = 3$ only when the downstream task is regime identification.

Cross-Asset Robustness and OoS Degradation on NVDA and JPM (Pipeline A). The Pipeline A cross-asset univariate results confirm generalisation across diverse risk profiles (SPY 95.5%, NVDA 96.7%, JNJ 99.4%, AAPL 98.3%, QQQ 95.5% IS KS). At $K = 18$, OoS degradation concentrates on NVDA (55.8%) and JPM (49.8%), a stationarity artefact: the IS-fixed transitions and emissions cannot capture sector-specific regime shifts (the AI-driven NVDA regime; the rate-repricing cycle for financials). The pattern persists at every K in the sweep, so it is not a model-selection issue; rolling-window re-estimation or Bayesian online learning of $\mathbf{T}$ is the natural remedy.

Why Copulas Beat SIM on Cross-Asset Fidelity (Pipeline B). The two copula constructions dominate the SIM factor baseline on per-asset KS (97.5% / 96.2% vs. 72.1%) and correlation reproduction (off-diagonal MAE 0.031 / 0.027 vs. 0.076); all three sit on the same Pipeline A CHMM-N marginals so only the coupling differs. Two structural reasons. First, rank reordering preserves marginals by construction (the simulated $\hat{G}^{(p)}_{j,1:T}$ is a permutation of the stand-alone CHMM path so its empirical CDF coincides with the CHMM marginal), whereas SIM's affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$ convolves the market factor with idiosyncratic residuals, distorting the marginal whenever $\beta_j \neq 1$ or residuals are heavy (JPM's $\hat{\beta} = 1.22$ produces the dramatic 31.5% IS KS). Second, the Student-t copula's edge over Gaussian on correlation MAE (0.027 vs. 0.031) reflects joint tail dependence: profile MLE selects $\nu^* = 6$, in the range typically reported for equity indices [22, 23], implying non-trivial tail-coupling that injects rare concordant extreme moves the Gaussian copula never produces. The same ordering holds in Alswaidan and Varner [12] with discrete marginals, evidence that the copula advantage is robust to the marginal family: practitioners can swap in richer marginals without revisiting the dependence structure.

Three-Way Operational Split: Distribution, Unconditional VaR, Conditional VaR. Synthesising the extended evaluation (Section 6.6), the SM ablation (Section 7.2), the conditional-VaR subsection (Section 7.3), and the QuantGAN / diffusion / MS-GARCH rows (Section 3.3), the recommendation for synthetic-data consumers splits along three axes. On *distributional fidelity* (indistinguishability on marginal and short-horizon joint metrics), **flat CHMM-t at $K = 18$** wins with the smallest MMD (2.0×10^{-5} IS), discriminator AUC closest to 0.5 (0.607), and cleanest monotone aggregational kurtosis decay; window-diffusion is competitive on the first two but collapses on global stylised-fact coverage ($\bar{pv}_{\mathrm{OoS}} = 0.204$). For *unconditional VaR calibration* (long-run breach rate under Kupiec), **two-regime MS-GARCH and SM-CHMM-N** both sit at the coarse-grid floor of 1% Kupiec ($\mathrm{LR}_{\mathrm{uc}} \approx 0.01$, breach rate exactly 1.0%); at $T_{\mathrm{OoS}} = 572$ this is close to point-mass identity on the achievable LR grid rather than a resolved head-to-head ranking between them, with MS-GARCH edging at 5% while SM-CHMM-N edges at OoS HAR-vol forecasting and supplies regime labels. For *regime-conditional VaR* (calibration without breach clustering), the **smoother-based (Viterbi-decoded) state-conditional quantile** of flat CHMM-t is the only combination in our panel that sits below the χ^2_1 Kupiec and Christoffersen thresholds at both $\alpha = 0.01$ and 0.05 on the 2024–2026 OoS window, using the same fitted model that leads the distributional axis. This is an IS goodness-of-fit diagnostic on the regime structure rather than a one-step-ahead backtest (the Viterbi state at time t depends on future observations), and at $T_{\mathrm{OoS}} = 572$ with $\alpha = 0.01$ the Christoffersen test sees only $\approx$ 5–7 expected breaches so small $\mathrm{LR}_{\mathrm{ind}}$ values should be read as "consistent with independence" rather than as resolved rankings. The operationally correct filter-based one-step-ahead backtest (forward posterior $P(s_t \mid r_{1:t})$ under IS-fit parameters, equation (18)) fails Kupiec for all three CHMM families on this OoS window (breach rate 0.2% vs. 1% target and 1.2–1.4% vs. 5% target, Table 16 filter rows): the filter mixture of heavy-tailed per-state emissions is systematically over-conservative, and Proposition 5 grounds the over-conservatism in the structural inequality $F^{-1}_{\sum_k \pi_k F_k}(\alpha) \leq \min_k F^{-1}_k(\alpha)$ that holds whenever the filter posterior places non-trivial mass on at least one heavy-tailed tail-state. Taken together, the smoother diagnostic validates the IS regime partition on the OoS stress window but the operational real-time estimator does not translate that regime information into a Kupiec-calibrated VaR. A consumer needing an operational three-axis pipeline today can assemble flat CHMM-t for paths and IS regime interpretation, plus MS-GARCH (or the SM-CHMM variant) for unconditional VaR; operational regime-conditional VaR is left open, with adaptive re-fitting, an explicit state-conditional variance overlay, and the ν_k-shrinkage direction below as the candidate remedies.

Limitations. Several limitations temper these conclusions. A *residual tail gap under CHMM-N* (5.02 simulated vs. 7.68 observed kurtosis at $K = 18$) remains; CHMM-t and CHMM-L close most of it within the same EM framework, but generalised hyperbolic or α-stable emissions would be the next step for the most extreme quantiles. A related constraint is *emission symmetry*: all three variants are symmetric within state, so SPY's negative skewness (-0.75) is recovered through asymmetric state occupancies and means rather than within-state asymmetry. The plug-in skew-emission ablation quantifies this: the symmetric CHMM-t at $K = 18$ already produces simulated IS skewness of -0.667, close to the observed -0.752; replacing each per-state symmetric emission with a Fernandez-Steel skew variant (per-state γ_k fit by weighted MLE on the EM posteriors with (μ_k, σ_k, ν_k) held fixed) brings simulated skewness to -0.728, with a small IS KS cost (96.4% $\to$ 95.0%). Crucially, the per-state γ_k values cluster tightly around 1 (range 0.96 to 1.02, median 1.00, zero of 18 states with $|\gamma_k - 1| > 0.05$), so multi-state symmetric mixing absorbs most of the observed asymmetry and the residual within-state-skew gap is genuinely small. A full joint EM with γ_k inside the M-step (rather than the plug-in used here) is a natural follow-up but is unlikely to materially change this conclusion. Our *stationarity assumption* is a second structural limitation, since $\mathbf{T}$ and the emissions are estimated on the IS window and held constant, so structural breaks (e.g. March 2020) may not be captured if the IS window lacks comparable events, the leading explanation for the OoS KS drop on NVDA and JPM. The single 572-day OoS window is complemented by the rolling-origin evaluation of Section 6.9, which runs five rolling one-year OoS windows across 2022–2026 and surfaces a direct counter-example: the 2022 OoS window on an 2014–2021 IS sees OoS KS collapse to 0.2–0.4% and Kupiec $\mathrm{LR}_{\mathrm{uc}}^{5\%} = 31.6$–$41.4$ because the rate-hike cycle is a structural break the IS does not contain. The 2023–2026 rolling windows sit in the 87–97% OoS KS range consistent with the paper's main single-window result, so the central stationarity limitation is that the CHMM scaffold is well-calibrated when the IS window contains volatility regimes comparable to those of the OoS window and fails predictably when it does not. The *KS test i.i.d. assumption* is relevant as well, since volatility clustering in simulated paths may slightly inflate KS pass rates relative to a block-bootstrap calibration [12]. The cross-asset study is restricted to a *small universe* of six tickers, and scaling to the 424-asset universe of Alswaidan and Varner [12] would require vine or factor copulas for tractability. Finally, our state selection is an *information-criterion rather than held-out* procedure, so a walk-forward log-likelihood would provide a complementary purely-predictive check on K.

9 Conclusion

We presented a family of continuous hidden Markov models (CHMM-N, CHMM-t, CHMM-L) trained by EM for generating synthetic financial time series, together with two cross-asset dependence extensions for multi-asset synthesis. All three variants share the same log-space forward-backward scaffold and quantile-based initialization; the M-step is the classical Baum-Welch update for Gaussian emissions, an ECM sequence with a golden-section search over ν_k for Student-t emissions, and closed-form weighted-median / weighted-MAD estimators for Laplace emissions. The central univariate finding is that each CHMM variant reproduces all three canonical stylized facts of equity returns (heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering) at a single moderate state resolution ($K = 18$, selected from a $\{3, 6, 9, 12, 15, 18, 21\}$ sweep by winning or tying on every distributional pass-rate metric) within a standard HMM framework, as an alternative to (rather than a replacement for) the HSMM [11] and jump-augmented [12] routes. The heavy-tailed CHMM-t and CHMM-L variants additionally close the residual kurtosis gap of the Gaussian CHMM-N.

Our experiments are consistent with the broader regime-switching literature in which a transition matrix with heterogeneous diagonal entries across more than two or three regimes produces a mixture of geometric sojourn-time distributions that approximates slow ACF decay. At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to sustain persistence, achieving ACF-MAE of 0.047–0.053 across the K sweep (comparable to GARCH(1,1) at 0.048) through a sum-of-exponentials approximation whose effect is comparable to the flexible sojourn-time distributions of hidden semi-Markov models [11]. This moderate-K route complements the HSMM [11] and jump-augmented [12] corrections to the Rydén et al. [10] $K = 2$–3 result rather than superseding them; the headline contribution of this paper is the integrative evaluation (unified EM scaffold, twelve-generator panel, regime-conditional VaR test), not a single-route resolution of the Rydén et al. limitation. Section 5 grounds the operational claims of this paper in stated propositions: the spectral ACF mechanism (Theorem 1) and the Rydén separation (Corollary 1) underlie the moderate-K ACF claim; ECM monotonicity (Proposition 1), identifiability (Proposition 2), and MLE consistency (Proposition 3) underwrite the EM estimator; rank-reordering marginal preservation (Proposition 4) anchors the copula-versus-SIM ordering on cross-asset KS; and the mixture-quantile inequality (Proposition 5) and the bin-centroid VaR floor (Proposition 6) explain the filter-VaR Kupiec failure and the discrete-baseline Kupiec failure respectively.

At the prior paper's state resolution ($K_{\text{disc}} = 90$) the discrete NJ and WJ variants of Alswaidan and Varner [12] match the CHMM family on coarse KS/AD pass rates, so the continuous-emissions advantage is a claim about tail structure and downstream VaR usability: simulated kurtosis near 3.5 against observed 7.68, aggregational-Gaussianity collapse at horizons $h = 10, 21$, and an over-conservative 5% VaR breach rate of 1.7% that rejects Kupiec with $\text{LR}_{\text{uc}} = 16.81$ ($p < 10^{-3}$), each of which the continuous CHMM family clears (Section 7.1).

The twelve-model benchmark of Table 5 confirmed that no single generator dominates all metrics, but the CHMM at $K = 18$ achieves the most balanced performance: 93.2–93.8% IS KS across the three emission families (vs. 25.9% GARCH, 97.9–98.3% for the $K_{\text{disc}} = 90$ discrete variants), ACF-MAE of 0.0507 under CHMM-N (vs. 0.0484 GARCH and 0.0598–0.0618 for the discrete variants), Wasserstein-1 of 0.101–0.103 under CHMM-t and CHMM-L (the smallest of any parametric model), and 100% IS quantile coverage. Cross-asset univariate generalization to NVDA (96.7% IS KS), JNJ (99.4%), JPM (97.8%), AAPL (98.3%), and QQQ (95.5%) confirmed adaptability to diverse equity risk profiles; OoS evaluation on held-out 2024-01-04 through 2026-04-20 data showed robust generalization on SPY/JNJ/AAPL/QQQ but substantial OoS KS drops on NVDA (55.8%) and JPM (49.8%), which we attribute to the 2024–2026 macroeconomic regime (the AI-driven high-volatility regime for tech, and the interest-rate-repricing cycle for financials) and which the walk-forward refit partially recovers.

The cross-asset extensions then showed that the CHMM marginals compose cleanly with standard cross-sectional dependence models. On the six-asset universe, both the Gaussian and Student-t copulas substantially outperformed the Single Index Model factor baseline: copulas achieved a median IS KS pass rate of 97.8%/96.0% versus 78.5% for SIM, and an off-diagonal correlation MAE of 0.031 (Gaussian) and 0.027 (Student-t) versus 0.076 for SIM. The Student-t copula's profile MLE selected $\nu^* = 6$ degrees of freedom, and the resulting tail-dependent copula produced the best correlation reproduction overall. These findings replicate the ordering reported by Alswaidan and Varner [12] with discrete HMM marginals, suggesting that the copula advantage is intrinsic to the rank-reordering construction and transfers across marginal families.

Option pricing, implied-volatility-surface calibration, volatility-index generalization, and leverage-effect modelling are explicitly out of scope for this paper and are deferred to future work.

Several directions for future work follow from these results. *Skew-heavy-tailed emissions* are the most immediate extension: this paper closes the kurtosis gap with Student-t and Laplace emissions, and skew-t or skew-Laplace emissions would additionally accommodate within-regime asymmetry and match the observed negative skewness of equity returns (-0.75 for SPY). *Time-varying transition dynamics*, estimated via rolling windows or Bayesian online learning of the transition matrix, would relax the stationarity assumption and should directly address the OoS cliff observed on NVDA and JPM over the 2024–2026 OoS window. *Full-universe multi-asset generation via vine copulas*, scaling the copula construction from six assets to the 424-asset universe of Alswaidan and Varner [12], requires vine or factor copulas for tractability and is a natural next step. *Principled state selection* is a further refinement: our $K = 18$ choice is a multi-metric score on top of a finite sweep, and combining it with BIC or held-out log-likelihood (walk-forward) would yield a formally principled selection that plugs directly into the appendix Table 21. Finally, *regime-aware dynamic asset allocation* is the most immediate downstream application of the synthetic-data engine, following the spirit of Bae et al. [3] by using CHMM-decoded states as conditioning variables in a dynamic allocation model and evaluating the realized utility of the resulting policy against policies calibrated on i.i.d. or GARCH generators.

Data and Code Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: `https://github.com/altashly1/CHMM-paper`. The core continuous HMM implementation (all three emission families, SIM, and Gaussian / Student-t copula modules) is available in the code repository: `https://github.com/altashly1/CHMM-Model`. The global random-seed policy for reproducibility is stated once in Section 3.5. The software environment is Julia $\geq$ 1.12 with package versions pinned in `Manifest.toml`; running `Pkg.instantiate()` inside the project directory reproduces the dependency graph, after which the analysis entry-point script regenerates every table and figure.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, implemented the Single Index Model and copula-based cross-asset extensions, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Validation Metric Details

A.1 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [57, 58] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{20}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{21}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [69],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| \, du. \tag{22}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{B} (\sqrt{p_i} - \sqrt{q_i})^2}, \tag{23}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 6.8), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P} \sum_{p=1}^{P} \frac{2}{d(d-1)} \sum_{1 \le i < j \le d} \left| \hat{\Sigma}^{(p)}_{\text{sim},\, ij} - \hat{\Sigma}_{\text{obs},\, ij} \right|.$$

A.2 Extended Evaluation: Detail Panels

Per-generator detail for the leverage, aggregational-Gaussianity, and joint p-value diagnostics summarised in Section 6.6 (headline panel: Table 7).

Table 17. Leverage effect (Pipeline A, SPY). Observed IS $\text{avg}_{k=1}^{20} \text{corr}(r_t^2, r_{t-k}) = -0.088$ with down-vs-up asymmetry 0.358; observed OoS averages are -0.066 and 0.478. p_{avg} is the two-sided simulation-based coverage p-value for the observed average over 1,000 paths.

Model	avg IS	asym IS	p_{avg} IS	avg OoS	asym OoS	p_{avg} OoS
Bootstrap	0.000	0.001	0.000	0.001	−0.006	0.000
Gaussian	−0.000	−0.001	0.000	−0.000	0.001	0.000
Laplace	−0.000	−0.000	0.000	−0.000	0.005	0.000
Discrete NJ	−0.010	0.287	0.000	−0.008	0.273	0.001
Discrete WJ	−0.010	0.290	0.000	−0.008	0.287	0.011
GARCH(1,1)	−0.001	0.064	0.014	−0.001	0.065	0.061
CHMM-N	**−0.038**	0.292	0.000	**−0.033**	0.287	**0.205**
CHMM-t	−0.025	0.281	0.000	−0.029	0.274	0.048
CHMM-L	−0.024	0.322	0.000	−0.025	0.310	0.019

A.3 VaR / ES Envelope Visualization (Companion to Table 13)

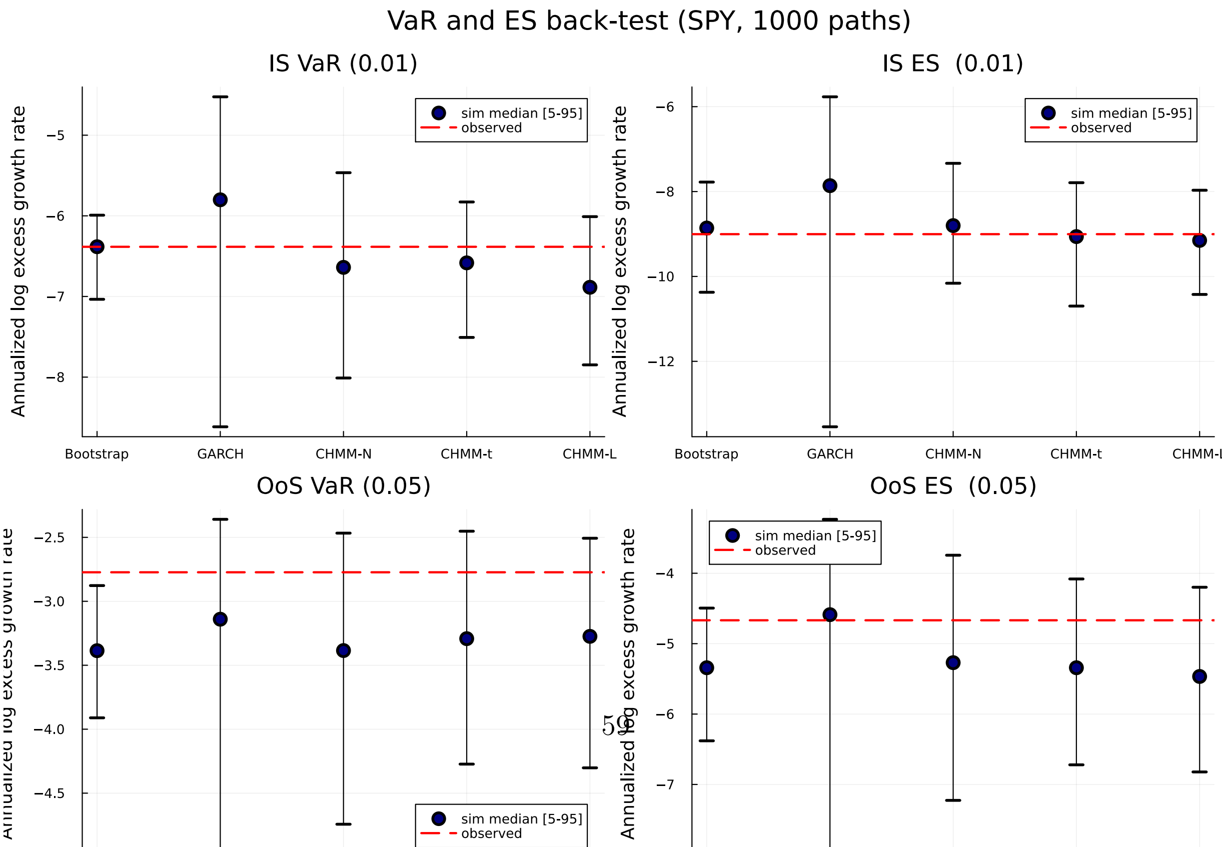
Figure 7 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 13.

Table 18. Aggregational Gaussianity (Pipeline A, SPY). Median simulated kurtosis at aggregation horizons $h \in \{1, 5, 10, 21\}$ over 1,000 paths. Observed IS kurtosis is $\{7.68, 4.34, 6.82, 2.63\}$; observed OoS kurtosis is $\{5.29, 1.98, 0.09, -0.16\}$.

Model	IS kurtosis $h=1$	$h=5$	$h=10$	$h=21$	OoS kurtosis $h=1$	$h=5$	$h=10$	$h=21$
Observed	7.68	4.34	6.82	2.63	5.29	1.98	0.09	−0.16
Bootstrap	7.44	1.31	0.54	0.13	5.24	0.77	0.04	−0.39
Gaussian	−0.01	−0.03	−0.09	−0.17	−0.05	−0.17	−0.30	−0.54
Laplace	2.86	0.51	0.16	−0.04	2.55	0.23	−0.15	−0.49
Discrete NJ	3.55	2.15	1.10	0.34	3.41	1.42	0.39	−0.29
Discrete WJ	3.50	2.16	1.16	0.42	3.34	1.41	0.47	−0.25
GARCH(1,1)	4.35	4.79	3.74	2.55	1.79	1.87	1.26	0.34
CHMM-N	4.88	3.70	3.27	2.38	4.08	2.38	1.60	0.40
CHMM-t	**8.46**	**2.65**	**2.06**	**1.35**	**5.55**	**1.89**	**1.15**	**0.21**
CHMM-L	6.33	2.45	1.80	1.02	5.49	1.97	1.04	0.07

Table 19. Joint simulation-based p-value coverage (Pipeline A, SPY). Each cell is the two-sided simulation-based p-value for the observed statistic under the generator. $\bar{pv}$ is the mean across {kurt, avg leverage, agg kurt $h = 5$, agg kurt $h = 21$}; larger $\bar{pv}$ means better joint coverage.

Model	p_{kurt}	p_{acf}	p_{lev}	p_{ak5}	p_{ak21}	$\bar{pv}$ IS	$\bar{pv}$ OoS
Bootstrap	0.894	0.000	0.000	0.012	0.005	0.228	0.466
Gaussian	0.000	0.000	0.000	0.000	0.002	0.000	0.116
Laplace	0.002	0.000	0.000	0.000	0.002	0.001	0.156
Discrete NJ	0.000	0.000	0.000	0.034	0.029	0.016	0.376
Discrete WJ	0.000	0.000	0.000	0.060	0.057	0.029	0.384
GARCH(1,1)	0.271	0.021	0.014	0.903	0.974	0.540	0.468
CHMM-N	0.009	0.000	0.000	0.659	0.894	0.390	0.539
CHMM-t	0.872	0.000	0.000	0.196	0.228	0.324	0.661
CHMM-L	0.369	0.000	0.000	0.061	0.130	0.140	**0.692**



B CHMM Algorithms and Derivations

B.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [18, 59]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \tag{24}$$

$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \dots, T, \tag{25}$$

$$\log \beta_T(k) = 0, \tag{26}$$

$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T{-}1, \dots, 1, \tag{27}$$

where $\operatorname{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, b_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, b_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{28}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{29}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \operatorname{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [14], Liu and Rubin [15] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{30}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{31}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee under Assumption 3, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 in Section 5 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{32}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [14], Liu and Rubin [15] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 8).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k}\exp(-|x-\mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k)), \qquad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\,|O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{33}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

B.2 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 is the unified EM procedure shared across all three emission families. It makes explicit the *shared scaffold* (quantile-based initialization, log-space forward-backward, transition update) and the three *family-specific branches* (Gaussian M-step for CHMM-N, ECM with one-dimensional golden-section ν-search for CHMM-t, closed-form weighted-median + weighted-MAD for CHMM-L); the branch points are lines 3, 12, and 28. Visualizing the three variants as a single algorithm with three M-step branches makes the architectural contribution of the paper (one EM engine, three interchangeable emission families) immediate from the pseudocode. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

B.3 CHMM-t Degrees-of-Freedom Diagnostics

This appendix backs the discussion in Section 8 of the CHMM-t IS kurtosis overshoot. Figure 8 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit on SPY IS (seed = 20260420). Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

Table 20 reports the seven-metric panel for CHMM-t fits whose ECM bracket lower bound is lifted from $\nu_{\min} = 2.1$ through $\{2.5, 3.0, 4.0\}$, plus the Gaussian limit ($\nu = \infty$, equivalent to CHMM-N). Lifting the bracket does not reduce the IS kurtosis overshoot: the simulated kurtosis moves within the 12.5–14.0 band across bracket floors and only collapses to 5.05 in the full Gaussian limit. The overshoot is therefore a feature of the Student-t emission family at $K = 18$ and not a lower-bracket artefact.

Table 20. CHMM-t $\nu_{\min}$ bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths).

$\nu_{\min}$	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
2.1	95.7	90.2	86.2	82.0	13.95	10.08	0.0546	0.100	0.0759
2.5	95.8	90.6	84.2	79.1	15.12	10.34	0.0547	0.100	0.0759
3.0	93.8	89.5	85.3	80.4	12.25	9.65	0.0548	0.099	0.0760
4.0	95.5	89.5	85.0	80.3	13.25	9.39	0.0550	0.101	0.0759
∞ (Gauss)	94.1	86.7	84.2	75.5	5.03	4.46	0.0510	0.110	0.0809
Observed	–	–	–	–	7.68	5.29	–	–	–

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 12, and 28 branch on the emission family $F \in \{\mathcal{N}, t, L\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t instantiates an ECM sequence whose second CM step updates ν_k by a one-dimensional golden-section search on the per-state Q-function.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \text{mean}(C_k), \quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \text{mean}(C_k), \quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6}), \quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \text{median}(C_k), \quad b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$.
8: **end for**
9: $T_{ij} \leftarrow 1/K; \quad \pi_k \leftarrow 1/K; \quad \mathcal{L}_{\text{prev}} \leftarrow -\infty$.
10: **EM Loop.**
11: **for** $n = 1$ to `max_iter` **do**
12:   **E-Step, per-family emission log-likelihood.**
13:   **switch** F
14:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
15:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
16:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
17:   **E-Step, shared log-space forward-backward.**
18:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
19:   **for** $t = 2$ to T, $j = 1$ to K **do**
20:     $\log \alpha_t(j) \leftarrow \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
21:   **end for**
22:   $\log \beta_T(k) \leftarrow 0$.
23:   **for** $t = T - 1$ down to 1, $i = 1$ to K **do**
24:     $\log \beta_t(i) \leftarrow \operatorname*{logsumexp}_j(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
25:   **end for**
26:   Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (28).
27:   **Shared transition update.** $T_{ij} \leftarrow \sum_t \xi_t(i,j) \,/\, \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
28:   **M-Step, per-family emission update.**
29:   **switch** F
30:     $F = \mathcal{N}$ {classical Baum-Welch [17].}
31:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
32:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
33:     $F = t$ {ECM of Peel and McLachlan [14], Liu and Rubin [15]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
34:       $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
35:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
36:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
37:       $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
38:     $F = L$ {closed-form weighted Laplace MLE.}
39:       $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
40:       $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
41:   **Convergence check.** $\mathcal{L}^{(n)} \leftarrow \operatorname*{logsumexp}_k(\log \alpha_T(k))$.
42:   **if** $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| <$ tol **then**
43:     **break**
44:   **end if**
45:   $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$.
46: **end for**
47: **return** $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$.}

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$
Ensure: Most probable state sequence $S^*_{1:T}$

```
for k = 1 to K do
    log δ_1(k) ← log(1/K) + log N(O_1 | μ_k, σ_k^2).
end for
for t = 2 to T do
    for j = 1 to K do
        vals(i) ← log δ_{t−1}(i) + log T_ij for all i.
        log δ_t(j) ← max_i vals(i) + log N(O_t | μ_j, σ_j^2).
        ψ_t(j) ← arg max_i vals(i).
    end for
end for
S*_T ← arg max_k log δ_T(k).
for t = T − 1 down to 1 do
    S*_t ← ψ_{t+1}(S*_{t+1}).
end for
return S*_{1:T}
```

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

```
for p = 1 to P do
    Sample initial state S_1 ~ Categorical(π).
    for t = 1 to M do
        Emit observation Ĝ_t^(p) ~ N(μ_{S_t}, σ^2_{S_t}).
        if t < M then
            Transition to next state S_{t+1} ~ Categorical(T_{S_t,·}).
        end if
    end for
end for
return {Ĝ^(p)_{1:M}}_{p=1}^P
```

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (PSD, unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T\times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T\times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow ZL^\top$
  if $\mathcal{C} =$ Gaussian **then**
      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
      Draw $W \sim \chi^2_\nu/\nu$ i.i.d. for $t = 1, \ldots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 3.
    Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \leq \cdots \leq \tilde{g}_{j,(T)}^{(p)}$.
    Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \ldots, T$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ for $t = 1, \ldots, T$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: IS return matrix $\mathbf{R} \in \mathbb{R}^{T\times d}$, market-column index m, market CHMM $\mathcal{M}_m$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

OLS fit. For $j \neq m$:
  $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m})/\text{Var}(\mathbf{R}_{:,m})$
  $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$
  $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}, \quad R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$
for $p = 1$ to P **do**
  Simulate market path $G_{m,1:T}^{(p)}$ from $\mathcal{M}_m$ via Algorithm 3.
  $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$.
  for $j \neq m$ **do**
    Draw residual indices $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \ldots, T\})$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

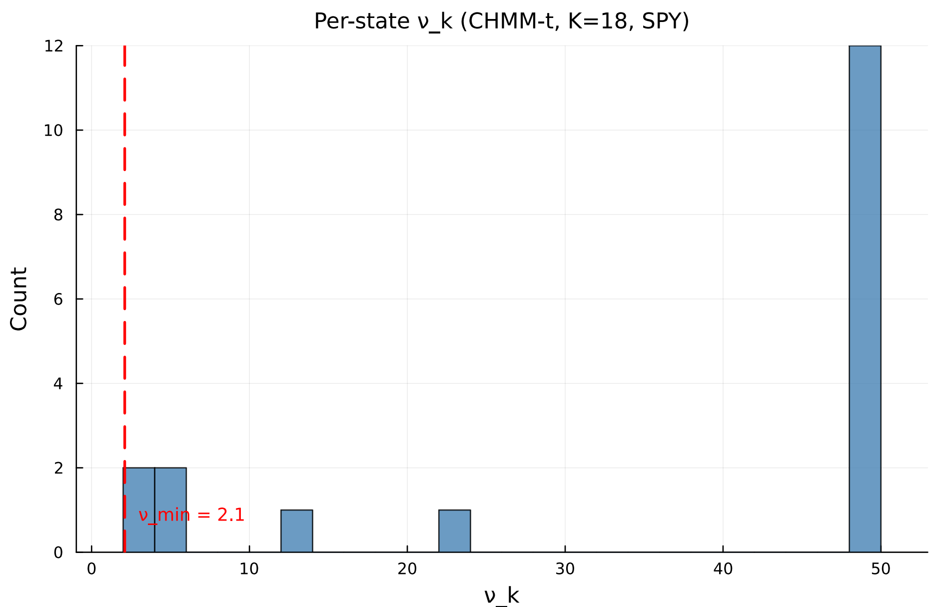


Figure 8. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

C State-Resolution and Multi-Emission Supplementary Material

C.1 Empirical Stylized-Fact Visualization (Companion to Table 2)

Figure 9 visualizes the stylized facts of SPY daily excess growth rates whose Jarque–Bera and Ljung–Box test statistics are reported in Table 2.

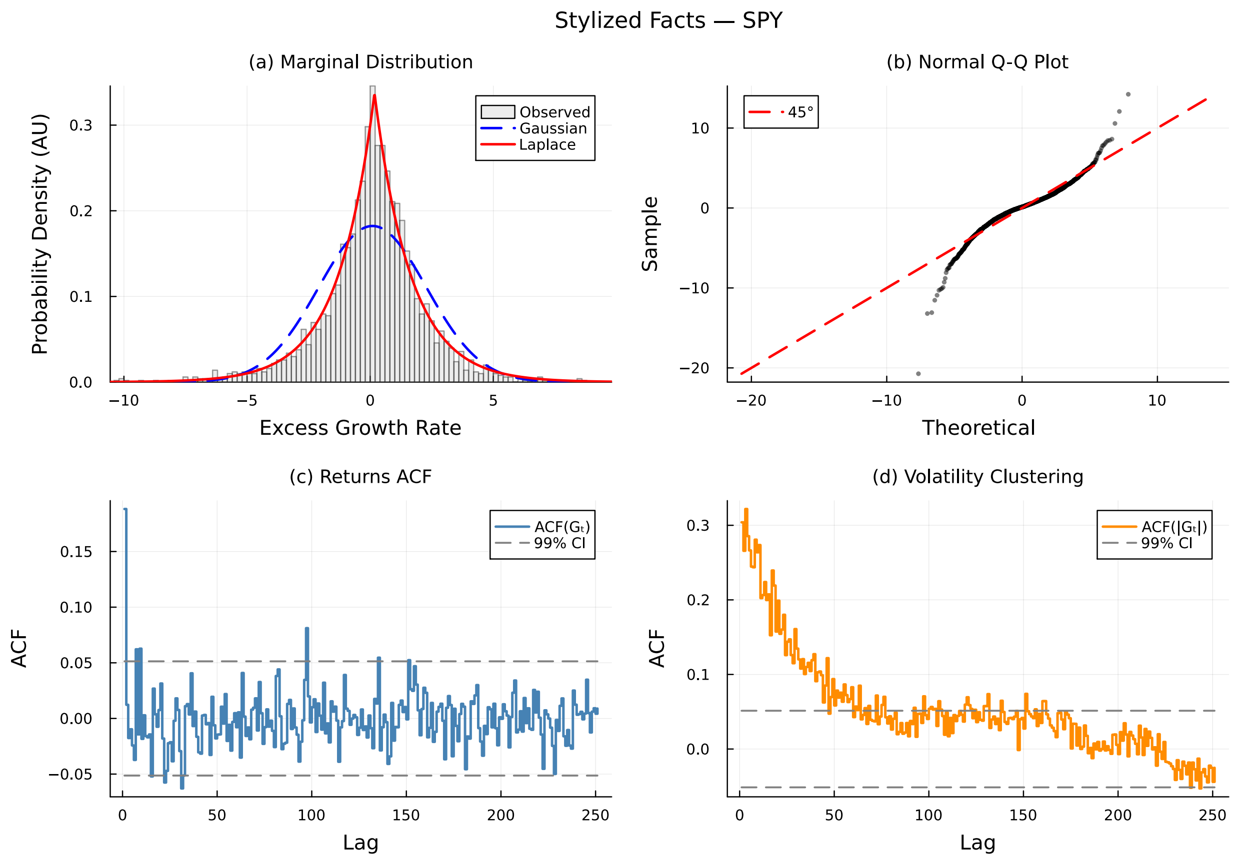


Figure 9. Empirical stylized facts of SPY daily excess growth rates (IS, 2014-01-03 through 2024-01-03). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

C.2 Full State-Resolution Sensitivity Table

Table 21 reports the full K-sweep referenced in Section 6.4 of the main text. Observed excess kurtosis is 7.68 IS and 5.29 OoS at all K.

Table 21. State resolution sensitivity for SPY CHMM-N (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 5 in the main text. IS coverage was 100% at every K; OoS coverage is reported separately.

	KS (%)		AD (%)		Kurtosis					
K	IS	OoS	IS	OoS	Obs	Sim	ACF-MAE	W_1	H	Cov OoS (%)
3	88.8	77.1	79.2	67.9	7.68	3.80	0.0466	0.130	0.0836	94.9
6	93.1	80.0	87.1	74.6	7.68	5.26	0.0499	0.113	0.0828	90.9
9	94.6	82.4	85.8	73.6	7.68	4.56	0.0494	0.116	0.0806	93.9
12	94.6	82.8	86.0	74.9	7.68	4.57	0.0495	0.112	0.0802	88.9
15	93.9	82.7	86.9	75.6	7.68	4.89	0.0505	0.116	0.0806	90.9
18	95.6	81.7	88.4	73.2	7.68	4.98	0.0509	0.110	0.0805	90.9
21	**96.1**	**86.6**	**89.0**	**78.4**	7.68	3.83	0.0532	0.111	0.0798	**100.0**

C.3 Multi-Emission State-Resolution Sensitivity

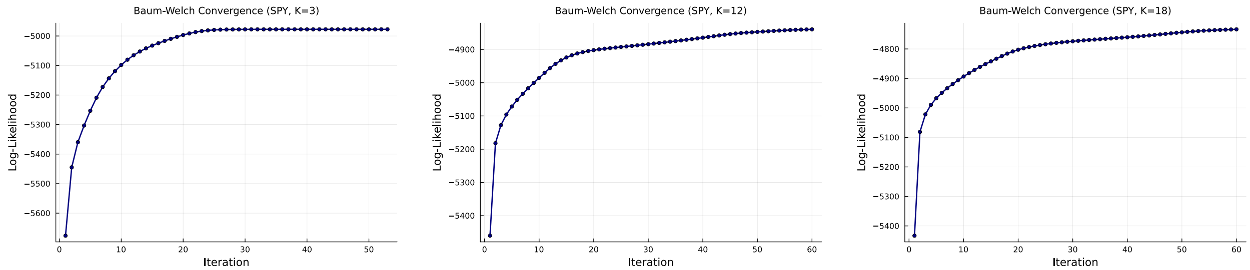
Table 22 extends Table 21 across the three emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t or Laplace emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost. Three patterns extend from Table 21 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all three families: CHMM-t reaches its highest IS KS at $K = 18$ (95.2%) and its highest OoS KS at $K = 18$ (85.1%), and CHMM-L reaches its highest IS KS at $K = 18$ (96.2%) and its highest IS AD at $K = 18$ (93.0%). Once a family enters the plateau the operating point is flat with respect to K, so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the ACF-MAE remains within the narrow 0.047–0.056 band observed for the Gaussian sweep, indicating that the volatility-clustering mechanism is a property of the regime structure rather than the emission family. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks around 5.0 at the high-K end, the Laplace sweep lands squarely in the 5.2–6.6 range (closest to the observed 7.68), and the Student-t sweep climbs sharply above 13 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, over-correcting IS kurtosis.

Table 22. Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale. Every metric is computed on the identical simulation pipeline used for Table 5. Observed excess kurtosis is 7.68 IS and 5.29 OoS.

		KS (%)		AD (%)					
Family	K	IS	OoS	IS	OoS	Kurt (IS sim)	ACF-MAE	W_1 IS	H IS
CHMM-N	3	89.8	76.3	81.3	67.7	3.81	0.0468	0.128	0.0835
	6	91.7	79.6	85.9	73.6	5.30	0.0497	0.115	0.0831
	9	92.8	82.2	84.7	75.5	4.55	0.0497	0.118	0.0810
	12	95.0	83.7	86.1	76.6	4.55	0.0497	0.114	0.0805
	15	95.2	83.5	86.9	76.3	4.91	0.0508	0.111	0.0802
	18	94.7	82.4	85.8	76.5	5.02	0.0509	0.111	0.0809
	21	**95.2**	**87.6**	**89.1**	**80.9**	3.80	0.0530	0.112	0.0802
CHMM-t	3	91.1	81.6	85.2	77.1	27.12	0.0540	0.109	0.0764
	6	92.2	82.1	86.1	76.6	20.80	0.0533	0.108	0.0768
	9	92.7	84.8	86.9	80.1	15.96	0.0537	0.104	0.0756
	12	94.1	83.1	88.3	80.6	14.68	0.0542	0.102	0.0757
	15	93.5	82.9	88.0	76.5	15.85	0.0537	0.103	0.0757
	18	**95.2**	**85.1**	**90.9**	**80.7**	14.40	0.0550	**0.101**	**0.0756**
	21	94.9	82.2	89.3	77.8	13.28	0.0540	0.103	0.0758
CHMM-L	3	82.4	63.6	82.9	66.9	5.24	0.0533	0.112	0.0853
	6	88.8	77.0	83.3	72.5	5.74	0.0511	0.113	0.0823
	9	90.0	77.9	84.1	74.8	6.05	0.0519	0.113	0.0823
	12	92.4	77.8	87.6	77.9	6.21	0.0540	0.106	0.0797
	15	92.6	79.4	87.0	77.6	6.06	0.0542	0.106	0.0795
	18	**96.2**	**80.9**	**93.0**	**80.6**	**6.60**	0.0564	**0.101**	0.0793
	21	90.7	80.7	87.3	78.8	5.99	0.0544	0.107	0.0809

C.4 Baum-Welch Convergence

Figure 10 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.



(a) $K = 3$ **(b)** $K = 12$ **(c)** $K = 18$ (selected)

Figure 10. Baum-Welch convergence for the SPY CHMM-N fit (Pipeline A, IS) at selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` $= 60$.

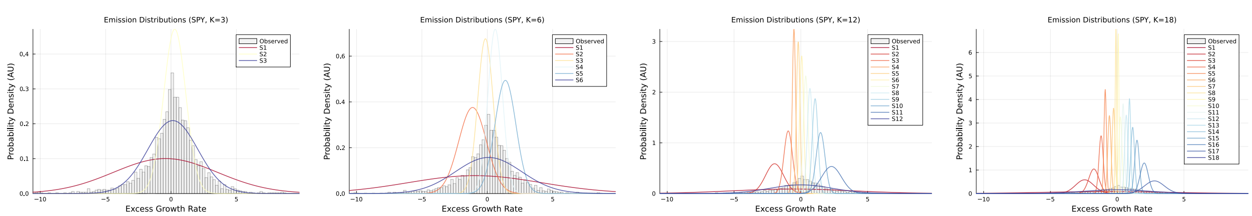
C.5 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K. Figure 11 compares the learned emission distributions for $K \in \{3, 6, 12, 18\}$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ, high σ), a calm state (near-zero μ, low σ), and a boom state (positive μ, moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels.

At $K = 12$–18, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.



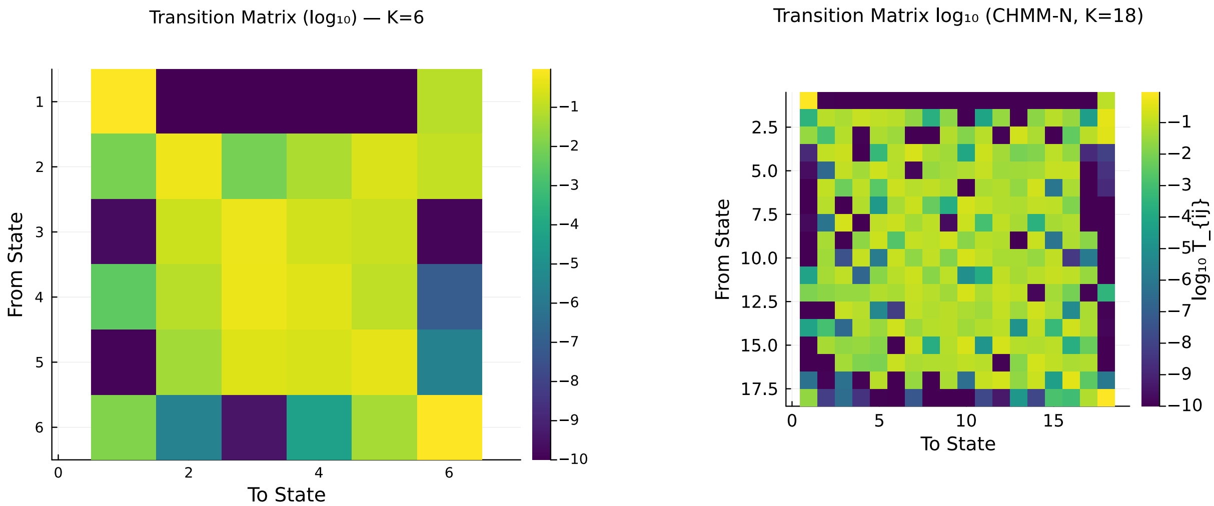
(a) $K = 3$ **(b)** $K = 6$ **(c)** $K = 12$ **(d)** $K = 18$ (selected)

Figure 11. Learned CHMM-N (Gaussian) emission distributions at different state resolutions, overlaid on the observed SPY IS return histogram (Pipeline A). Higher K values produce finer decomposition with dedicated states capturing tail behavior.

C.6 Transition Matrix Structure

Figure 12 shows the learned transition matrices for $K = 6$ and $K = 18$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (for example, from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 8: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.



(a) $K = 6$

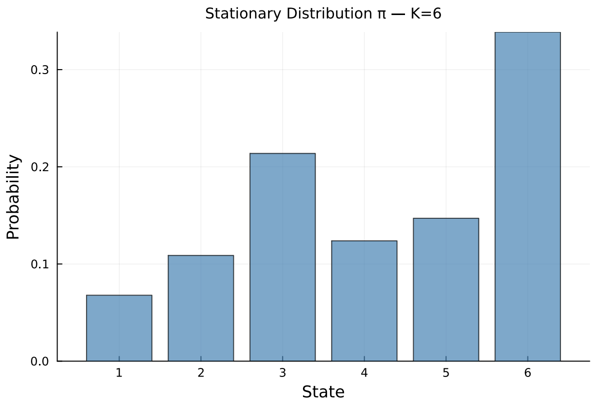
(b) $K = 18$ (selected)

Figure 12. Learned CHMM-N transition matrices for the SPY IS fit (Pipeline A) at $K = 6$ and $K = 18$ in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

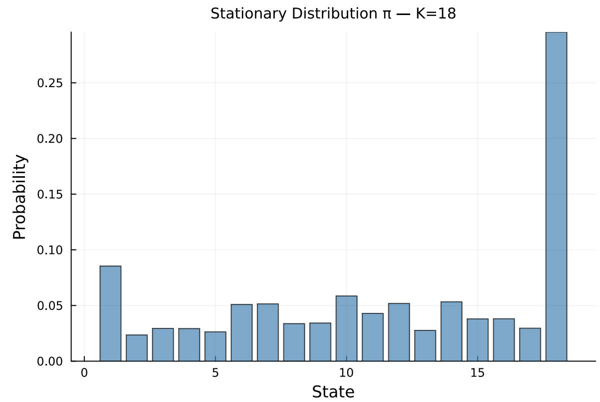
C.7 Stationary Distributions and Residence Times

Figure 13 shows the stationary distributions $\bar{\boldsymbol{\pi}}$ for $K = 6$ and $K = 18$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

Figure 14 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 18$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps. Central states exhibit the longest residence times (5–20 days), while tail states have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. Notably, even without a jump mechanism, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$–0.5) to produce multi-day tail episodes. This contrasts with the discrete model of Alswaidan and Varner [12], where tail-state residence times were too short to sustain clustering without the Poisson jump extension.

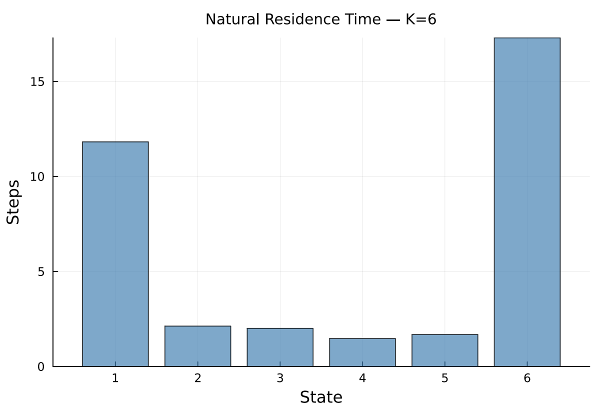


(a) $K = 6$

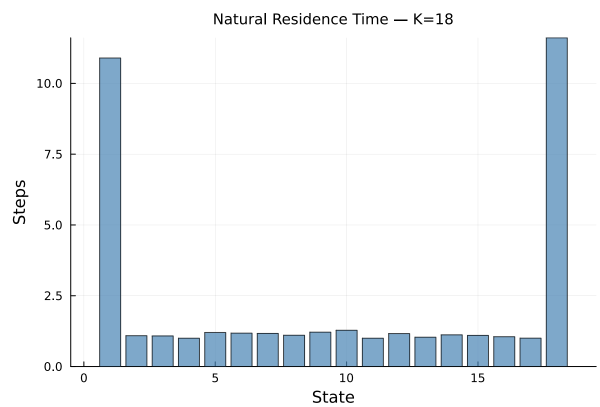


(b) $K = 18$

Figure 13. Stationary distributions $\bar{\boldsymbol{\pi}}$ for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.



(a) $K = 6$



(b) $K = 18$

Figure 14. Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

C.8 IS Comparison at Non-Selected State Resolutions

Figures 15–18 show the IS model comparison panels for $K \in \{3, 6, 12, 21\}$, complementing the $K = 18$ comparison in Figure 4 of the main text. Across all resolutions, the CHMM's simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel. $K = 9$ and $K = 15$ panels are omitted because they interpolate smoothly between the shown resolutions; their numerical metrics are in Table 21.

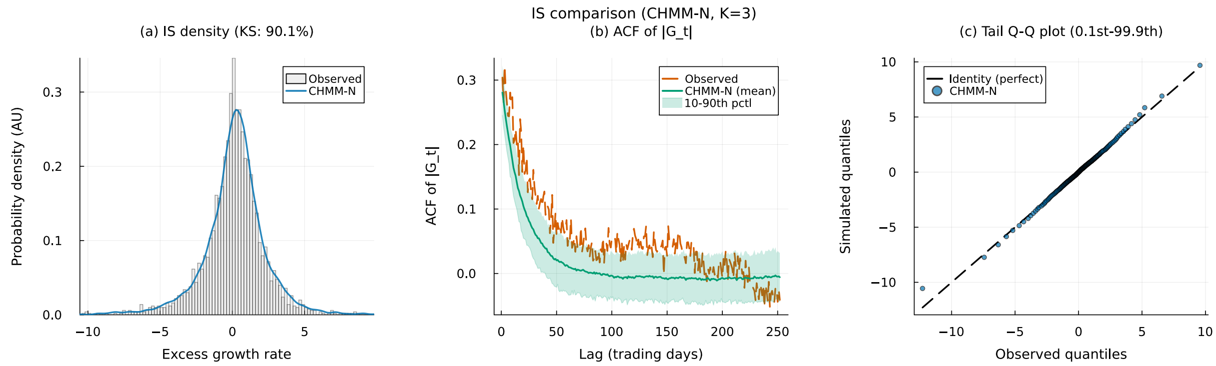


Figure 15. IS SPY CHMM-N comparison at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4). The $K = 18$ counterpart is Figure 4 of the main text.

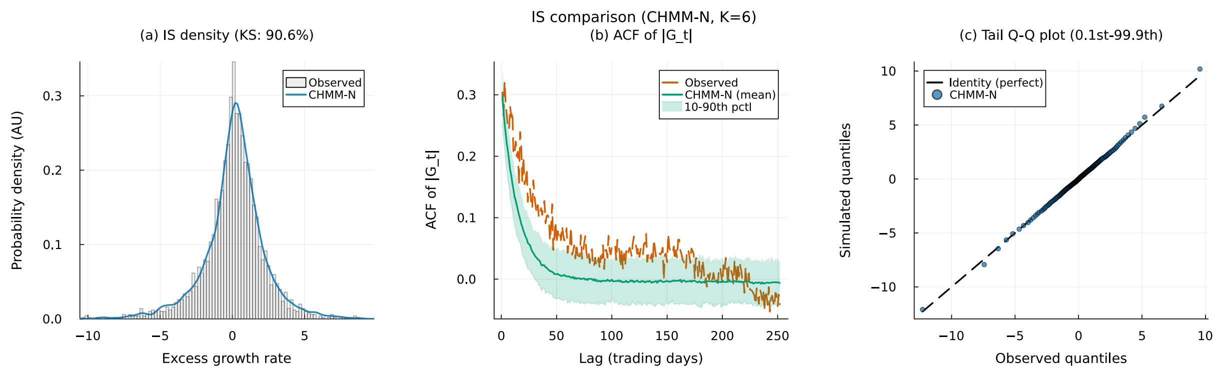


Figure 16. IS SPY CHMM-N comparison at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

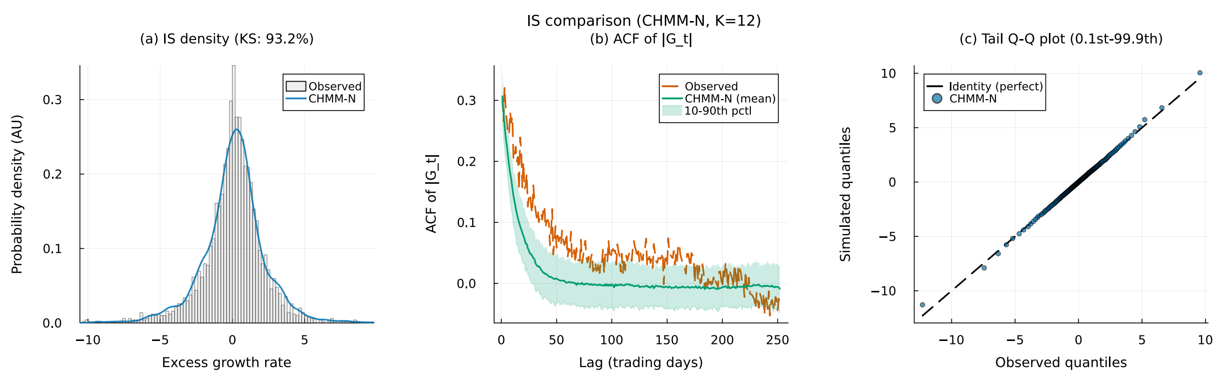


Figure 17. IS SPY CHMM-N comparison at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

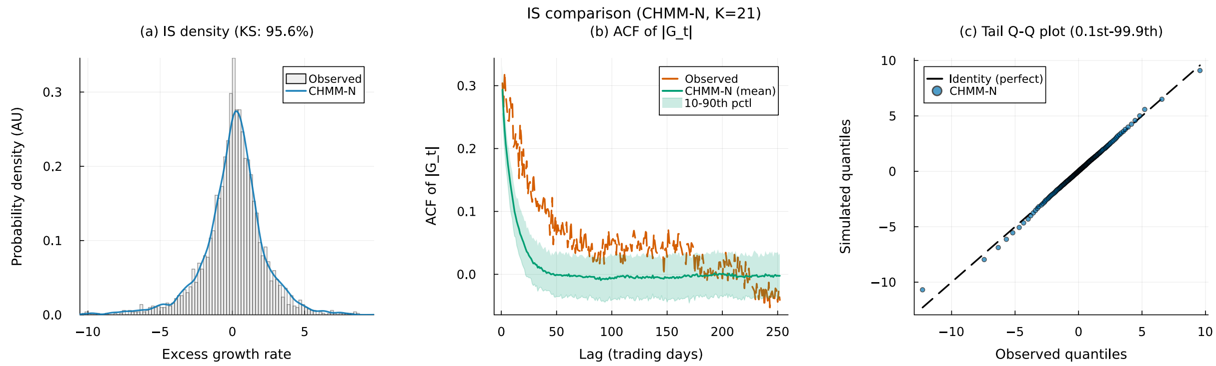


Figure 18. IS SPY CHMM-N comparison at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

C.9 OoS Validation at Non-Selected State Resolutions

Figures 19–22 show the OoS validation panels for $K \in \{3, 6, 12, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 5 of the main text. As with the IS panels, $K = 9$ and $K = 15$ are omitted; numerical OoS KS pass rates are in Table 21.

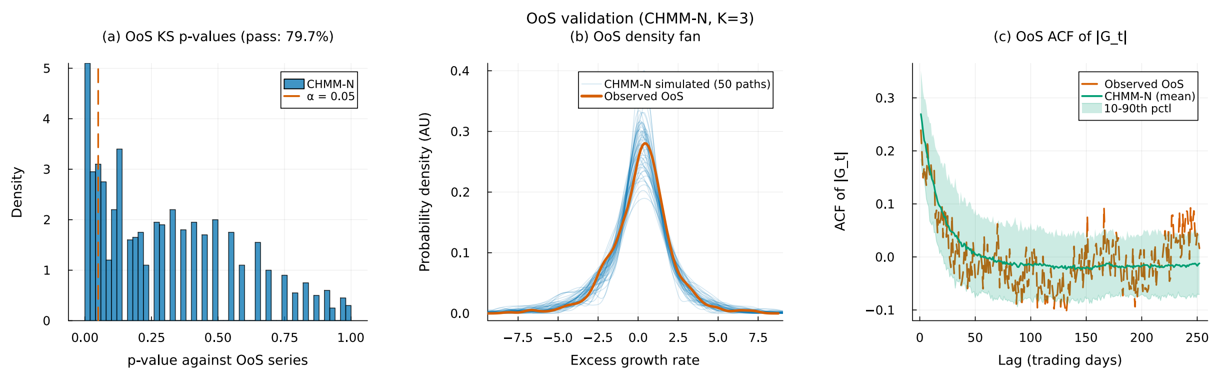


Figure 19. OoS SPY CHMM-N validation at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5). The $K = 18$ counterpart is Figure 5 of the main text.

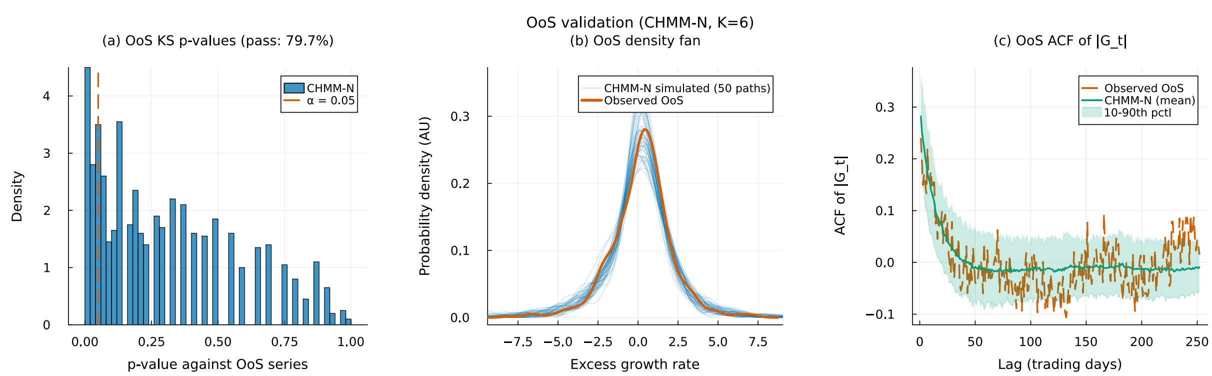


Figure 20. OoS SPY CHMM-N validation at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

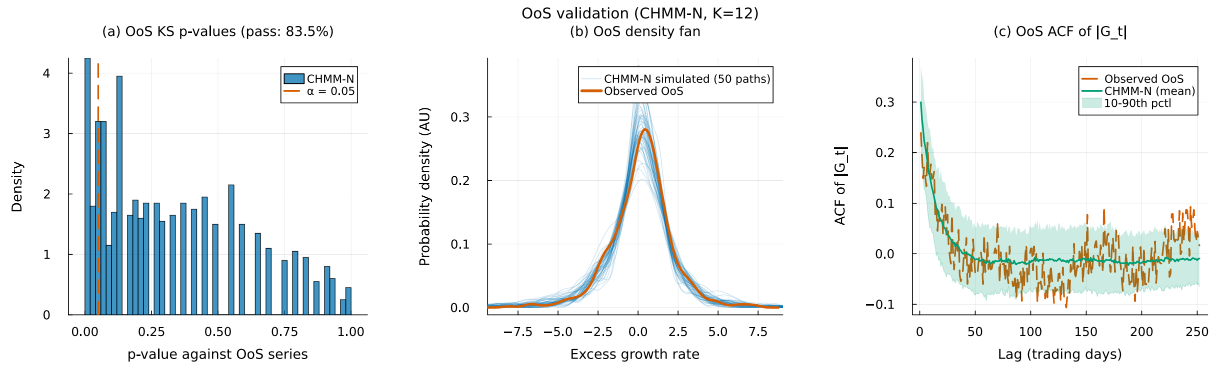


Figure 21. OoS SPY CHMM-N validation at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

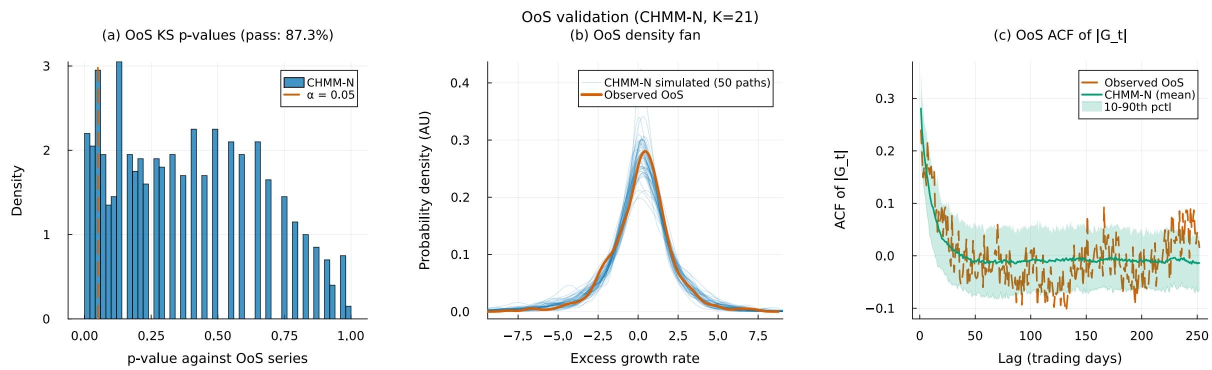
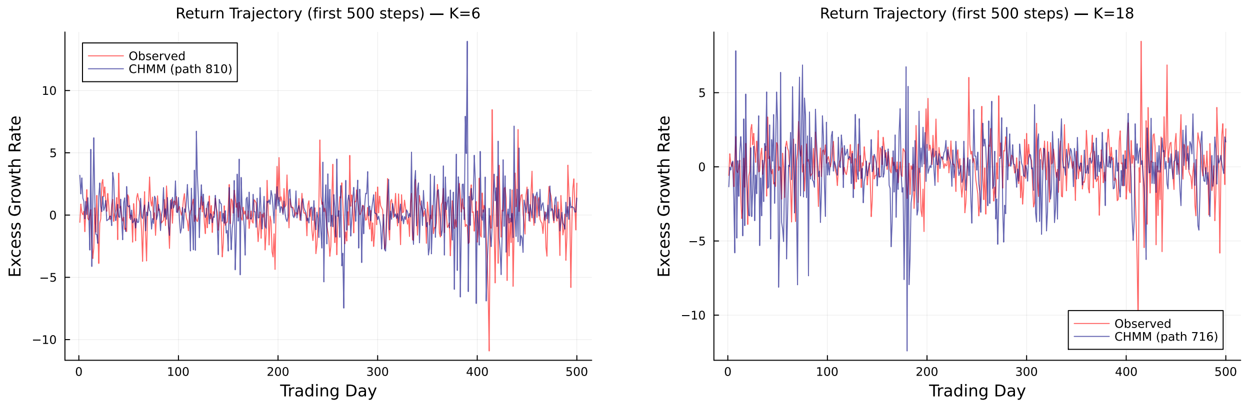


Figure 22. OoS SPY CHMM-N validation at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

C.10 Example Simulated Trajectories (Illustrative, IS-Fitted CHMM-N)

Figure 23 shows example simulated return trajectories from the SPY CHMM-N at $K = 6$ and $K = 18$, illustrating the qualitative dynamics produced by the model. Each panel overlays one unconditional simulated path (blue; the sampled-path index is labelled in the figure legend) against the first 500 trading days of the observed IS SPY excess growth-rate series (red). The two lines do not need to agree pointwise (simulated and observed hidden-state sequences are drawn independently); the intended read is qualitative, showing that both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events.



(a) $K = 6$

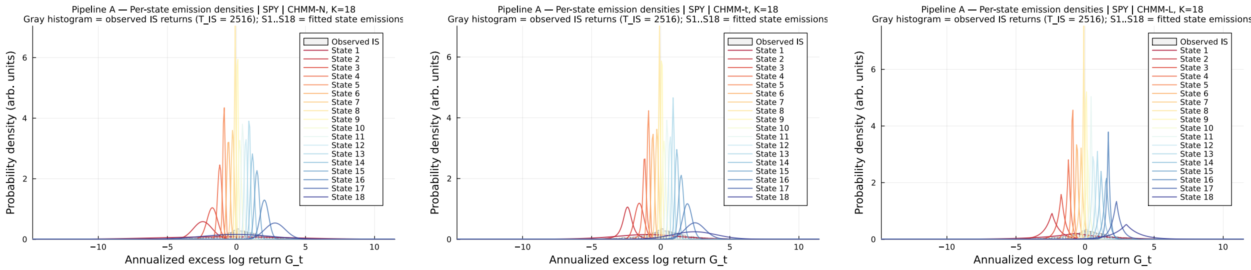
(b) $K = 18$ (selected)

Figure 23. Example simulated return trajectories from the SPY CHMM-N (Pipeline A, IS fit, first 500 simulated steps). Blue: one unconditional simulated path (index labelled in the figure legend); red: first 500 trading days of the observed IS SPY excess growth rate series. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns. This figure is qualitative: simulated and observed state sequences are independent, so point-by-point agreement is not expected.

C.11 Multi-Emission Figures at $K = 18$

This appendix reproduces the standard diagnostic suite (emission PDFs, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$; transition matrices and stationary distributions are discussed in prose only below, since they are visually indistinguishable across families at $K = 18$. The Gaussian counterparts are already shown as Figures 3, 4, 5, 10, 12, 13, and 14 in the main text and above in this appendix. The figures confirm that the shared forward–backward scaffold produces internally-similar regime structure across the three families: the near-diagonal transition band, the K-component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

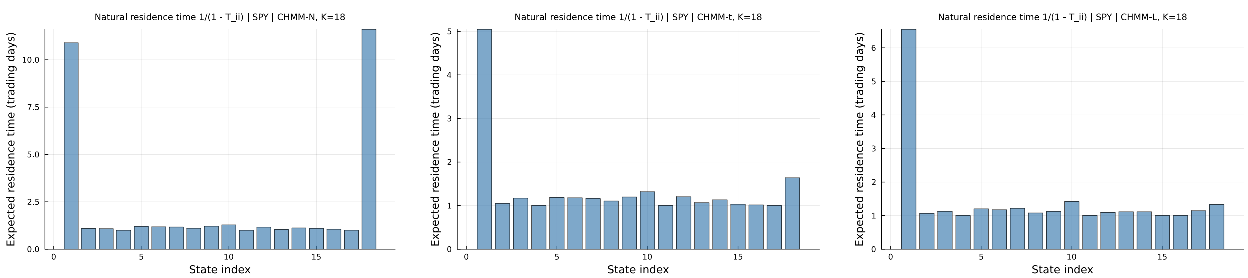
Emission PDFs. Figure 24 compares the learned $K = 18$ emissions for the three families. The CHMM-t panel's per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel's Laplace components have sharper peaks and slower exponential tail decay than the Gaussians.



(a) CHMM-N (Gaussian) **(b)** CHMM-t (Student-t) **(c)** CHMM-L (Laplace)

Figure 24. Learned emission distributions at $K = 18$ across the three emission families (CHMM-N Gaussian, CHMM-t Student-t with per-state ν_k, CHMM-L Laplace), overlaid on the observed SPY IS return histogram (Pipeline A). Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM.

Transition matrices and stationary distributions. The $K = 18$ transition matrices and stationary distributions learned under CHMM-t and CHMM-L are visually indistinguishable from their CHMM-N counterparts (right-hand $K = 18$ panels of Figures 12 and 13): the near-diagonal band, the magnitude of self-transition probabilities, and the mass concentration at central states are features of the regime structure, not the emission family. This is consistent with the ACF-MAE being stable across the family axis (Table 22). The per-family residence times $\tau_k = 1/(1 - T_{kk})$, which magnify small differences in the self-transition probabilities each family recovers, are compared in Figure 25.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L

Figure 25. Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ for the SPY IS fit (Pipeline A) across the three emission families (CHMM-N Gaussian, CHMM-t Student-t, CHMM-L Laplace).

IS comparison. Figures 26–27 reproduce the three-panel IS comparison (density, ACF of $|G_t|$, tail Q-Q) at $K = 18$ for the two heavy-tailed families; the CHMM-N panel is Figure 4 in the main text.

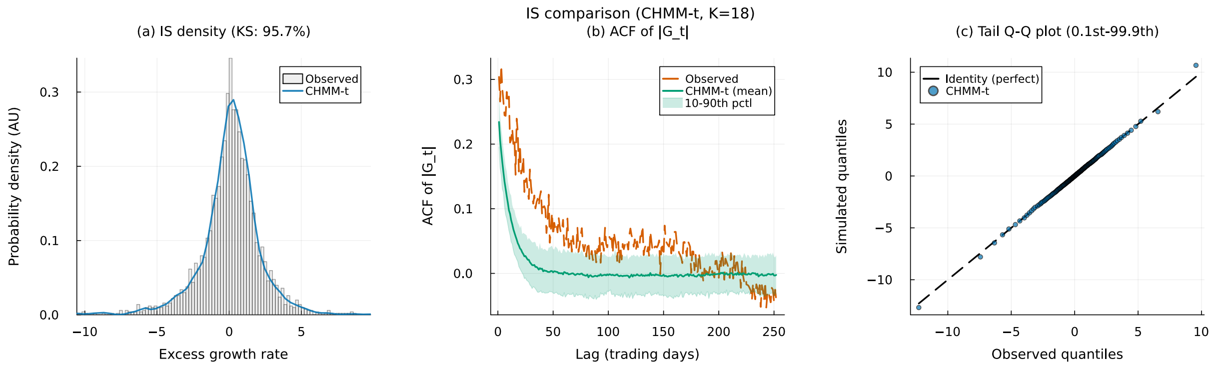


Figure 26. IS SPY model comparison at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) marginal density with KS pass rate, (b) ACF of $|G_t|$, and (c) tail Q-Q plot. The CHMM-N counterpart is Figure 4; the CHMM-L counterpart is Figure 27.

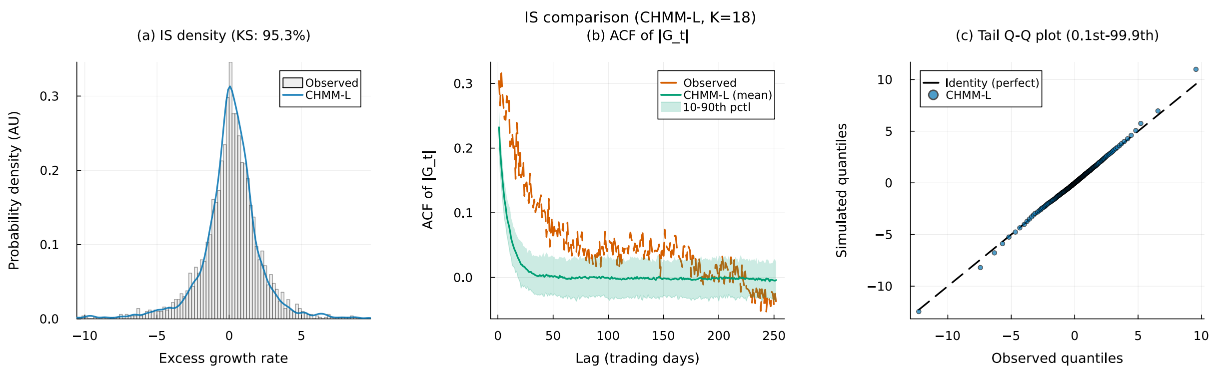


Figure 27. IS SPY model comparison at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 26.

OoS validation. Figures 28–29 show the OoS validation panels for CHMM-t and CHMM-L at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text (simulated 6.40 vs. observed 6.24), and the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian.

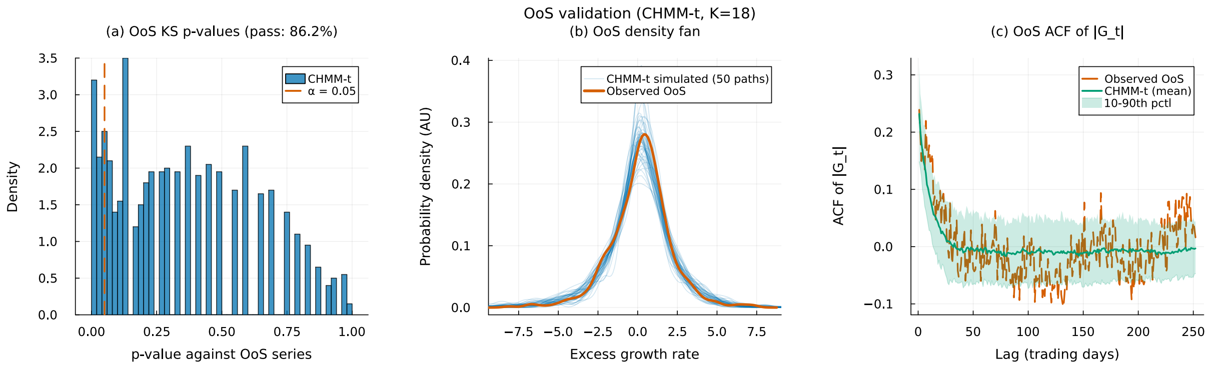


Figure 28. OoS SPY validation at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) KS p-values, (b) OoS density fan, and (c) OoS ACF of $|G_t|$. The CHMM-N counterpart is Figure 5; the CHMM-L counterpart is Figure 29.

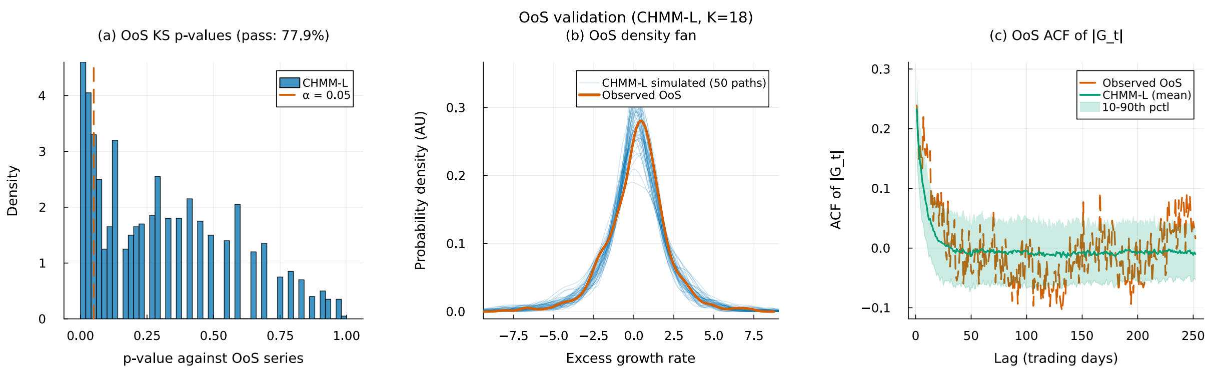
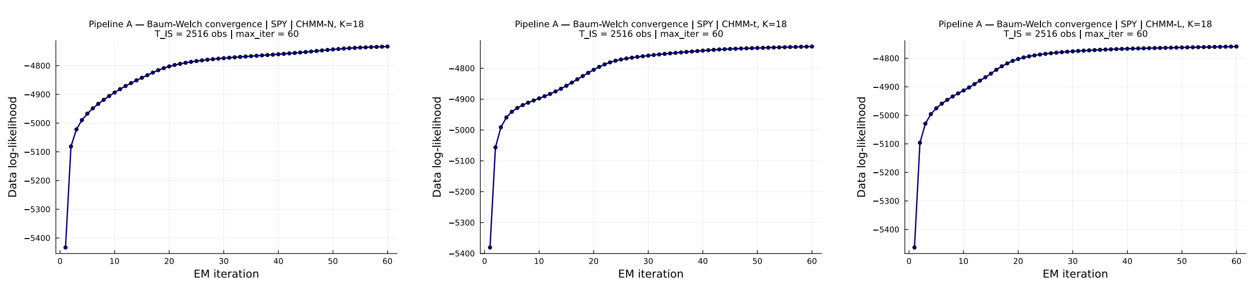


Figure 29. OoS SPY validation at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 28.

Convergence. Figure 30 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all three families. The CHMM-t and CHMM-L curves inherit the monotone increase of the classical EM guarantee; the CHMM-t curve has a slightly longer pre-plateau phase because each ECM sub-iteration refines ν_k with a golden-section search.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L

Figure 30. Log-likelihood histories for the SPY IS fit (Pipeline A) at $K = 18$ across the three emission families (CHMM-N Gaussian classical EM; CHMM-t Student-t ECM with per-state ν_k golden-section sub-step; CHMM-L Laplace weighted-EM). All three curves are monotonically non-decreasing and plateau well before the `max_iter` $= 60$ budget.

C.12 Rydén K = 2 Replication

Table 23 reports the direct replication of Rydén et al.'s 1998 $K = 2$ setting on SPY IS under both initialization strategies, as discussed in Section 8. The quantile-based initialization used in the main body is row 1; five independent random-seed initializations are rows 2–6. Random init here draws per-state (μ_k, σ_k) from a moderate perturbation of the sample moments, which is representative of the random-init practice common in the 1990s HMM literature.

Table 23. Rydén replication: $K = 2$ CHMM-N on SPY IS under quantile vs random initialization (seed sub-index varies the random init).

Init	KS IS (%)	AD IS (%)	ACF-MAE	Kurt IS	KS OoS (%)
Quantile (main body)	79.0	71.5	0.0501	2.57	72.4
Random seed = 1	78.7	70.7	0.0502	2.57	73.2
Random seed = 7	75.3	69.6	0.0498	2.56	74.0
Random seed = 42	77.6	70.0	0.0498	2.59	71.8
Random seed = 100	76.2	69.4	0.0500	2.59	72.3
Random seed = 2024	76.4	69.8	0.0499	2.54	72.9

The key observation is that ACF-MAE is essentially constant at ≈ 0.050 across all six rows, matching the $K = 18$ CHMM-N of the main body. IS KS pass rate remains below the $K = 18$ operating point (75.3–79.0%) because two Gaussian components cannot tile the full multi-regime return distribution. The combination is consistent with the reading that the binding low-K constraint in the Rydén et al. [10] setup is distributional rather than temporal: the ACF is already well reproduced at $K = 2$, and it is distributional fidelity that drives the preference for $K = 18$.

D Cross-Asset Dependence Supplementary Material

D.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 6.8 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 6.7) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [21, 41] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [39].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi \tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [23, 56]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\begin{aligned}\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; &\log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\log\Gamma\!\left(\tfrac{\nu}{2}\right) - d\,\log\Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2}\log|\Sigma| \\ &- \tfrac{\nu+d}{2}\,\log\!\left(1 + \tfrac{\mathbf{x}^\top\Sigma^{-1}\mathbf{x}}{\nu}\right) + \sum_{j=1}^{d}\tfrac{\nu+1}{2}\,\log\!\left(1+\tfrac{x_j^2}{\nu}\right).\end{aligned} \tag{34}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (34) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 6.8 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 24 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 6.8. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation in equation (5) re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 9.

Table 24. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$). Coefficients $\hat{\alpha}, \hat{\beta}$ and R^2 are reported for the factor equation used in equation (5).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

D.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 31 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 6.8. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature.

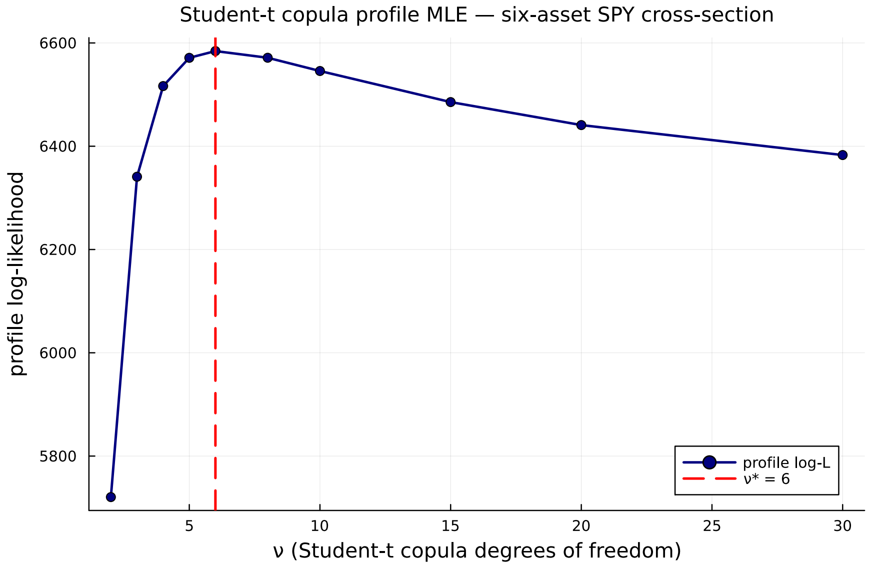


Figure 31. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

D.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 9)

Figure 32 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 6.8. The panel ordering is SIM, Gaussian copula, Student-t copula (each reported quantitatively in Table 9), plus a truncated C-vine variant with SPY as root for visual comparison. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.

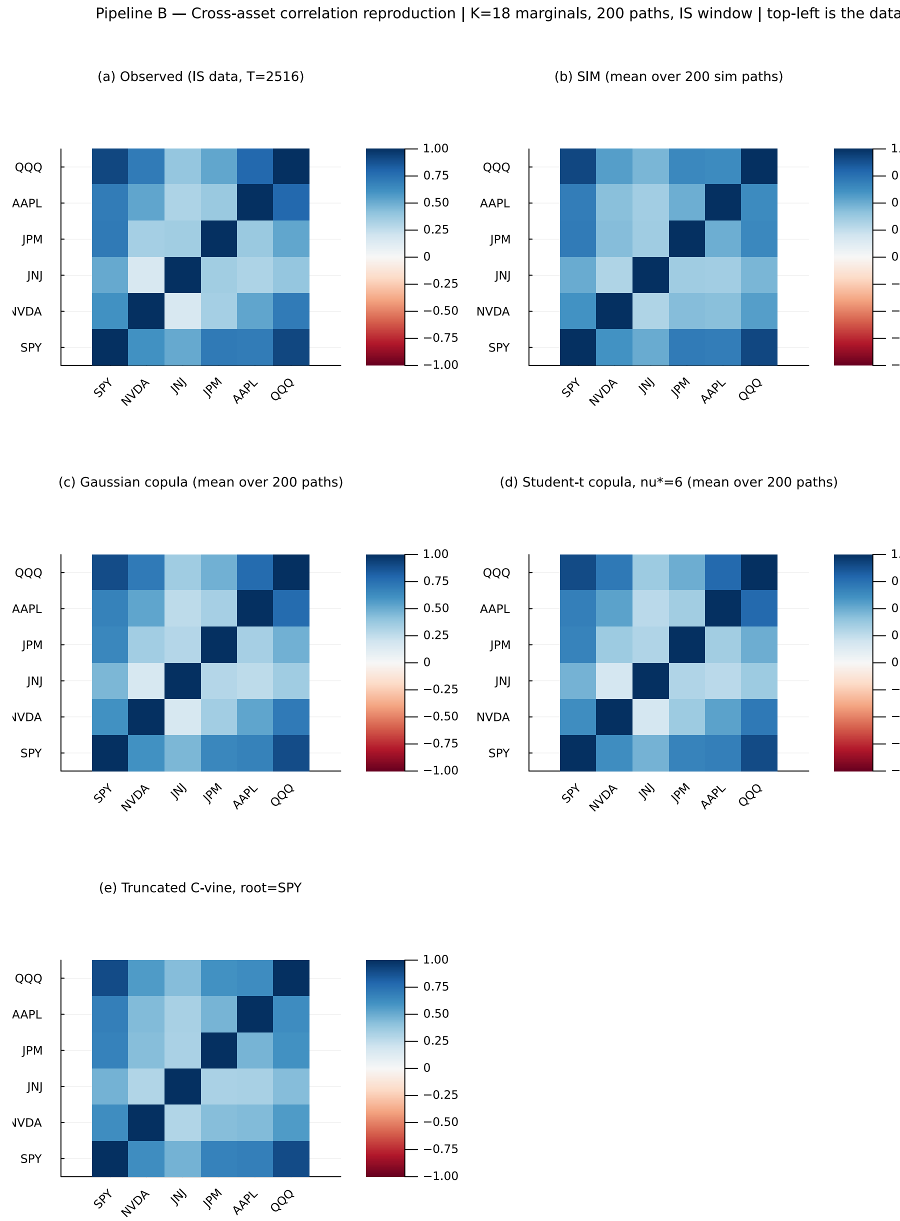


Figure 32. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 9). Panel (a), top-left: *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\mathrm{IS}} = 2{,}516$). Panel (b), top-right: path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c), middle-left: path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d), middle-right: path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e), bottom-left: path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 8; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

D.4 Per-Asset KS Bar Chart (Pipeline B, Sanity Check for Table 9)

Visualisation of the per-asset KS pass rates from Table 9; the SIM-vs-copula gap on JPM and AAPL is the visual headline.

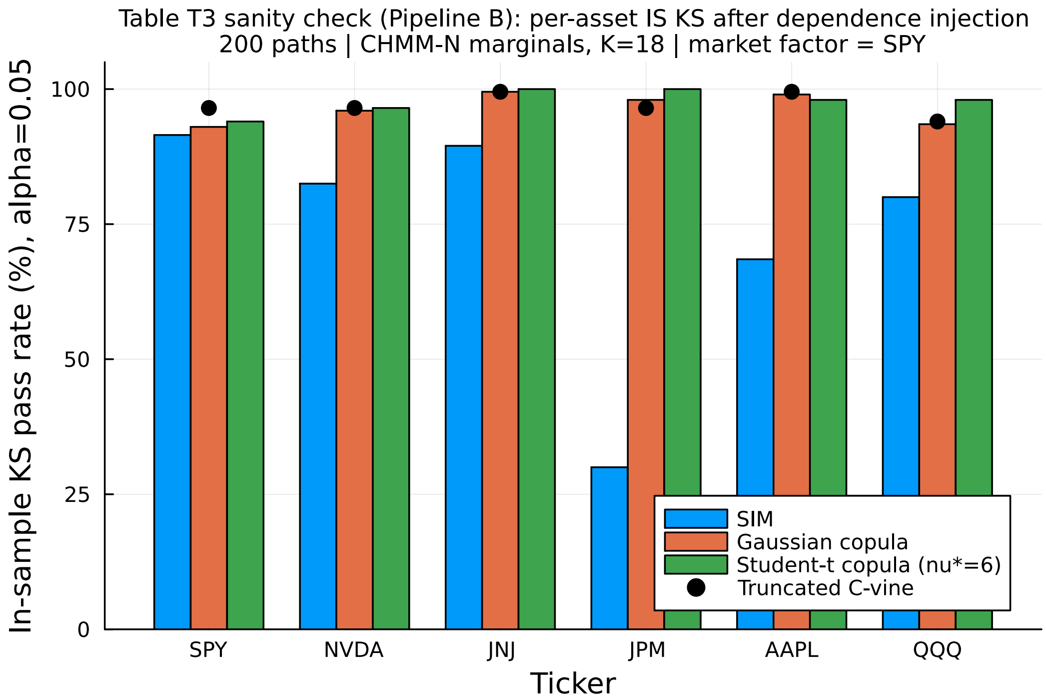


Figure 33. Per-asset IS KS pass rates after dependence injection (Pipeline B sanity check for Table 9). For each of the six tickers and each of the three dependence constructions (SIM, Gaussian copula, Student-t copula with $\nu^* = 6$), the bar reports the percentage of 200 simulated paths whose marginal distribution passes the two-sample KS test against the observed IS series at $\alpha = 0.05$. The market-factor asset SPY is approximately tied across constructions (it is the SIM engine). For non-market assets the copulas dominate because rank reordering preserves each CHMM marginal exactly; the SIM's affine propagation visibly degrades JPM and AAPL.

E Baseline and Per-Ticker Supplementary Material

E.1 OoS KS Test-Power Calibration

Table 25 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 6.5. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\mathrm{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 25. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\mathrm{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\mathrm{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\mathrm{IS}}, \hat{\sigma}_{\mathrm{IS}}$) at length T_{IS} (negative control)	0.0

Table 26. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$. "asymp pass" is the original Section 6.3 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383. CSV at `results/robustness/ks_block_bootstrap.csv`.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.8	0.812	0.371	0.999	98.2	99.0	99.6
GARCH(1,1)	25.4	0.042	0.000	0.201	**5.6**	**11.2**	**16.2**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.059	0.987	**82.4**	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 6.5 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

E.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM, GARCH, and discrete HMMs all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$" reduces a continuous p-value to a one-bit summary. Table 26 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

Two patterns emerge that the asymptotic pass-rate metric of the main panel does not surface. First, the cross-generator ordering on the mean p-value scale (bootstrap 0.81, CHMM-t 0.57, CHMM-N 0.55, CHMM-L 0.50, GARCH 0.04) is much more compressed for CHMM-vs-bootstrap than the 94.8–99.8% asymp pass rates suggest: the CHMM family lands at mean p-value ~ 0.5, the natural value for paths that are statistically indistinguishable from the observed series. Second, GARCH(1,1)'s drop from 25.4% asymp $\to$ 5.6–16.2% block-aware confirms concern (i): the asymptotic test is mildly lenient on volatility-clustering generators with the wrong marginal, and the block-aware test catches GARCH's misspecification more sharply. The CHMM family, by contrast, sees only a small block-bootstrap penalty (94.8% $\to$ 88–92%), indicating that its asymptotic-test passes are not a clustering-induced artefact.

E.3 Bandwidth, Bin-Count, and Multi-Seed Robustness

Fixed observed-sample MMD bandwidth. The original Section 6.6 computes MMD with median-heuristic bandwidth on the pooled (observed + simulated) window matrix, so the kernel scale γ varies with the generator under evaluation. We refit MMD across the full extended generator panel under a single bandwidth γ_{obs} computed once on 500 observed standardised 20-day windows from SPY IS. The legacy and fixed MMDs differ only in the third or fourth decimal across all ten generators in our reproduction (full numbers in `results/robustness/mmd_fixed_bandwidth.csv`); the cross-generator ranking under the fixed bandwidth is preserved. The bandwidth-heuristic concern is therefore real in principle but small in practice for this panel.

$K_{\text{disc}} = 13$ centroid ablation. The Bin-T NJ row of Table 5 changes both the bin count ($K_{\text{disc}} = 90 \to 13$) and the emission family (centroid $\to$ bin-conditional Student-t). To isolate the bin-count effect we add a parallel $K_{\text{disc}} = 13$ row with the standard centroid emissions: IS KS 0.0%, OoS KS 35.4%, IS AD 0.0%, OoS AD 63.4%, simulated IS excess kurtosis 0.63, ACF-MAE 0.0596. At $K_{\text{disc}} = 13$, centroid emissions cannot tile the SPY IS distribution at all (IS KS collapses to zero); the Bin-T NJ row's IS KS of 95.4% at the same bin count is therefore driven by the within-bin Student-t emissions, not by the bin count. The within-bin tail mass is the dominant contributor to that row's calibration. CSV at `results/robustness/kdisc13_centroid_ablation.csv`.

Multi-seed Monte Carlo on headline metrics. The paper reports headline statistics under a single global seed. We re-run the CHMM-N / -t / -L pipeline at $K = 18$ across ten alternative global seeds (base 20260422, increment 100) with $N_{\text{paths}} = 250$ per seed and report mean $\pm$ std for the headline metrics in Table 27. Seed-to-seed variability is small: IS KS std is 1.2–1.8%; OoS KS std is 1.8–3.6%; simulated IS kurtosis std is 0.06 for CHMM-N, 0.05 for CHMM-L, and 1.95 for CHMM-t (reflecting the well-known instability of the per-state ν_k ECM at moderate K, already discussed in Section 8). Kupiec $\text{LR}_{\text{uc}}^{1\%}$ has std up to 1.25 for CHMM-N because the breach count moves on a coarse 0–10 grid at $T_{\text{OoS}} = 572$. The single-seed headline numbers in Table 5 (e.g., CHMM-N IS KS 93.8%, simulated kurt 4.99) sit comfortably inside the multi-seed bands. CSV at `results/robustness/multiseed_headline.csv`.

Table 27. Multi-seed Monte Carlo on headline metrics. Ten global seeds; $N_{\text{paths}} = 250$ per seed (reduced from the headline panel's 1,000 for tractability across 30 fits, so the seed-to-seed std reported here is an upper bound on the std the headline panel would show at $N_{\text{paths}} = 1{,}000$).

Model	IS KS %	OoS KS %	sim Kurt	ACF-MAE	$\text{LR}_{\text{uc}}^{1\%}$	$\text{LR}_{\text{uc}}^{5\%}$
CHMM-N	94.7 ± 1.2	82.6 ± 1.8	5.03 ± 0.06	0.0467 ± 0.0006	2.12 ± 1.25	3.83 ± 0.00
CHMM-t	94.6 ± 1.2	85.6 ± 1.9	13.95 ± 1.95	0.0518 ± 0.0003	0.54 ± 0.15	3.19 ± 0.34
CHMM-L	94.4 ± 1.8	80.8 ± 3.6	6.71 ± 0.05	0.0537 ± 0.0003	3.26 ± 0.00	2.71 ± 0.59

E.4 Walk-Forward Rolling-Window Re-Estimation (Pipeline A)

This robustness check lives inside Pipeline A: the CHMM-N is still fit per ticker independently, only the fit window rolls forward. It does not introduce cross-asset coupling, and therefore should not be interpreted alongside the Pipeline B results in Section 6.8. Table 28 reports the quarterly walk-forward re-estimation results referenced in Section 6.7. The walk-forward protocol refits the CHMM-N every 63 trading days (one quarter) across the 572-day OoS window, feeding the realized returns of the prior quarter into the training set before each refit. The rolling protocol recovers a modest ~ 1 percentage point on JNJ and a substantial ~ 15 percentage points on JPM relative to the fixed-IS baseline, confirming that the stationarity assumption is the principal contributor to the JPM OoS gap while JNJ's OoS gap is already small under the fixed fit.

Table 28. Walk-forward (quarterly CHMM-N refit) vs fixed-IS CHMM-N on JNJ and JPM (Pipeline A, per-ticker independent fits; walk-forward only rolls the fit window, it does not introduce cross-asset dependence). $K = 18$, 1,000 simulated paths per (ticker, mode), seed = 20260420. IS window 2014-01-03 through 2024-01-03; OoS window 2024-01-04 through 2026-04-17 ($T_{\text{OoS}} = 572$). The walk-forward mode refits the CHMM-N every 63 trading days across the OoS window, feeding each completed quarter's realized returns back into the training set before the next refit. Columns report percent KS / AD pass rate at $\alpha = 0.05$, excess kurtosis, ACF-MAE of $|G_t|$, Wasserstein-1, and Hellinger distance.

Mode	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
JNJ fixed-IS	99.3	98.2	94.0	92.1	5.46	4.98	0.0322	0.095	0.0786
JNJ walk-forward	99.3	98.2	94.9	91.3	5.46	5.14	0.0322	0.095	0.0786
JPM fixed-IS	96.6	94.4	49.0	48.6	7.33	6.31	0.0449	0.165	0.0875
JPM walk-forward	96.6	94.4	64.3	63.6	7.33	5.96	0.0449	0.165	0.0875

E.5 Block-Bootstrap Baseline

Table 29 reports the full seven-metric panel for the stationary block bootstrap at block length $b = 10$ trading days, the temporally-aware non-parametric baseline added to Table 5.

Table 29. Stationary block bootstrap (block length $b = 10$ trading days) on SPY (1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	99.2	96.4	7.07	0.0568	0.092	0.0534	100.0
OoS	87.5	86.4	5.34	0.0512	0.225	0.1570	86.9

E.6 Discrete HMM with Bin-Conditional Student-t Emissions

Table 30 reports the full seven-metric panel for the Bin-T NJ variant referenced in Section 6.3: the discrete NJ transition matrix with $K_{\text{disc}} = 13$ quantile bins, but with each bin emitting from a bin-conditional Student-t density rather than from the bin centroid.

Table 30. Discrete HMM with bin-conditional Student-t emissions on SPY (no jumps, $K_{\text{disc}} = 13$, 1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	95.4	92.1	26.18	0.0619	0.116	0.0929	89.9
OoS	86.7	93.2	10.76	0.0538	0.177	0.1614	93.9

The bin-conditional Student-t fit picks $\nu \approx 2.71$ for the two extreme bins (bins 1 and 13, corresponding to the lower and upper quantile tails of the IS distribution) and $\nu = 50$ (upper bracket) for the eleven central bins, which explains both the large IS simulated kurtosis (26.18) and the very thin-scale central bins ($\sigma \sim 0.06$–0.28).

E.7 GRU Deep-Generative Baseline

Architecture and training details for the GRU baseline used in Section 6.3 are summarised here.

- **Architecture.** A single-layer Gated Recurrent Unit (GRU) with hidden dimension $H = 32$ followed by a Dense head with two outputs $(\mu_t, \log \sigma_t)$. Input is a single feature (the standardised return) over a context window of $w = 20$ trading days.
- **Loss.** Per-window negative log-likelihood of a Gaussian next-step density, equation (3).
- **Optimiser.** Adam, learning rate 1.0×10^{-3}.
- **Training schedule.** 20 epochs, full pass per epoch over $T_{\mathrm{IS}} - w$ training windows, randomly shuffled at each epoch.
- **Numerical stability.** $\log \sigma_t$ is clamped to $[-6, 4]$ inside the loss to prevent gradient explosion when the network attempts to assign vanishingly-small variance to extreme observations.
- **Pre-processing.** The input return series is standardised by its IS mean and standard deviation; predictions are re-scaled back at sampling time.
- **Synthesis.** The trailing w IS returns seed the recurrence; for each generation step the predicted Gaussian is sampled, the sample is appended to the context window, and the procedure rolls forward. The first 64 steps are discarded as burn-in.
- **Implementation.** Julia using `Flux.jl` [47] with the `Optimisers.Adam` state. Random seed for weight initialisation, training shuffling, and per-path sampling: 20260420, with deterministic per-path sub-seeds.

The trained model is callable through the same `simulate_gru(model, n_steps; seed_window)` interface as the CHMM and discrete-HMM models, so the seven-metric panel evaluates the GRU on byte-identical windows to the other generators.

Table 31. GRU deep-generative baseline on SPY (single-layer, hidden $H = 32$, window $w = 20$, 20 epochs, Adam $\eta = 10^{-3}$, 1,000 paths, seed = 20260420). Final training NLL: 0.2005.

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	18.1	13.7	2.85	0.0518	0.196	0.0966	37.4
OoS	52.5	55.0	2.21	0.0500	0.293	0.1603	69.7

E.8 Per-Ticker OoS Price-Simulation Figures and Full Path-Level Metrics (Unconditional CHMM)

This appendix collects (i) the full per-family path-level probabilistic-forecast metrics deferred from Table 12 in Section 6.10, and (ii) the per-ticker price-fan and terminal-distribution figures for the five non-body tickers NVDA, JNJ, JPM, AAPL, and QQQ. All figures use the same plotting conventions as Figure 6 in the main body: blue shaded bands are the 5–95% and 25–75% quantile envelopes of the 500 simulated paths, blue solid line is the simulated median, red solid line is the observed OoS price track. Terminal-histogram panels show the cross-path distribution of $\widetilde{P}_T$ at T_{OoS} in gray, with the observed terminal P_T^{obs} as a red vertical line and the simulated median as a blue dashed line. For brevity, the figures show CHMM-N only (the representative family; differences across the three emission families in the price-level metrics are small, see Table 32).

Table 32. Full path-level probabilistic-forecast metrics for OoS equity-price simulation, all (ticker, family) cells ($K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$; global seed policy in Section 3.5). $\overline{C}_{0.90}$: fraction of OoS steps at which the observed price falls inside the simulated 5–95% band (calibration); $\mathrm{MAPE}_{0.50}$: mean absolute percentage error between observed path and simulated median (sharpness); $\overline{\mathrm{CRPS}}$: horizon-average bias-corrected sample CRPS of Gneiting and Raftery [65]; $\overline{\mathrm{CRPS}}/S_0$: CRPS scaled by S_0 to remove price-level effects. Lower MAPE and CRPS, higher $\overline{C}_{0.90}$, are better. **Headline:** no CHMM emission family dominates on all four metrics across all six tickers; differences between CHMM-N, CHMM-t and CHMM-L live mostly in the return tails and propagate only weakly into aggregated price-level scores.

Ticker	Family	$\overline{C}_{0.90}$	$\mathrm{MAPE}_{0.50}$	$\overline{\mathrm{CRPS}}$	$\overline{\mathrm{CRPS}}/S_0$
SPY	CHMM-N	1.000	0.100	39.25	0.0835
	CHMM-t	1.000	0.112	42.66	0.0908
	CHMM-L	1.000	0.102	38.68	0.0823
NVDA	CHMM-N	0.698	0.339	30.62	0.6424
	CHMM-t	0.636	0.383	34.77	0.7293
	CHMM-L	0.666	0.411	36.58	0.7674
JNJ	CHMM-N	1.000	0.101	12.55	0.0780
	CHMM-t	1.000	0.098	12.62	0.0784
	CHMM-L	1.000	0.101	12.65	0.0786
JPM	CHMM-N	1.000	0.195	32.23	0.1881
	CHMM-t	1.000	0.241	40.38	0.2357
	CHMM-L	1.000	0.211	35.25	0.2057
AAPL	CHMM-N	1.000	0.108	21.30	0.1156
	CHMM-t	1.000	0.075	18.90	0.1025
	CHMM-L	1.000	0.077	18.82	0.1021
QQQ	CHMM-N	0.998	0.073	30.48	0.0763
	CHMM-t	0.995	0.079	31.63	0.0792
	CHMM-L	1.000	0.118	41.12	0.1029

E.9 Expanded GARCH-Family Baselines

We add five conditional-volatility baselines to the single GARCH(1,1) Gaussian row of Table 5 and the two-regime MS-GARCH row of Table 7. All are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\mathrm{IS}} = 2{,}516$); simulation follows the same recursion used in estimation. Full results are in Table 6; model specifications are below.

EGARCH(1,1) Gaussian. The exponential GARCH of Nelson [70] evolves $\log \sigma_t^2$ in levels:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha \left(|z_{t-1}| - \sqrt{2/\pi} \right) + \gamma z_{t-1}, \qquad z_t = (r_t - \mu)/\sigma_t, \quad z_t \sim \mathcal{N}(0,1). \tag{35}$$

The log-variance formulation removes the GARCH positivity constraints, and the leverage term γz_{t-1} captures the asymmetric response to negative returns.

GJR-GARCH(1,1) Gaussian. The threshold GARCH of Glosten et al. [71] / Zakoïan [72] augments GARCH(1,1) with an indicator on negative innovations:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}\{\varepsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2, \tag{36}$$

with $\varepsilon_t = r_t - \mu$. Stationarity requires $\alpha + \beta + \gamma/2 < 1$; positive γ indicates leverage.

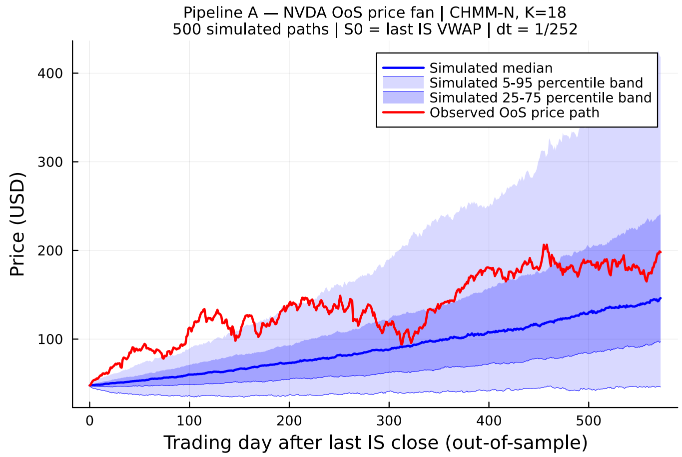
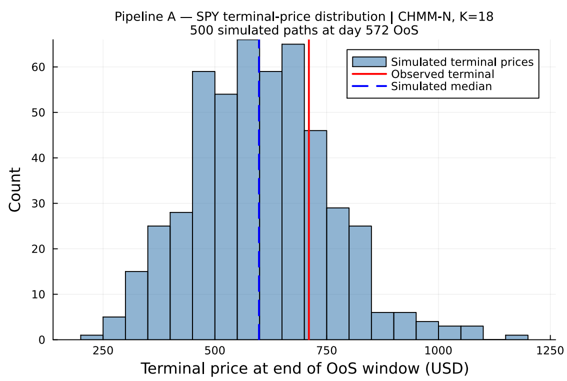
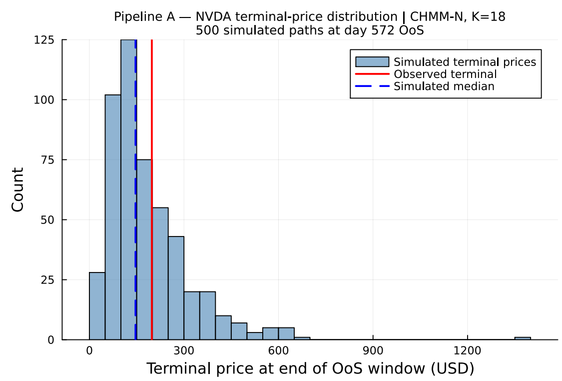


Figure 34. NVDA OoS price fan, CHMM-N ($S_0 = 47.67 \to P_T^{\text{obs}} = 198.07$, a 4.15× rally). The observed path exits the 25–75% interquartile band by mid-horizon but remains inside the 5–95% envelope; horizon-wide coverage under CHMM-N is 0.70 (CHMM-t: 0.64, CHMM-L: 0.67; Table 32), the lone calibration outlier in the panel.



(a) SPY terminal.



(b) NVDA terminal.

Figure 35. OoS terminal-price distributions under CHMM-N for SPY and NVDA. The right-skew is characteristic of log-returns aggregated to prices; the SPY terminal sits close to the simulated median, while the NVDA terminal sits in the upper-right of the simulated distribution (the realised rally is in the upper tail of the CHMM-N terminal density).

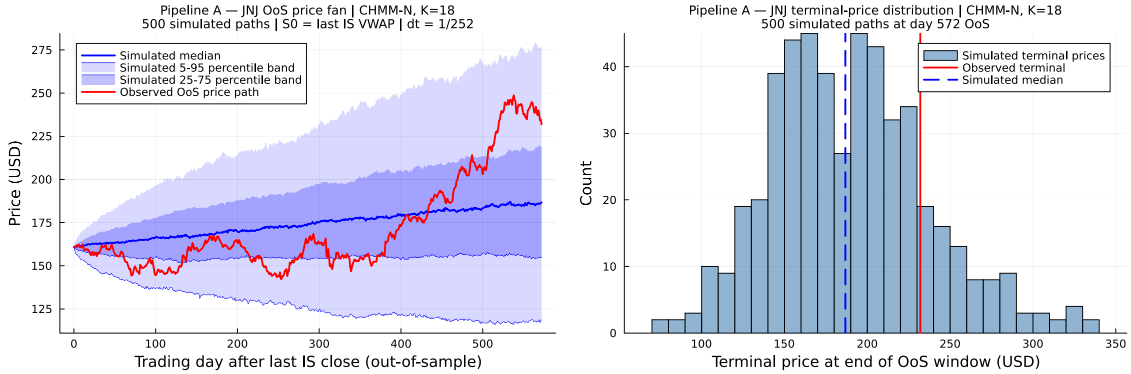
GARCH(1,1) Student-t innovations. A direct tail-aware competitor to CHMM-t: the GARCH(1,1) variance recursion is unchanged, but innovations follow a standardised Student-t with $\nu > 2$ degrees of freedom rescaled to unit variance. The log-likelihood uses the Student-t density:

$$r_t - \mu = \sigma_t \cdot z_t, \qquad z_t = \eta_t \sqrt{(\nu - 2)/\nu}, \qquad \eta_t \sim t_\nu. \tag{37}$$

The fitted $\hat{\nu} = 6.89$ on SPY IS sits inside the moderate-tail regime the equity-risk literature typically reports.

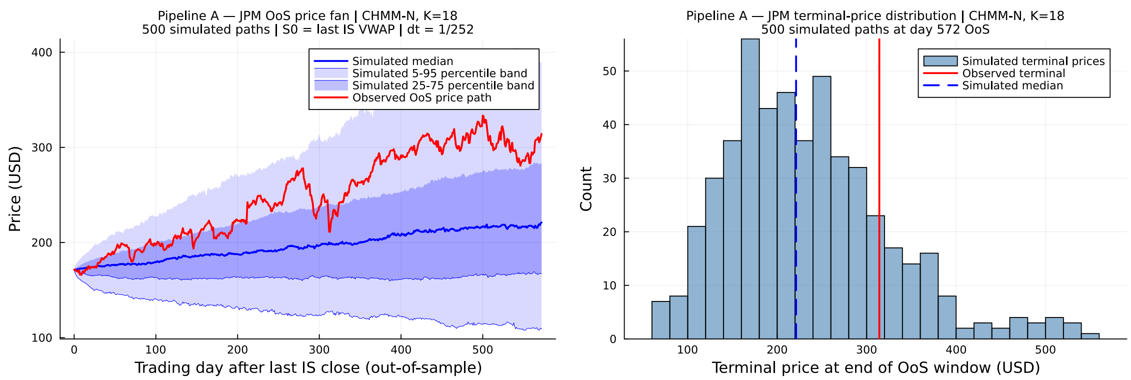
HAR-RV on daily squared returns. The heterogeneous-autoregressive model of Corsi [73] models realised variance as a mixture of daily, weekly, and monthly components:

$$\log \mathrm{RV}_t = \beta_0 + \beta_d \log \mathrm{RV}_{t-1} + \beta_w \log \overline{\mathrm{RV}}_{t-5:t-1} + \beta_m \log \overline{\mathrm{RV}}_{t-22:t-1} + \eta_t. \tag{38}$$



(a) Price fan. **(b)** Terminal distribution.

Figure 36. JNJ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as in Figure 6).

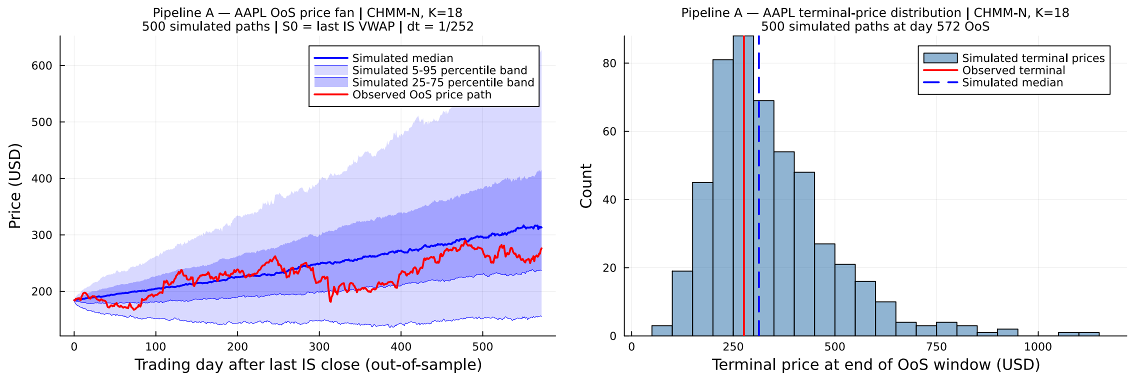


(a) Price fan. **(b)** Terminal distribution.

Figure 37. JPM OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as in Figure 6).

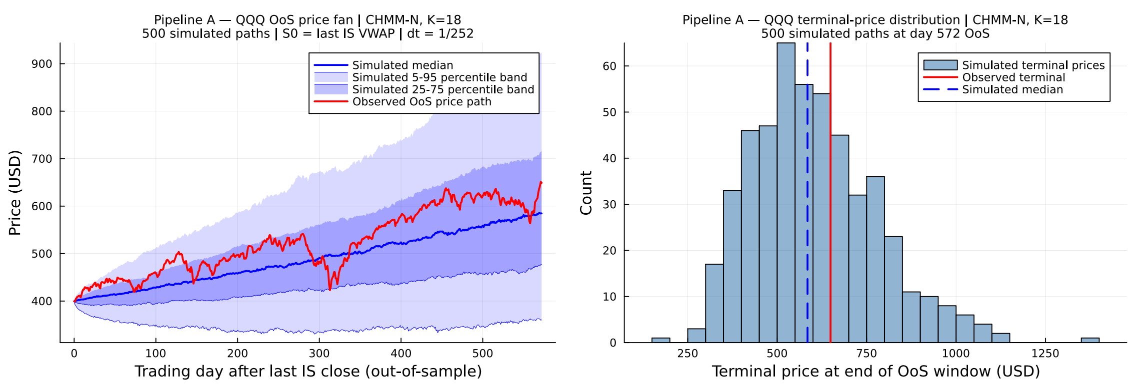
In the absence of intraday bars we use daily squared demeaned returns as the RV proxy; simulation samples $\eta_t \sim \mathcal{N}(0, \hat{\sigma}^2_\eta)$ and rolls forward the recursion, then draws $r_t - \mu = \sqrt{\mathrm{RV}_t}\, z_t$ with $z_t \sim \mathcal{N}(0, 1)$.

MS-GARCH $K = 3$. The three-regime extension of the Markov-switching GARCH(1,1) in Section 3.3. Parameter layout is three regime triples $(\omega_k, \alpha_k, \beta_k)$ plus six transition-matrix logits (softmax-normalised per row, enforcing a non-trivial diagonal); fit by grid-initialised Nelder-Mead as in the $K = 2$ case. States are canonicalised by ascending unconditional variance, producing the calm / medium / stress decomposition $\sigma = [0.10, 1.77, 10.04]$ on SPY IS.



(a) Price fan. **(b)** Terminal distribution.

Figure 38. AAPL OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as in Figure 6).



(a) Price fan. **(b)** Terminal distribution.

Figure 39. QQQ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as in Figure 6).